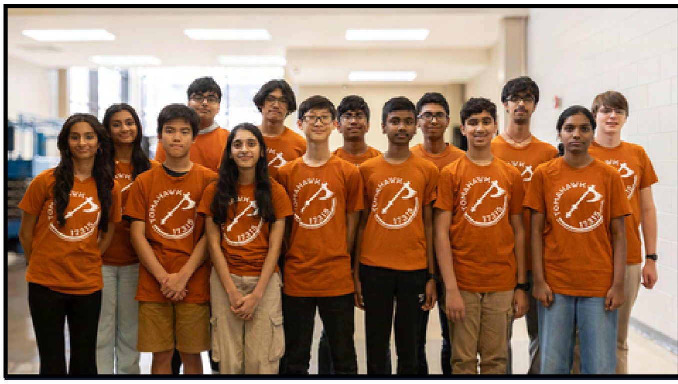
Tomahawk
17315

Hardware Portfolio

2025-2026

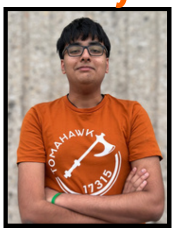
OUR TEAM



We are team #17315, Westwood Tomahawk, from Westwood High School based in Austin, Texas. The 15 of us came together with different backgrounds from rookies to years of experience. In our 5th year as a competitive team, we seek to empower young minds both locally and internationally by introducing them to FIRST, aid and nurture our fellow FIRST members, and provide others the same opportunities and resources that we are lucky to have access to.

Deekshita

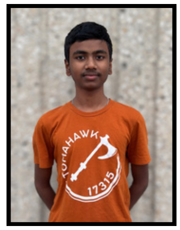
Ritvij



Aditya

Isaac

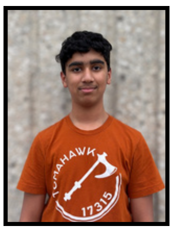
Karthik



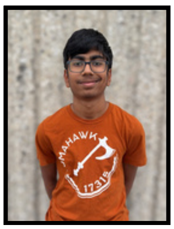
Prahastha

Aron

Minessh



Shrikar



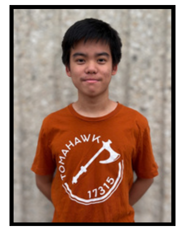
Saloni

Sanaika

Jusuf

Arjun

Daniel



The Hardware TEAM

Karthik B.

2 Years in FTC

About Me

I am a sophomore on Team 17315 Tomahawk. I joined FTC for my hobby of building random things that are annoying, but fun to work with, like the robots in FTC. I also joined FIRST because of its great community. I really like to play video games in my free time.

Deekshita S.

2 Years in FTC

About Me

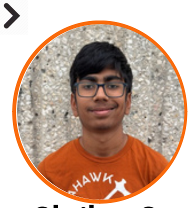
I am a sophomore on Tomahawk. I joined FTC to be able to improve my design and programming skills, and enjoy 3D printing in my free time. This year I hope to expand my design skills and explore more complicated mechanisms!

Arjun A.

2 Years in FTC

About Me

I joined FTC Team 17315 Tomahawk as a sophomore at Westwood HS to push the boundaries of STEM as well as to learn more about Outreach and Hardware. FTC has given me a platform to build on my knowledge, learn from teammates, and mentor others in pursuing their STEM goals.

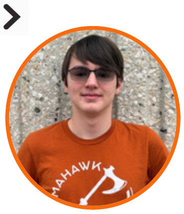


Shrikar S.

1 Year in FTC

About Me

I am a freshman on Team 17315 Tomahawk. In Tomahawk I am driver 1 and on hardware. I joined FTC because of my passion for robotics. I also am on team 99206V, which is a part of Vex V5. I also like to code and play chess in my free time.



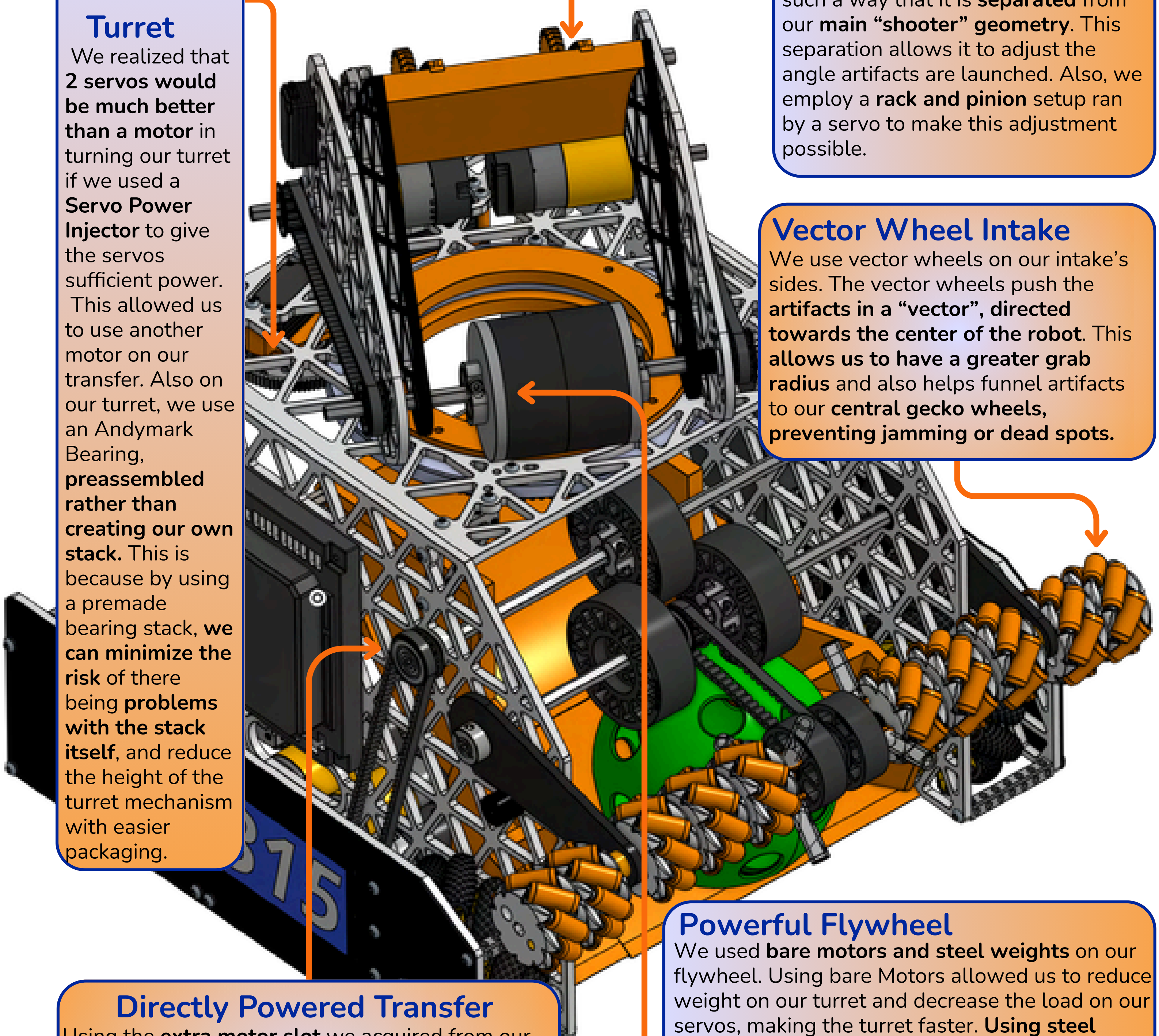
Aron H.

2 Years in FTC

About Me

I am a sophomore on Tomahawk. I love taking up random CAD, 3D printing and coding projects in my free time. At Tomahawk, I hope to push the possibility of FTC design to its limits this year while having fun with my teammates!

OUR ROBOT



Servo Powered Turret

We realized that **2 servos would be much better than a motor** in turning our turret if we used a **Servo Power Injector** to give the servos sufficient power. This allowed us to use another motor on our transfer. Also on our turret, we use an Andymark Bearing, **preassembled rather than creating our own stack.** This is because by using a premade bearing stack, **we can minimize the risk** of there being **problems with the stack itself**, and reduce the height of the turret mechanism with easier packaging.

Adjustable Hood

We have our **hood geometry** made in such a way that it is **separated** from our **main "shooter" geometry**. This separation allows it to adjust the angle artifacts are launched. Also, we employ a **rack and pinion** setup ran by a servo to make this adjustment possible.

Vector Wheel Intake

We use vector wheels on our intake's sides. The vector wheels push the **artifacts in a "vector", directed towards the center of the robot**. This **allows us to have a greater grab radius** and also helps funnel artifacts to our **central gecko wheels, preventing jamming or dead spots.**

Directly Powered Transfer

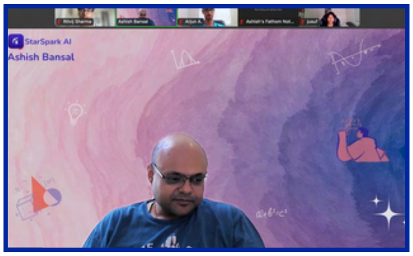
Using the **extra motor slot** we acquired from our turret, we used it to make **separately powered intake and transfer mechanisms.** This would allow us to **control when the artifacts entered the outtake**, giving us more control. This was an issue that we faced in our previous iterations **which led to more difficult driving and inconsistent shooting.**

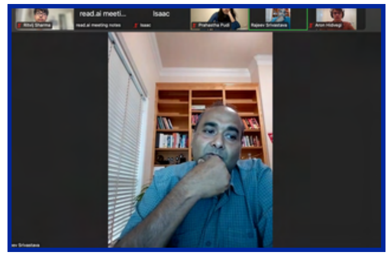
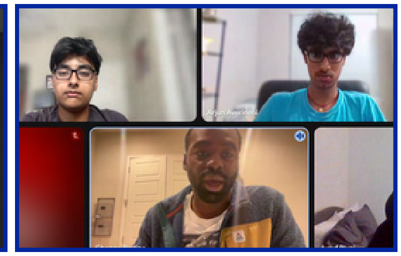
Powerful Flywheel

We used **bare motors and steel weights** on our flywheel. Using bare Motors allowed us to reduce weight on our turret and decrease the load on our servos, making the turret faster. **Using steel weights was a trade-off** between ***higher spinup time and reduced recovery time,*** *due to a higher* ***moment of inertia***. We chose to use them to make our **robot more consistent and reduced time to shoot 3 artifacts.**

Meet SandStorm

Design Process Creation

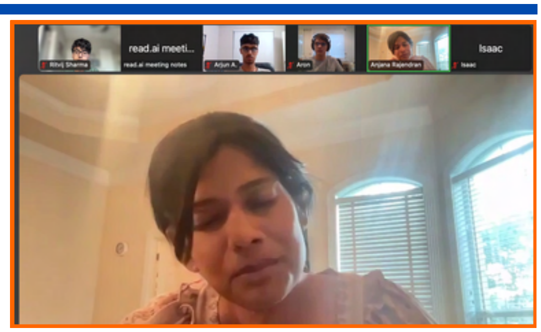



Throughout the season, we learned numerous industry practices and methods through meetings and visits with industry professionals. These experiences not only helped us collaborate more effectively as a team, but also exposed us to real-world engineering processes that strengthened our own design workflow. In addition, guidance from the mentors we worked with throughout the season introduced us to professional standards and best practices. Below are some of the key practices we learned through these interactions and experiences that we chose to implement during our season.

Tyrex Tour

At this tour, we were able to learn about the machines Tyrex used to stress test components under different conditions such as heat, water, salt, cold, and rocky motion. Mr. Coffman suggested to us from this tour that although it may be difficult to test parts with machines, it would be a good idea to **stress test before building** using stress testing softwares such as **Autodesk Simulation as well as physical tests on our prototypes.** We saw this in action at the tour itself as we saw Tyrex parts being put in high pressure or high temperature chambers to test their resilience.

Stanford Alumni Meeting



In this meeting with Mrs. Rajendran, we learned about the **importance of planning** before in detail to avoid issues down the road, for us we felt we could apply this by making a **very well constructed cad model** of the robot before building it , accounting for **every single screw** to ensure issues wouldn't show up later. Also, she taught us about the **"Pareto Principle"**, a principle which she told us plagues many even in her own industry where **80% of the work gets done by 20% of the people.** Mrs. Rajendran suggested that to counter this problem, we should ensure **equal distribution of work** as well as a good environment.

Microsoft Tour

We were given the opportunity to tour the Microsoft Building and speak with their employees on the industry processes that they implement to work better together. One of the employees, Mr. James explained to us how their team attempts to implement **NUDDs(New, Unique, Different, and Difficult)** to help optimize their design. Mr. Santosh, our industry mentor also explained how they implemented **BDRP in their team specifically(Brainstorm, Design, Refine, Perform)** along with NUDDs to allow to create a more organized structure for creating products. Mr. Prakash, their Product Manager also explained how similar to FTC's constraints, his job was to take data from customers and use that to **create constraints so the engineers would create better products.**

Design Process Creation

Tesla Gigafactory Tour

In our visit to the Tesla Gigafactory in south Austin, we were able to see the manufacturing process occur in front of us. There we learned about Tesla's organization of production where our tour guide Mr. Matthew explained how **Tesla implements SPARC which is a customer based design process meant to beat their competitors in the market** by orienting their cars to be fully developed from customer feedback. The cars customizability also makes it so the car is able to cater to different customers needs differently. At the Gigafactory, we were also able to learn about Tesla's **MAYA (Most Advanced Yet Acceptable)** policy in terms of product design in their **customer centric approach**.

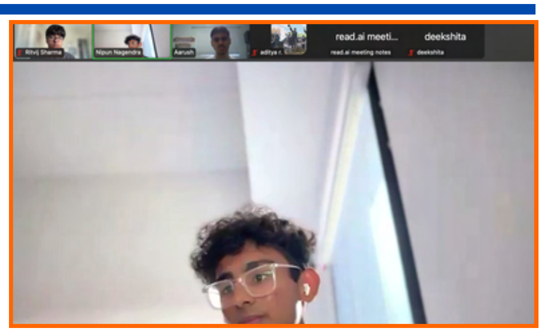
Indeed Building Tour

We were able to visit the Indeed Building in our community and were able to meet with Mr. Tim Kao. Mr. Kao explained Indeed's strategy to solving issues using strategies such as **Swarming** (Everyone helps give input when a critical issue arises and works together to quickly resolve it). Mr. Kao also taught us about how Indeed tried to implement **RAID** (Risk, Assumptions, Issues, and Dependencies) analysis into their planning whenever they needed. From Mr. Kao we were also able to understand key points of problem analysis, To **understand your own strengths and weaknesses, your opponents strengths and weaknesses, and the resources you have available and making the best possible out of the situation.**

TMI Tour

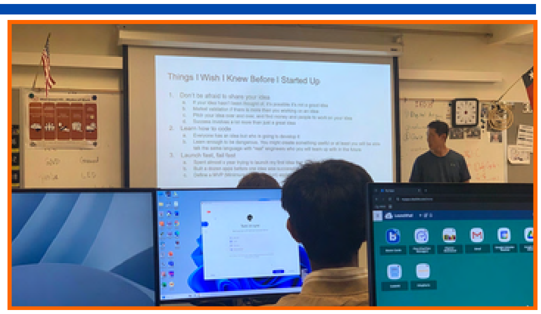
We were able to visit the Texas Material Institute facility at the UT Campus. There, we met with **Doctor Ioana Gearba-Dolocan and Doctor Andrei Dolocan**. These two professors explained they worked together in the Material institute laboratories using many **different types of microscopes to scan material surfaces and view the material at an atomic or even chemical level.** Dr. Ioana explained to us how scientists would **use the transfer of electrons on the materials to see the quality of the material.** This tour also taught us the importance of material testing and truly understanding even the **chemical and physical qualities of the materials we were to use to better understand their ability to sustain hits and overall last longer.**

Nipun Nagendra Meeting



We were then able to meet with Nipun Nagendra, 6210 Stryke alumni. Nipun told us about his own experiences as the Software Lead and Project Manager on Stryke. He also gave us advice on design saying that last year his team followed **KAIZEN**, a Japanese phrase that means "change for the better". Nipun explained that his team implemented this idea by using it to identify weaknesses with their robot where they could improve. **KAIZEN** also emphasizes the **importance of constant testing and performance reviews** to be the main driver for your products improvement.

John Hwang Meeting



During our meeting with John Hwang, **founder of several companies including Sidestep**, we learned about key principles he relied on throughout his career as both an employee and entrepreneur. One of the main ideas he emphasized was **KISS(Keep It Simple, Stupid), a design philosophy that prioritizes simplicity to avoid unnecessary complexity when solving problems.** He also highlighted the importance of **creating flexible, modifiable designs that allow for efficient iteration as improvements or changes are needed.**

Design Process Creation

Design Process

Using our **learnings** from some of our Outreach Connect Meetings, we decided to create our seasons design process that we wanted to follow that could combine the best of all worlds from all of these design principals that we had learned to ideally create an **optimal** design process which would ensure our bot was **unique**, **modular**, and **consistent**.

Plan

Inspired from our Tesla tour, Microsoft tour, Indeed building visit, and our meeting with Mrs. Rajendran, we knew that planning would be essential. Using **Tesla's SPARC mindset**, we attempted to plan around **our drivers' and coders' skills** and desires similar to how Tesla planned around their **customers**. We also **defined** our **constraints** from the start, based on the time and resources we had available. **The Pareto Principle** was also something we wanted to avoid. Inspired by Microsoft we used **NUDD (New, Unique, Different, Difficult)** ideas as a core part of our planning phase.

Design

From our meetings, we were inspired by Tesla's **MAYA principles** and applied them to our design process. We aimed to incorporate scoring features that distinguished our robot while ensuring simplicity and consistency in performance. This also aligned with the principle we learned in our meeting with Mr. Hwang's **KISS principle.** Another principle we hoped to implement was **efficient CADing**. Last year we had to make **many adjustments because our CAD was not detailed**. This year our hope is to be able to avoid that by **designing everything, down to the last screw.**

Simulate

Our next step was to simulate everything we do beforehand. Not only to create all of our components in a Computer Automated Design software (CAD) but to also **simulate stress and tension** where it was applicable. Although we weren't able to stress test every aspect of our robot, we attempted to design our parts to be sturdy and to last. Our team policy was to **stress test everything that was made of wood or aluminum,** as those were the components that were the most susceptible to damage. This was **inspired from our teams tour of Tyrex** where we saw their usage of stress test machines.

Build

The next step we had in our design process was the actual build phase. this phase was designed to assemble together the components that we designed. Our main goal from this phase was to assemble the robot in such a way that it wouldn't break.To achieve this, we used ***loctite*** and other thread lockers to ensure the screws would be secure. Also in this step, we decide to have a **hardware check** the day before our competition to identify any potential risks on our robot and correct them as soon as we could. This precaution we hoped would allow for **more consistency in performance.**

Test

The final step of our design process is testing, which plays a critical role in improving both our designs and overall robot performance. **Inspired by Tesla's review-based design process**, we continuously evaluate our mechanisms during and after production. Because components are almost always in use on the robot, testing is ongoing rather than a single step. Following the **principles of Kaizen**, we regularly **identify areas for improvement** and iterate by **returning to the design or build phase** whenever performance falls below our standards.

Summary

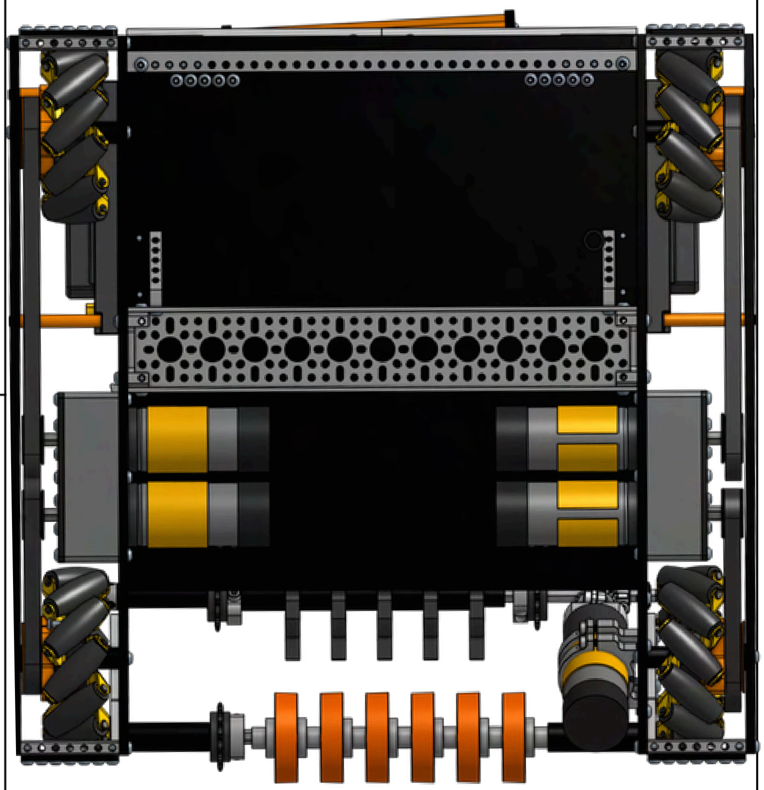
In this section you can see a **quick snapshot of our robot and some of its best features** as well as how **some aspects changed throughout the season**.

The features that we highlight are:

- **Drivetrain**
- **Intake**
- **Transfer**
- **Outtake**

We hope that this will provide some background information before diving into **the step by step evolution of our robot**.

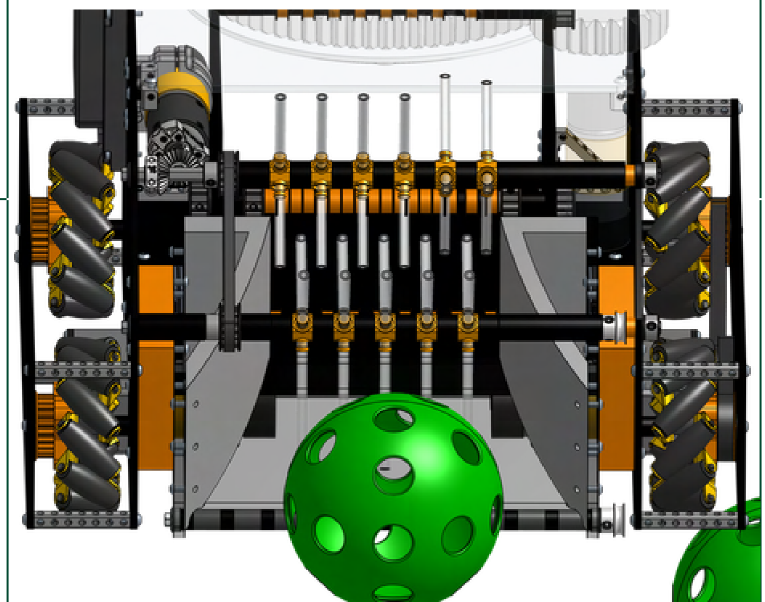
Drivetrain

DESCRIPTION		
What	Our drivetrain is made up of wooden plates with 4 **wheels driven by 4 motors using belted motion transmission**. Another aspect of our robots drivetrain is that we have **two channels that connect the two inner plates** for better stability, **one in the middle and one in the back**. We also have 3D printed blocks between the chassis and motor plate to have greater motor rigidness.	
Why	Despite the evolution of our other mechanisms, the overall structure and style of our **Chassis** stayed **relatively the same** throughout the course of the season. This was due to the chassis' relative consistency and effectiveness, which we saw no reason to change throughout the season.	

Intake

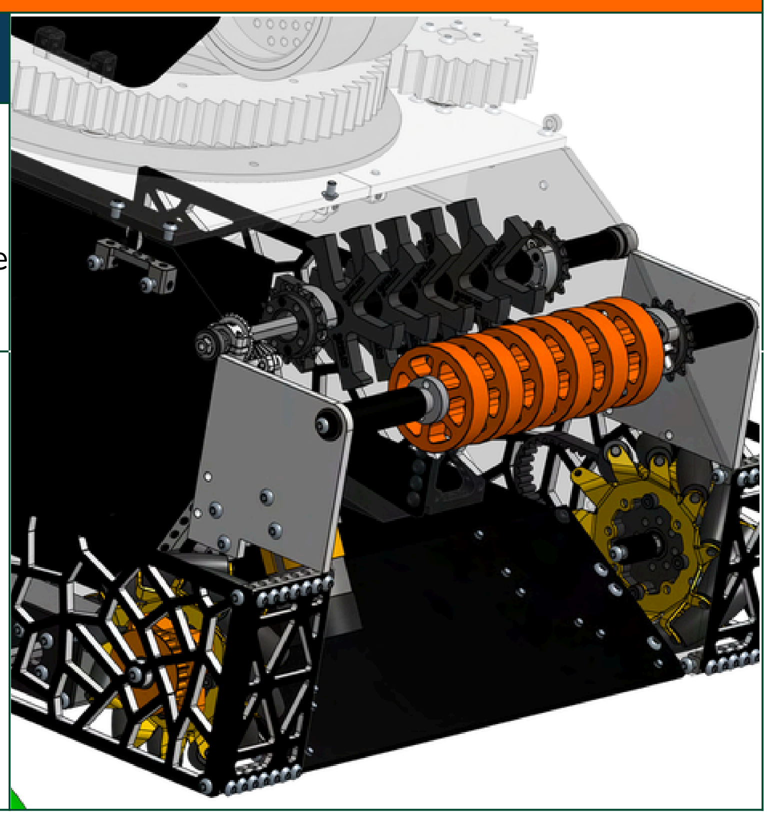
VERSION 1.0

DESCRIPTION	
Pros	Our Intake was made up of a simple ramp attached to the front of the robot with an **intake made up of 2 stages of tubing**. We made this design simple in order to get consistency first, and we planned on improving it in the future.
Cons	Although it was a simple design, a problem that would arise was that the **tubing could easily get stuck in each other due to being so close** together. Also, this design was unable to pick up artifacts and **be within the 18 by 18 by 18 frame**. Thus, we had to change how we wanted to intake to be able to fit within the restrictions and fix the issues with the tubing easily getting stuck.



VERSION 1.25

DESCRIPTION	
Pros	Our Intake was made up of a simple ramp attached to the front of the robot with an **intake made up of 2 stages of complaint wheels and tubing, for better grip.** Since we were changing our intake quite a bit, we wanted to keep it simple.
Cons	Although it was a simple design, a problem that would arise was that **the artifacts would sometimes get stuck between the intake and the transfer** (either the left part or right). This was greatly **hurting our intake cycles and was making us really inconsistent** at intaking.

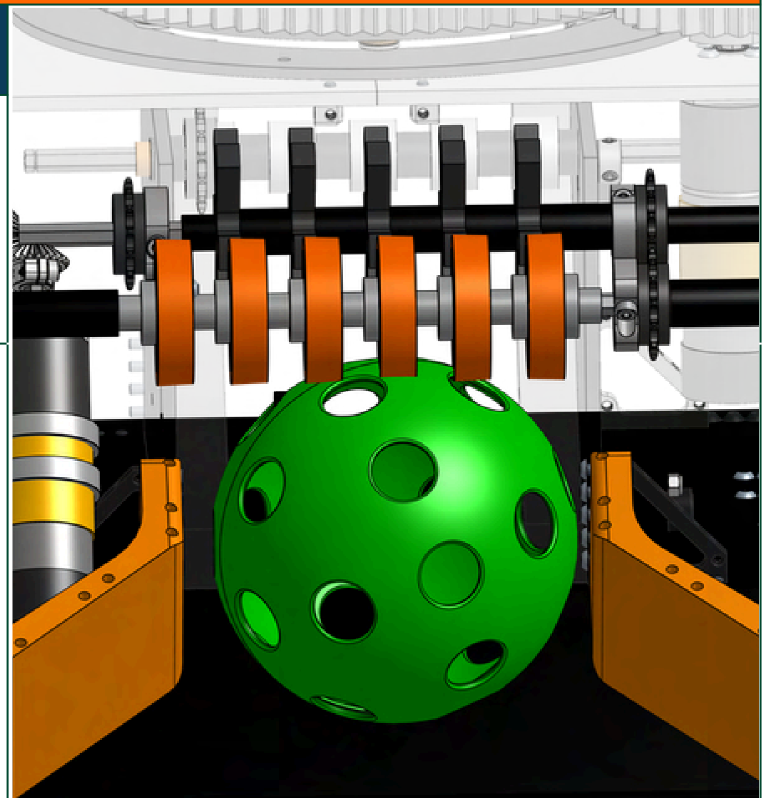


Intake

VERSION 1.50

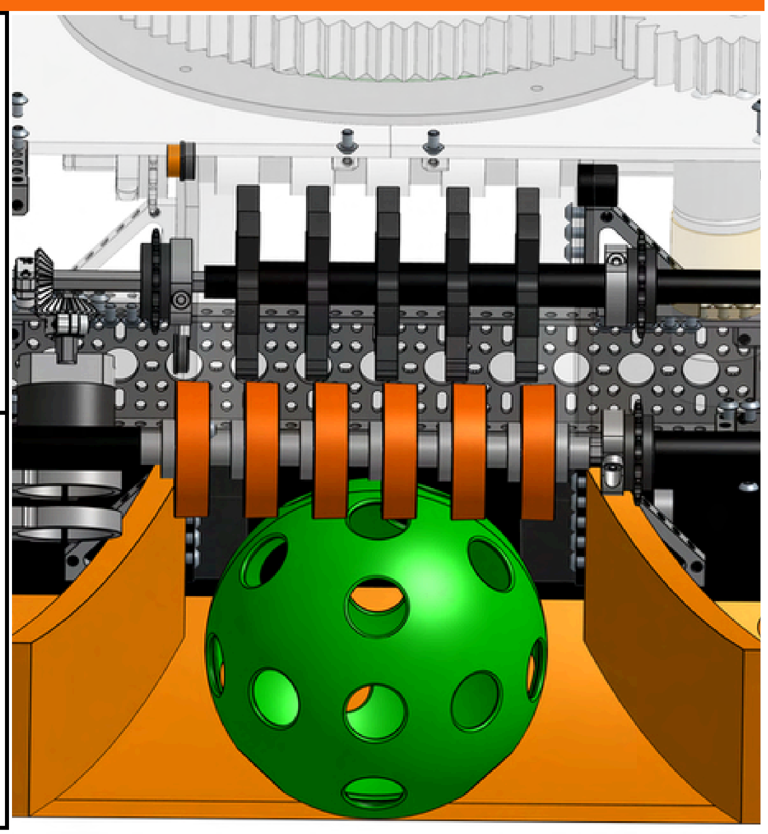
DESCRIPTION

Pros	In our last iteration, we had identified our **major issue to be being unable to intake the artifacts consistently.** To alleviate this issue, **we added guides on both sides of our ramp**, to vector the artifacts smoothly up the ramp.
Cons	After installing the guides onto the ramp, **we realized that since we had created the height of the guide to be greater than half the size of the artifact** to ensure it couldn't roll away, **it was out of the 18in. x 18in. x 18in. FTC Robot requirement** now. This led us to have to think of a way **in which the guides would fit and would also be effective at guiding the artifact.**



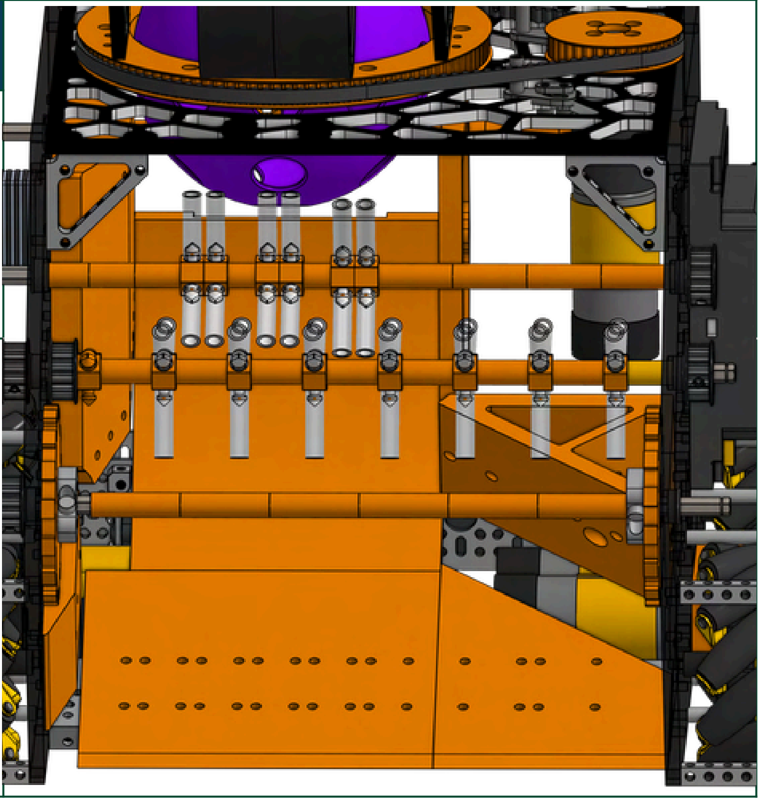
VERSION 1.75

Pros	After testing the previous ramp, we redesigned the geometry so it could **fit between the transfer and intake** within the 18-inch limit. This involved **increasing the curvature of the ramp** and attaching it lower to meet the 18 × 18 × 18 in. requirement. This geometry made it **easier for us to intake** and thus score faster. We hoped this would **decrease** our **cycle** times.
Cons	We realized soon that the intake speed could still improve. We wanted our intake to **grab the artifacts in under 1 second** and this intake took almost **2 seconds to fully control the artifact**. This change would be crucial to earn the artifact Ranking Point which was important to overall rankings.

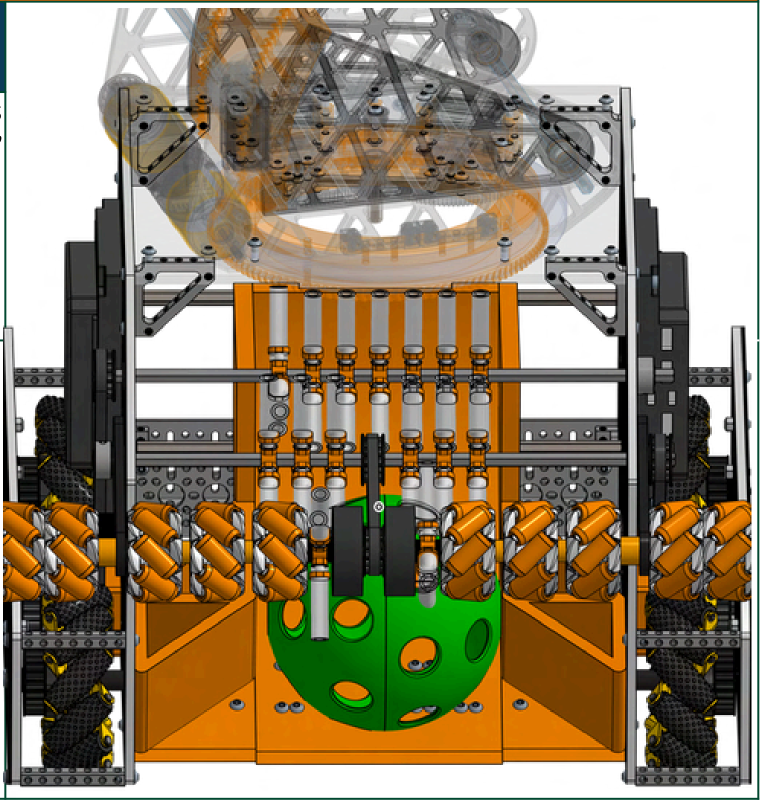


Intake

VERSION 2.0

DESCRIPTION		
Pros	The usage of rubber bands on our intake allowed the **artifact to compress much better** to the shape of the **rubber bands** and would allow for greater grip on the artifact. This led to faster transfer times and overall **greater efficiency.**	
Cons	Although the rubber bands worked great, we quickly realized that the **rubber bands** were **prone to snapping** and catching on other robots, as well as catching on the transfer, making them a weakness. Also, the **rubber bands** would sometimes pull the artifacts into **“dead spots”** which led to jamming of artifacts. To avoid these issues in the future, we decided to redesign once again.	

VERSION 3.0

DESCRIPTION		
Pros	We switched to vector and gecko wheels to improve our intake. The vector wheels’ 45-degree rollers helped center artifacts more effectively, we hoped that this upgrade would also eliminated our jamming issues. that we faced earlier.	
Cons	After using vector wheels, they initially seemed to work well however, we soon **realized that the vector wheels even with tubing didnt have good grip**. The ramp geoemtry we soon realized was also faulty as it make it very hard for our driver to intake two artifacts at once. We knew that in order to score well, we would have **to edit our guides and ramp to be able to prevent these jams**.	

Intake

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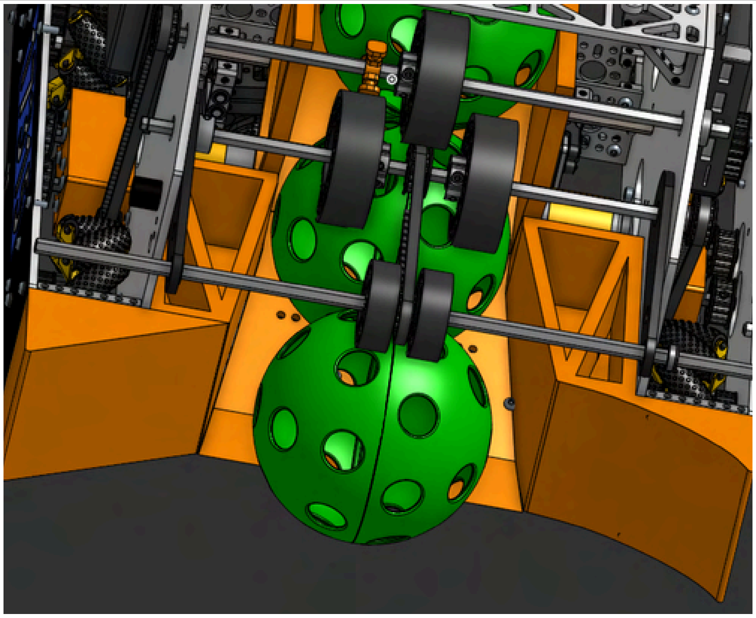
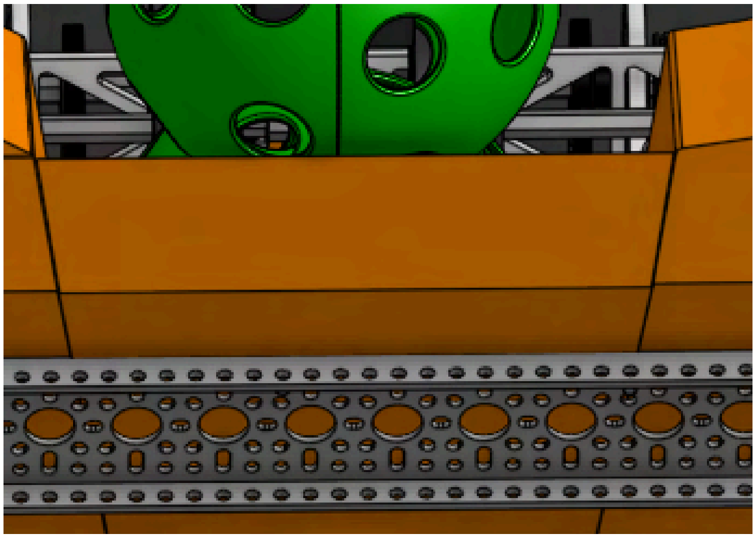
<tr><th colspan="2">DESCRIPTION</th><td rowspan="2">
</td></tr>
<tr><td>CURRENT VERSION</td><td>In our third iteration for our shooter, we changed how our hood worked and allowed is to rotate freely on the flywheel axle. This was because it increased the stability in our hood, for the shooter in our previous iteration, had faced issues with the bearings shifting around from tolerances. We also changed our turret to be rotated by an Andymark bearing, a change we made due to the greater consistency of a premade industry crafted bearing compared to creating your own using a bearing stack. Additionally, we manufactured steel weights to increase the rotational inertia of our flywheel which improved consistency and reduced recovery time allowing us to “rapid fire” artifacts together</td></tr>
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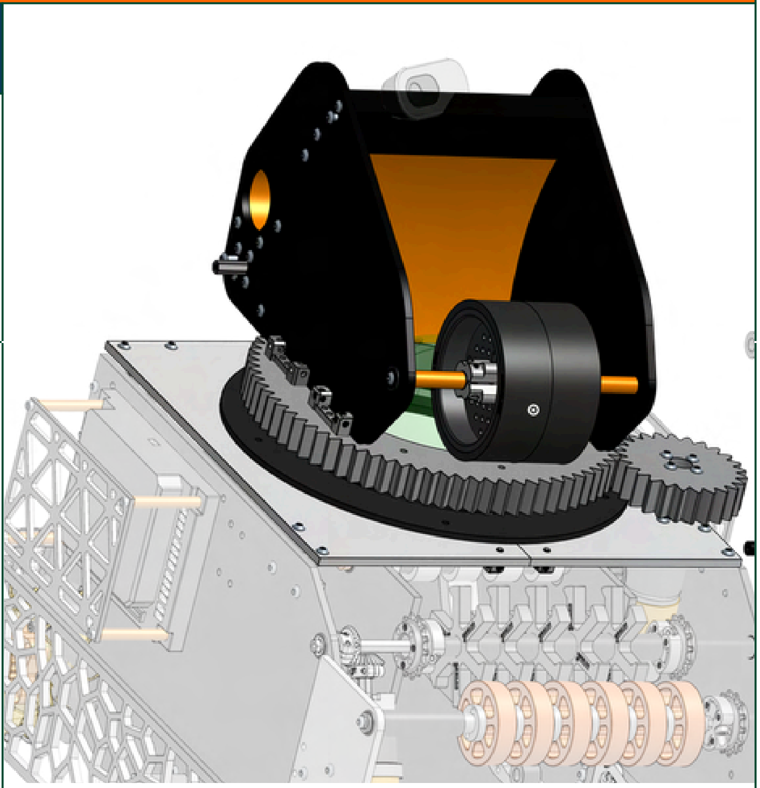
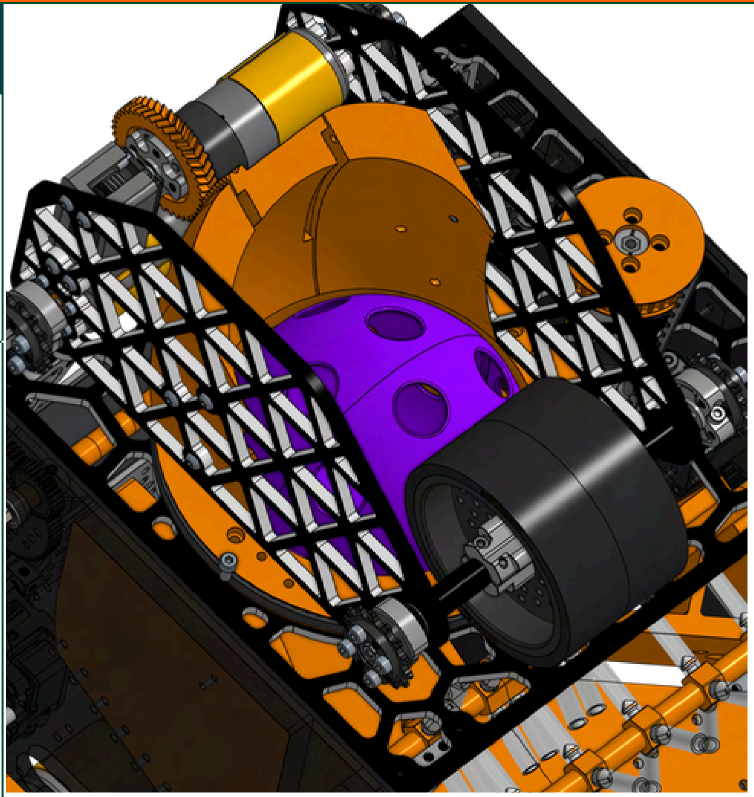
Outtake

VERSION 1.0		
DESCRIPTION		
Pros	We combined two Rhino wheels into a single flywheel for **greater contact and inertia, achieving longer shots**. Testing showed that 8 mm of compression gave the best range, and a **180° rotating turret** design provided flexible aiming.	
Cons	Although this was a good initial design, we decided later in the season that we wanted our shooter to be able to **"auto aim" onto the goals from anywhere in the shooting zones**. To do this we needed to redesign the **outtake completely, making the hood adjustable** and also improving the efficiency of our turret system as well.	

VERSION 2.0		
DESCRIPTION		
Pros	We made our new shooter to be shorter than before **but also have an adjustible hood** to be able to shoot from any area. Our hood was made **using bearings to allow movement** and a servo using a **rack and pinion system to drive it**. We also changed our **turret to be Belt Driven.**	
Cons	One issue that we faced with the redesign was one that we had underestimated the angle that would be needed for the hood, leading to high arc trajectories. Another issue that we faced was that the hood was prone to shifting at some points, because it was being held in by the servo in contact which was attached to flexible wood.	

Outtake

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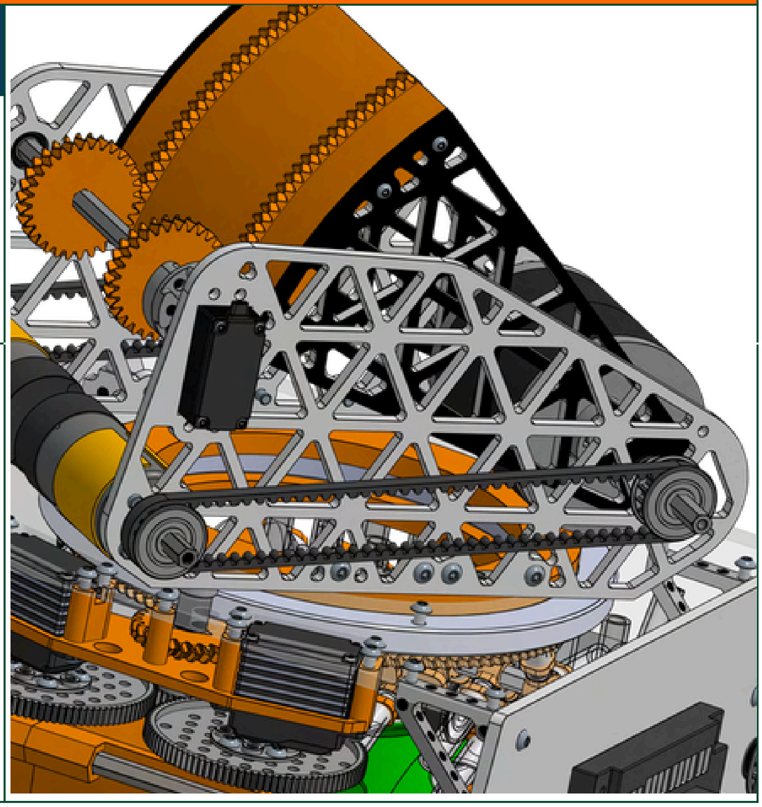

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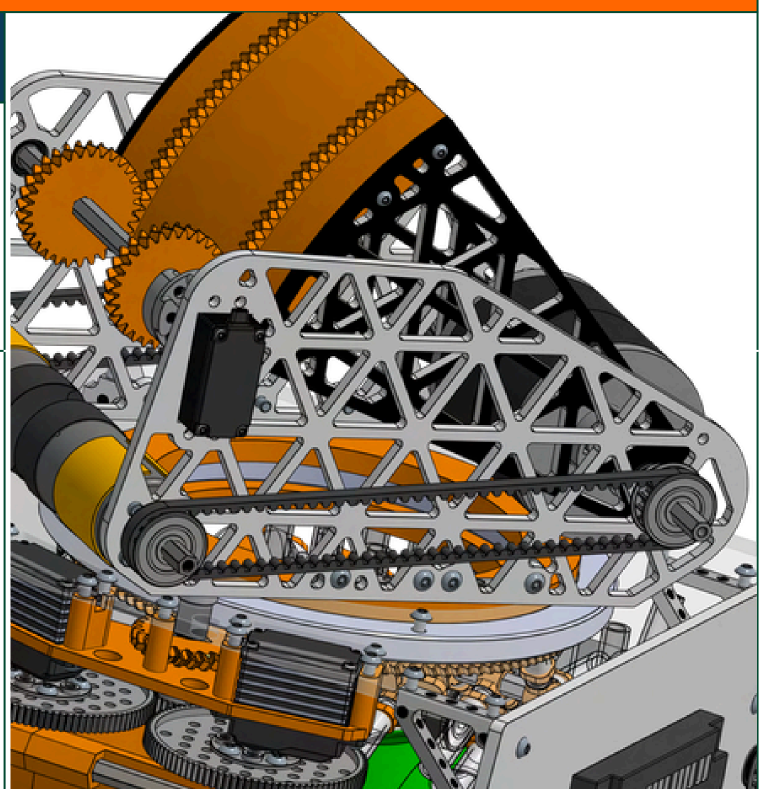
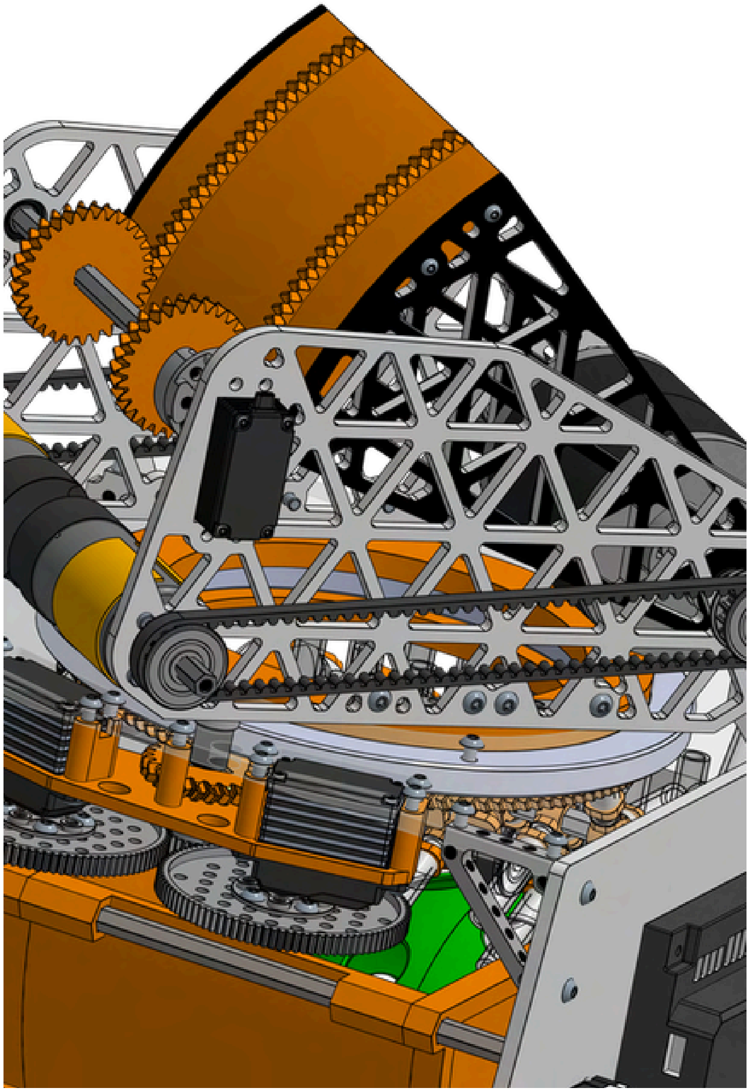


Outtake

VERSION 3.0		
DESCRIPTION		
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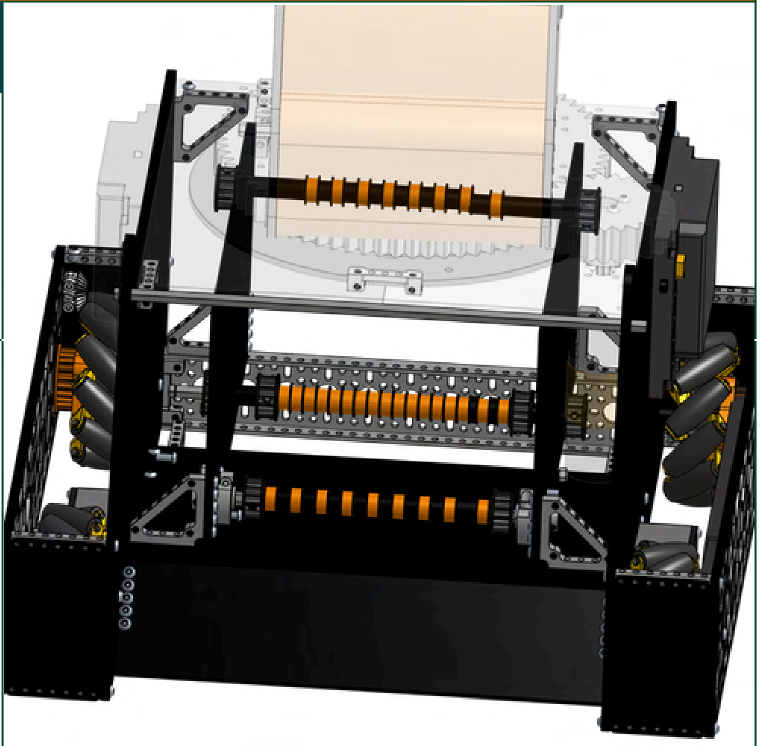
Transfer

VERSION 1.0

DESCRIPTION

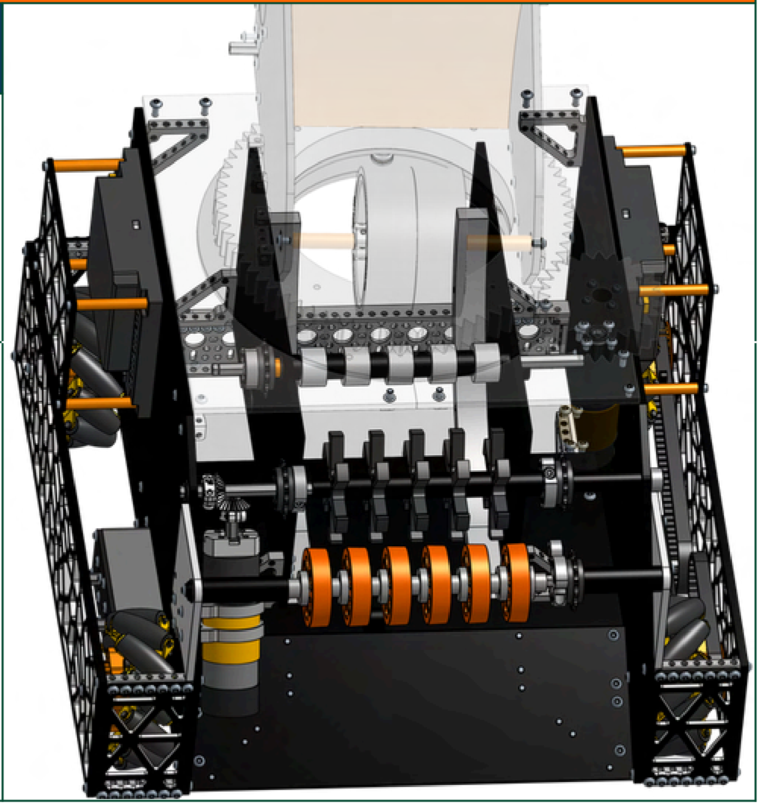
Pros	For our transfer, we decided to originally have a **rubber band transfer** sort of system where the **artifacts would go on a sort of rubber band conveyer** from where it would transfer from the intake to the outtake.
Cons	Before testing, we realized that this **solution would be very costly** in terms of the **bill of materials** and figured there could be better ways to make the transfer. Another concern we had was that the **artifacts would be unable to fit in the actual shooter hole**. When the artifacts was able to fit, the angles of the stages of the **transfer was too high for the artifacts to be able to transfer well** and thus we looked for another way to transfer that would be **less costly and more efficient**.



VERSION 2.0

DESCRIPTION

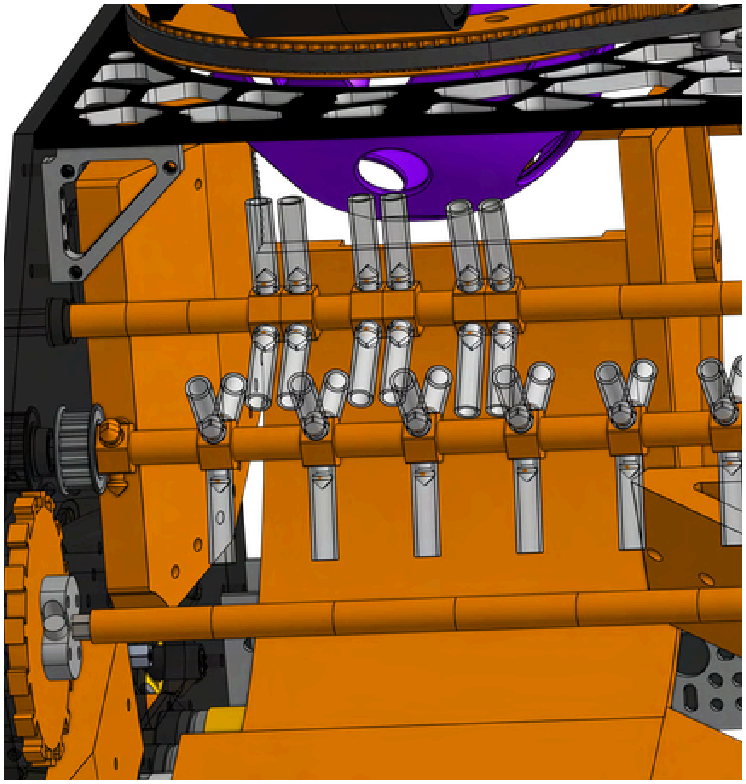
Pros	For our second transfer iteration, we implemented a **rail-and-roller design**. Since standard GoBilda rollers didn't fit, we used custom TPU rollers to guide the artifacts. The system is powered by a **sprocket-and-chain setup** driven by the **intake motor**.
Cons	Although initially this design worked better than the first idea, we quickly realized that this design also had its own **flaws**. Due to the **small size of the tpu rollers**, transfer took a more time. Additionally, TPU did not have as good grip as silicone and often slipped. We decided we wanted to redesign the transfer to allow us to get the artifacts out faster.



Transfer

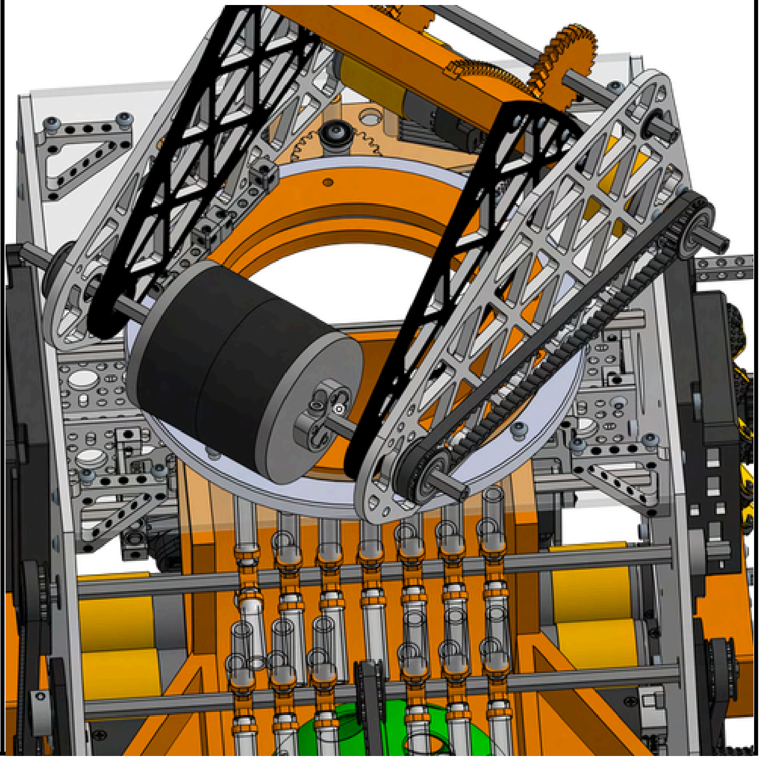
VERSION 3.0

DESCRIPTION	
Pros	For our third transfer iteration, we created a **new transfer geometry**. which was much more calculated and consistent. **Earlier we faced issues with compression and pushing the artifact up to the actual outtake** but this new geometry made **transfer much easier.**
Cons	The design was great for our League Meet 3. However, we realized that since we needed to **change our intake geometry to vector wheels** (centers the artifacts), we needed to **change the transfer geometry to be centered.** We also wanted the transfer to be **one part and not multiple guide parts for better rigidity.**



VERSION 4.0

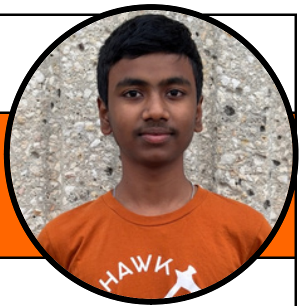
CURRENT VERSION	For the fourth iteration of the transfer system, we **changed our intake style to allow the artifacts to be centered**(due to the vector wheels geometry which allowed the balls to be vectored / moved to the center). This **change meant there was no need to send artifacts to the left which had kept our transfer unstable** and needing additional guides. Instead, **we centered our transfer and made it almost completely symmetric** since the new vector wheel geometry made it so the artifacts would go right to the center. This transfer change allowed us to be much more efficient due to the artifacts having to travel much less distance compared to before.



Kickoff Ideas

Brainstorm | CAD | Prototype | Build | Review

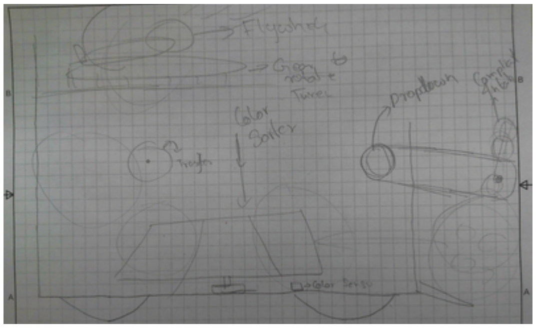
09/9/25	Game Reveal

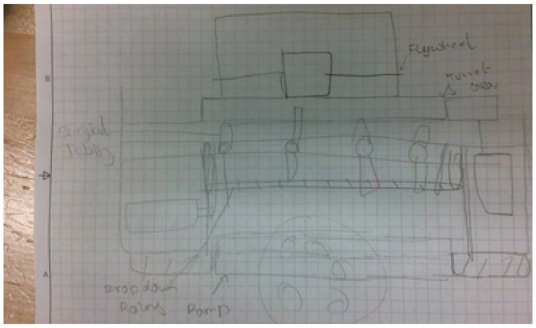


Karthik Borusu

Intake	My initial idea for the intake was a **drop-down intake system** that works like an excavator, **lowering to the ground** and scooping artifacts into the robot. I felt this was the best approach because the **artifacts are spherical** and **could easily roll away** if handled with a traditional roller or surgical tubing intake. By containing them right as they are collected, the **drop-down intake provides more control** and makes gathering artifacts more reliable.
Transfer	Since the intake is a drop-down, the transfer would be simple and probably just used for **color sorting**. The transfer would be a sort of spindexer that would spin in the center that with the assistance of a **color sensor** and **servos** would choose which artifact to launch based on the pattern, spin the indexer and release it out.
Outtake	After analyzing the game by watching the game reveal video, my first idea was a **tri-flywheel** shooter (Flywheel in the left, right, and top), so that the artifact can be shot from the **farthest launch zone**. This was the plan because all the other robots are going to crowd the closer launch zone, and with the crowd, it will be really hard to aim. Thus, my hope is that by launching with 3 wheels from farther away, **we can stand out among other teams** and be able to demonstrate to a potential alliance captain our ability to be able to shoot from both zones that we would be a good partner.

Sketches





The above sketches display the designs for the shooter, intake, and transfer mechanism.

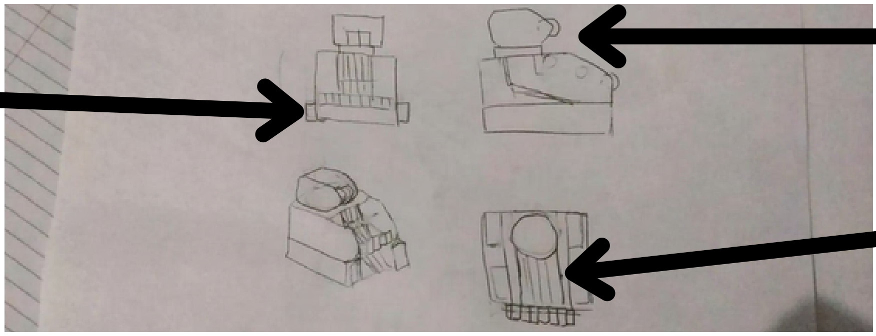
Kickoff Ideas

Brainstorm | CAD | Prototype | Build | Review

09/9/25	*Game Reveal*
Deekshita Sivakumar	
Intake	My first thought for the intake was a **compliant intake** with arms. Such designs were common in older games like Skystone and Relic Recovery, and the large game piece size showed similarities. We brainstormed designs similar to **11115 Gluten Free's** intake from their offseason project. We felt that a **compliant intake would allow us to be able to compress the balls better** and have **more stability** compared to surgical tubing which could break.
Transfer	My original idea for transfer was a **rubber band powered conveyer belt because such designs were common in FRC** and I saw it being used in prior FTC games (Ultimate Goal). Most of our initial ideas came from **finding similarities between Decode and other shooter applications.** I thought that since FRC has found success with a rubber band **"conveyer belt"** transfer, that it could potentially be a **"meta"** for this season that many teams would use.
Outtake	For outtake we drew inspiration from an old off season project from **11115 Gluten Free**. We wanted to replicate how they packaged their motors and wiring. For our overall design, we decided to prototype a singular axle flywheel with a compliant back to aid compression. We thought that this design would be more **efficient compared to VEX designs** we had seen earlier such as dual flywheels. This was because the **dual flywheels weren't mechanically linked** compared to the flywheel and were much **more difficult to package due to the space the dual flywheels system took up**.

Sketches

Gecko Wheels / Compliant Pieces



Singular Flywheel

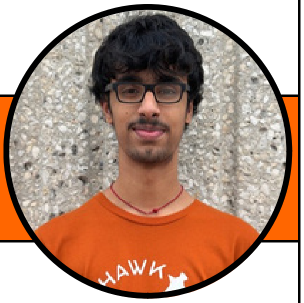
Rubber Band Transfer

The above sketches display the designs for the Shooter, intake, and transfer mechanism.

Kickoff Ideas

Brainstorm | CAD | Prototype | Build | Review

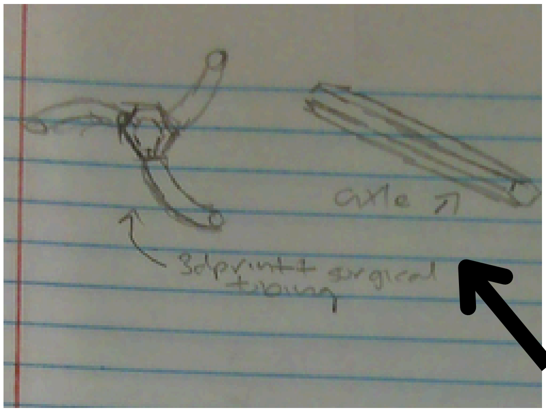
09/9/25	*Game Reveal*

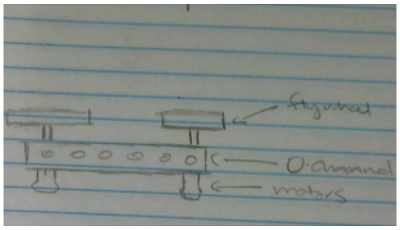


Arjun Averineni

Intake	My concept for shooter first began with a **consideration for a catapult**, however, this initial concept later evolved into my take on the Flywheel outtake which would rotate around a platform after considering the inaccuracy and reduced speed of the initial catapult design. **My initial Flywheel outtake prototype consisted of a wide channel** with mounted motors on either sides that would power the Flywheels on the **left and right of the gap the artifacts passes through.**
Transfer	My original idea for intake consisted of a large axle that would be powered by a GoBilda 1150 rpm motor. **In exact increments, 3D Printed cores would be placed which would be used to attach the surgical tubing**. I initially preferred using a surgical tubing because it is grippy and has a greater range, therefore eliminating the need to push the bot into the artifact in order to intake balls.
Outtake	Initially, I was leaning towards a transfer that would make use of 3d printed spools that **the rubber bands wrap around to behave like a belt**. In the beginning I preferred this idea because **rubber bands as grippy while a rotating transfer would also offer efficient movement of the artifact within the bot.** However, with consistent testing of out initial prototype, my team and I noticed how the **rubber bands would tend to snap** or come off the spool, reducing the reliability of this option. However, I still believed that this idea could work out based off of the FRC designs from previous years which had done well using this design.

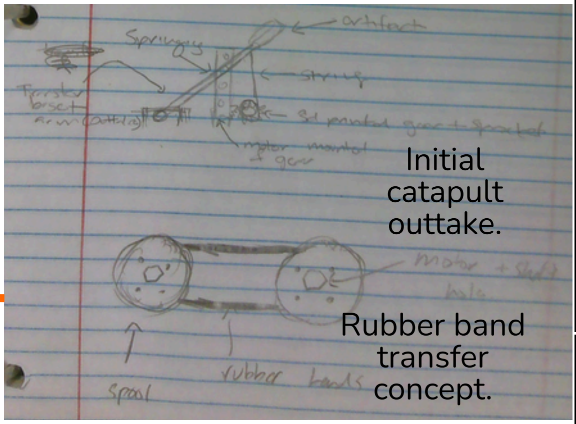
Sketches





Above is the flywheel outtake.

Represents the axle and the surgical tubing + 3d printed part

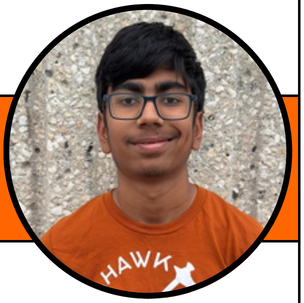


Initial catapult outtake.

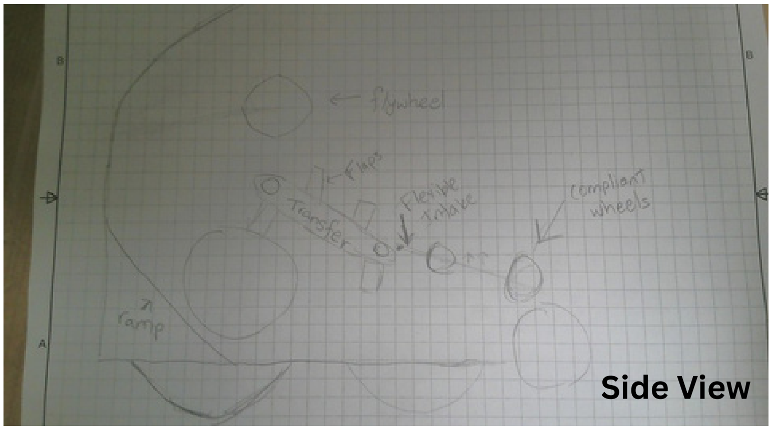
Rubber band transfer concept.

Kickoff Ideas

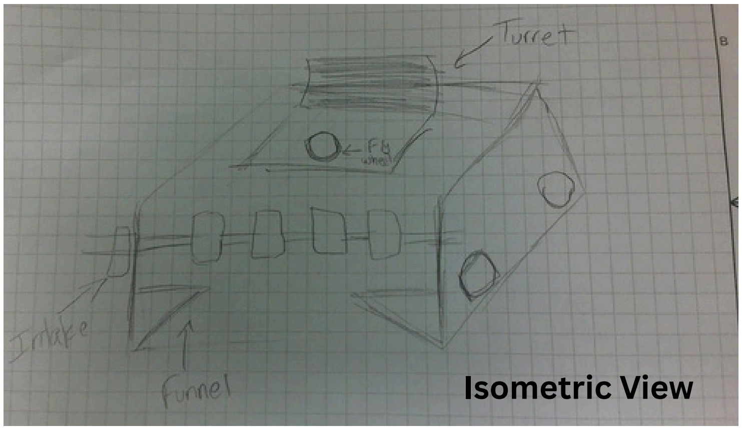
Brainstorm | CAD | Prototype | Build | Review

09/9/25	*Game Reveal*
Shrikar Sudarshan	
Intake	My idea for the intake was to **use a set of gecko wheels mounted on a pivoting or sliding mechanism** that can move up and down, allowing the system to adapt to different artifact positions and maintain consistent contact during collection. **The vertical motion could be controlled using a simple linkage driven by a small motor**, or an elastic suspension system. This would enable the intake to flex when it contacts a artifact or the ground and then return to its neutral position.
Transfer	My plan for the transfer is to use two parallel axles **connected by a continuous chain with flaps that move artifacts** from the intake into a storage or staging area. At the back of the system, **a dedicated roller would sort artifacts by color using feedback from an optical sensor.** A tensioner keeps the chain running smoothly, and the **flaps are made from a slightly flexible material so the artifacts are carried securely without being crushed.**
Outtake	The flywheel system would consist of **two counter-rotating wheels made from a high-friction material**, powered by 6000 RPM motors to consistently launch the artifacts. **A stiff servo or pneumatic kicker would feed artifacts from the transfer into the flywheels at the right moment**. For accurate targeting, the flywheels would be **mounted on a turret driven by a 312 RPM motor, allowing the system to rotate smoothly and aim at different goals.** The turret would rely on a camera to detect and track targets, while odometry data **would be used to account for robot movement and adjust shot timing and velocity.**

Sketches



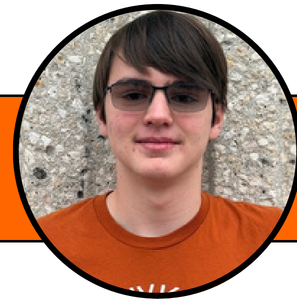
Side View



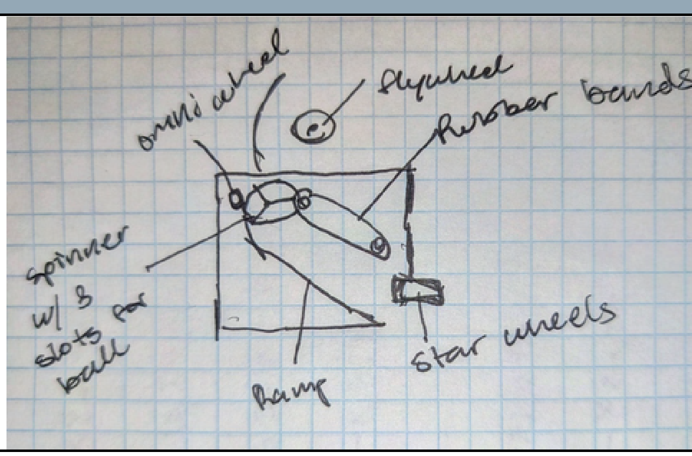
Isometric View

Kickoff Ideas

Brainstorm | CAD | Prototype | Build | Review

09/9/25	*Game Reveal*
Aron Hidvegi	
Intake	I would design a **compliant intake with two stages**. A compliant intake uses two **squishy rubber wheels on either side of the artifact** that spin inwards. Compliant intakes are typically **useful for picking up larger game pieces**, such as this year's game pieces. Additionally, with the no extension rule this year, it is vital for our intake to fit into **as small of an area as possible** while maintaining consistency, making compliant intake perfect for our robot.
Transfer	For transfer, **a spinning carousel (or "merry-go-round")** allows us to store and sort up to three artifacts at once. **The disk is divided into sections so each artifact can be tracked and fired in a controlled order,** which is especially useful for meeting end-of-match scoring requirements in both AUTO and TELEOP. **This flat, rotating storage design is also very space-efficient, helping us make the most of the limited room on the robot** despite the large size of the game pieces.
Outtake	A **single, high-speed flywheel paired with a curved, adjustable-angle hood creates a simple but powerful shooter design.** The flywheel provides the necessary launch speed and backspin, **while the curvature of the hood helps guide and stabilize the artifact as it exits**, improving consistency from shot to shot. **Choosing a non-turreted shooter allows us to focus on reliability and ease of tuning**, which is especially important early in the competition season. By **limiting mechanical complexity, this design is easier to build, test, and iterate on, while still offering enough adjustability** to handle different shooting distances effectively.

Sketches



Kickoff Ideas

Brainstorm | CAD | Prototype | Build | Review

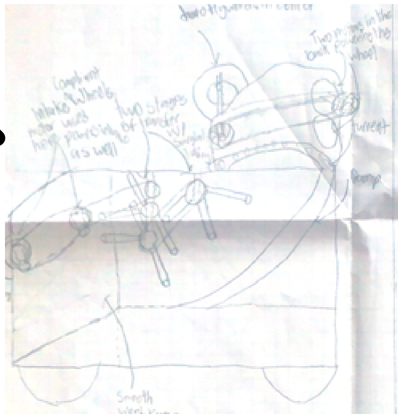
09/9/25	*Game Reveal*

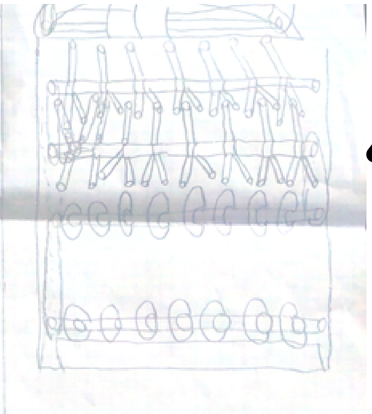
Ritvij Sharma

Intake	My idea for a potential intake was to have surgical tubing on a potential drop down intake. The idea is that when we get close to an artifact, we can drop down the intake powered by linkages to be able to **sort of "trap" the artifact** to be able to intake. I was also thinking about adding more **tubing parallel to the chassis to better guide the artifacts**. Also, I was thinking of making multiple stages of intake tubing to be able to better guide the artifact to the transfer.
Transfer	Finally, there was the transfer stage. My main idea for the transfer stage was to **have multiple stages on intake tubing run by two motors and would be chained all the way to the head of the drop down intake** in multiple stages to help spread the power output from the motors to both the intake and the transfer. My idea then was for the intake tubing to ramp up to the actual shooter in which the artifact would be pushed into the flywheel which would lead the flywheel to shoot it rapidly.
Outtake	My Idea for a shooter was a two flywheel system in which the artifact would shoot out. My idea was to have a rotatable shooter(Potentially a turret system) along with a flywheel powered by a **singular 6,000 rpm motor**. This would allows us to score much faster and much more efficiently. Also, in my design I hope to be able to make the hood of the shooter be able to be easily **adjust the angle using a rack and pinion type system powered by a servo**. This would allow for the projection of the artifact to be able to adjust which would thus give us more options on what angle tos hoot at depending how far or close we were to the goal.

Sketches

Side view





front view

Kickoff Ideas

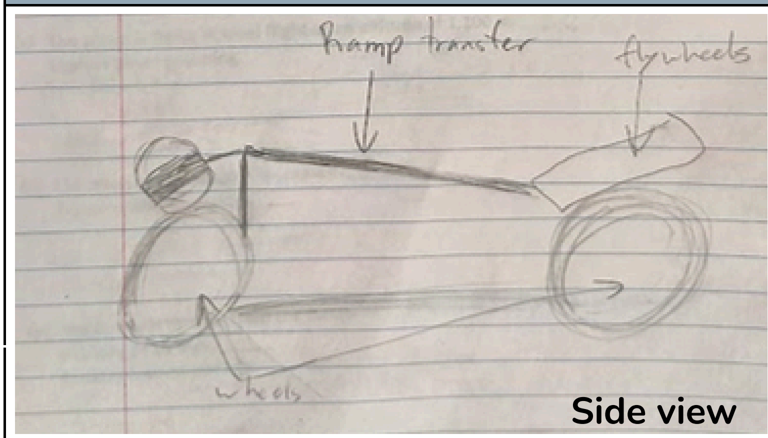
Brainstorm	CAD	Prototype	Build	Review

09/9/25	*Game Reveal*

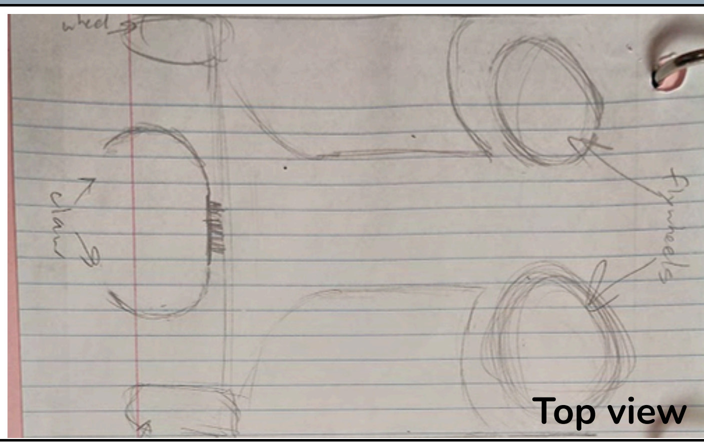
Isaac Kao

Intake	Because of the **claw's success in the Into the Deep season**, I thought a round claw would work well. The **claw would be 3D printed, and lined with a somewhat compliant**, grippy material so that the claw can hold onto the artifact without it falling out. **It would flip up using servos and drop the artifact into a the transfer.**
Transfer	The transfer would be **short and gravity-powered to keep things simple**. A hood would cover the top to prevent artifacts from falling or flying out during movement. **If artifacts start escaping through the front intake, a small servo arm could be added to keep them contained.** To avoid feeding artifacts into the outtake before it's up to speed, **a servo-controlled channel would hold them until the outtake is ready.**
Outtake	I initially thought of a **two-flywheel launcher design**. It would use a central channel to **guide the artifact into position** before feeding it into two flywheels spinning in opposite directions for a consistent launch. **A servo-powered pivot could allow the shooter to aim at both the big and small triangles**, making it more **compatible with different field layouts and alliance partners.** The design would likely **use mostly 3D printed parts** for better control and quick iteration, **with wood plates for the bottom** to provide extra strength and stability.

Sketches



Side view



Top view

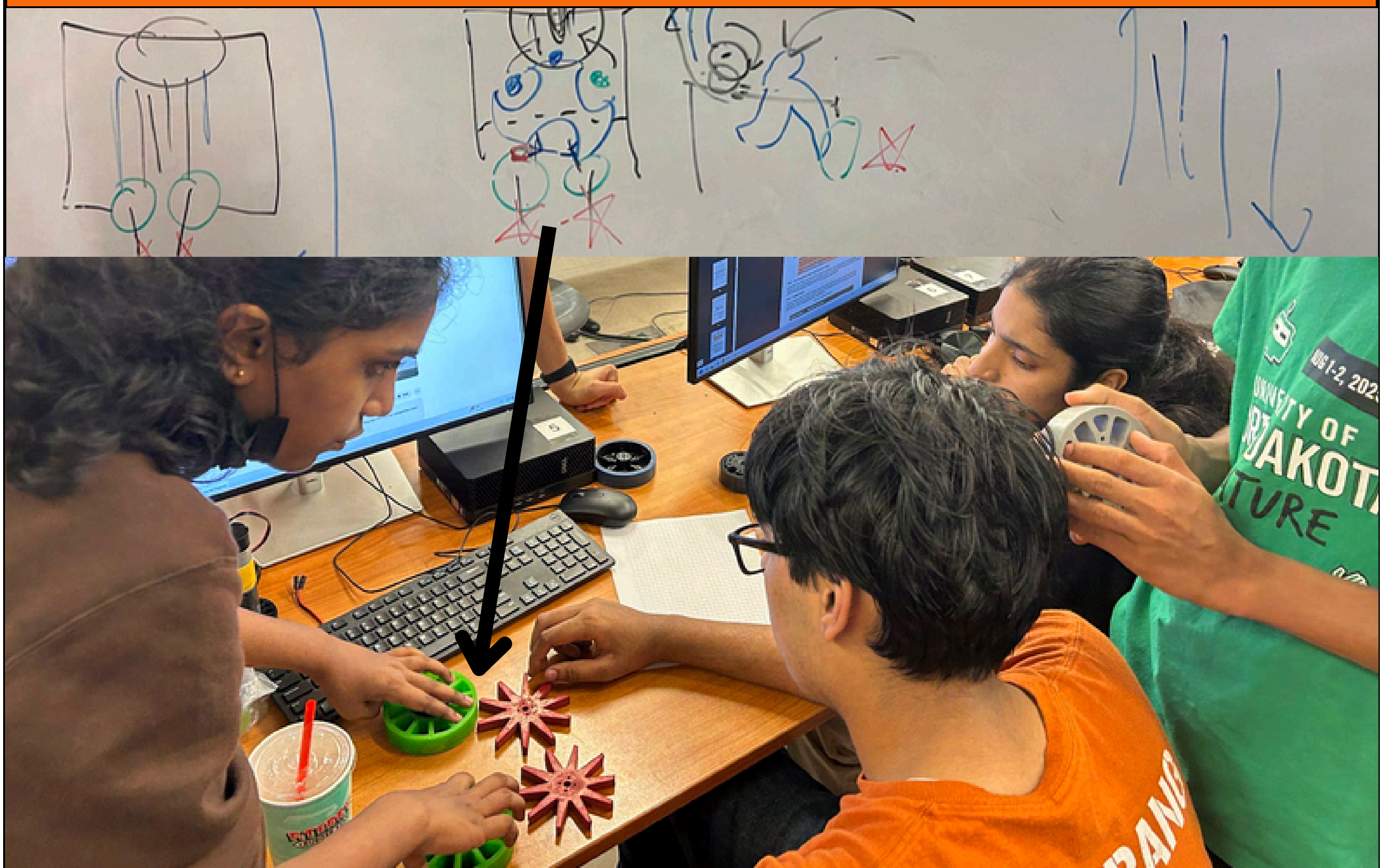
Brainstorming Week

Brainstorm | CAD | Prototype | Build | Review

Dates	*09/11/25- Brainstorming Week*

Intake

Intake Brainstorming



Above are pictures of us brainstorming intake ideas the meeting after kickoff. We wanted to originally use compliant intake pieces as we felt they would be very useful in holding the artifacts and thus started planning about how we could use them.

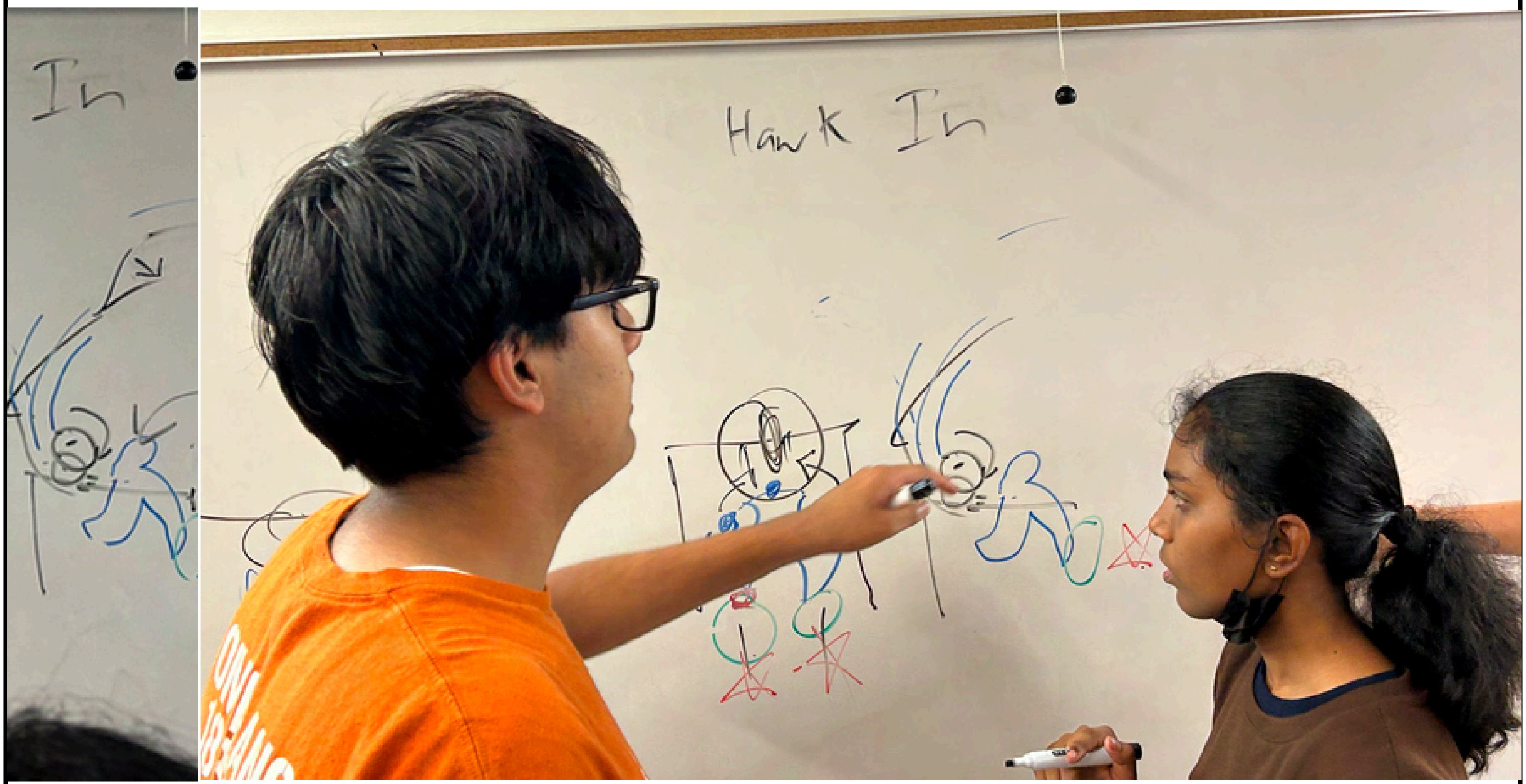
Brainstorming Week

Brainstorm	CAD	Prototype	Build	Review

Dates	*09/11/25- Brainstorming Week*

Transfer

Transfer Brainstorming



At the same meeting, we also began to brainstorm transfer. We wanted to create a transfer that would be able to sort as well as be able to be fast to get it to the outtake. Thus, we began to come up with the idea of a spinning wheel in the middle from which the intake could drop into and be able to sort the artifacts in a certain way to be able to get the pattern points in both the autonomous period and the end of the TeleOp period.

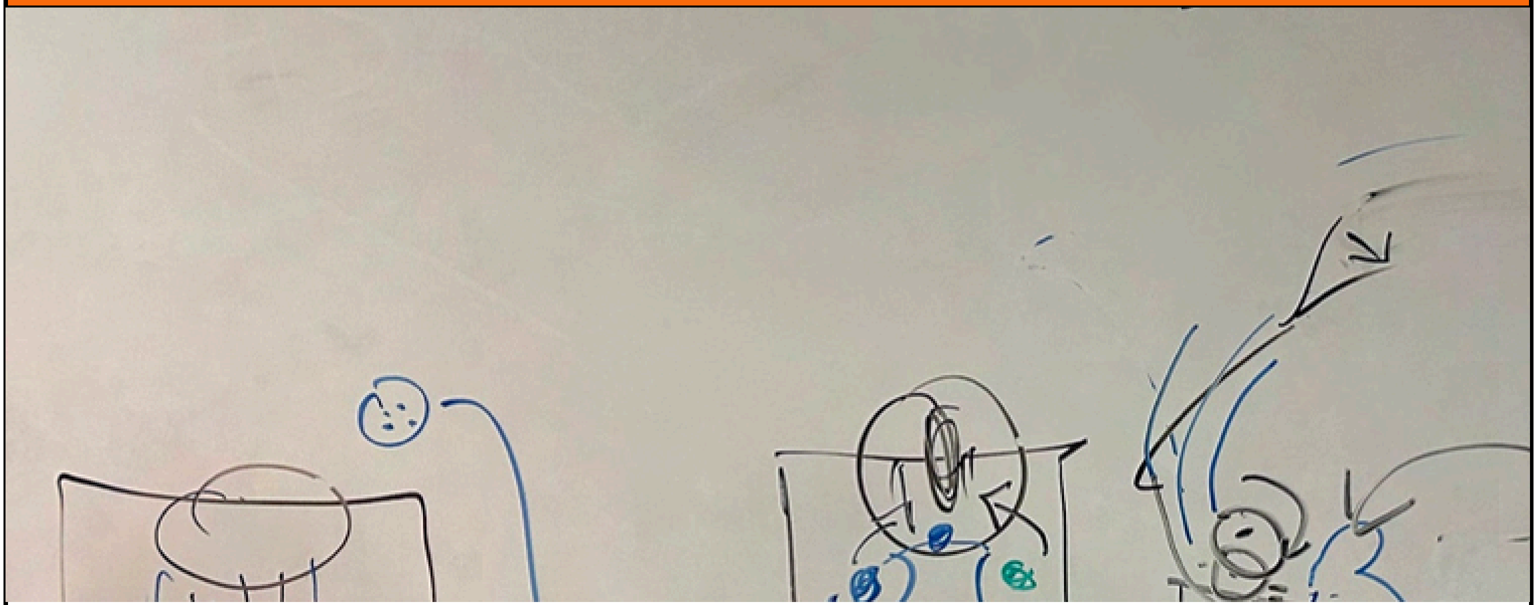
Brainstorming Week

Brainstorm | CAD | Prototype | Build | Review

Dates	09/11/25- Brainstorming Week

Outtake

Outtake Brainstorming



At this meeting, we decided to start brainstorming ideas for our Outtake. We took inspiration from FRC teams from the 2017 season in which the game was very similar; to shoot artifacts into a steam stack. We saw that many FRC teams would use a singular flywheel with high compression in order to shoot the dozens upon dozens of artifacts. Farther in our research, we realized in order to shoot from the farther end of the field, we would need two motors power. During our research, We also realized that we would need 2 motors in order to shoot from the smaller triangle.

Decision Matrix

Brainstorm | **CAD** | **Prototype** | **Build** | **Review**

Date	***09/14/25- Brainstorming Week 1***
PURPOSE	We are rating the functionality of the **outtake** using a point system that will allow us to **evaluate which design is the most efficient** for our robot.

Outtake Decision Matrix

Type	Simplicity	Size	Functionality	Cost	Total
Dual Motor Flywheel	3.0	4.0	5.0	4.0	**16.0**
Dual Flywheel	3.0	1.0	2.0	2.0	**8.0**
Singular motor Flywheel	3.0	3.0	2.0	4.0	**12.0**

We rated our designs on 5 criteria from 1 to 5 (1 being the worst and 5 being the best). The best design was determined by the outtake with the most points.

Decision Matrix

Brainstorm | **CAD** | **Prototype** | **Build** | **Review**

Date	***09/14/25- Brainstorming Week***
PURPOSE	We are rating the functionality of the **intake** using a point system that will allow us to **evaluate which design is the most efficient** for our robot.

Intake Decision Matrix

Type	Simplicity	Size	Functionality	Cost	Total
Surgical Tubing	4.0	3.0	3.0	4.0	**14.0**
2 stage compliant wheel intake	1.0	2.0	2.0	2.0	**7.0**
Drop down 4bar with wheels, surgical tubing behind it	4.0	3.0	2.0	4.0	**13.0**

We rated our designs on 5 criteria from 1 to 5 (1 being the worst and 5 being the best). The best design was determined by the intake with the most points.

Decision Matrix

Brainstorm | CAD | Prototype | Build | Review

Date	*09/14/25- Brainstorming Week*
PURPOSE	We are rating the functionality of the **transfer** using a point system that will allow us to **evaluate which design is the most efficient** for our robot.

Transfer Decision Matrix

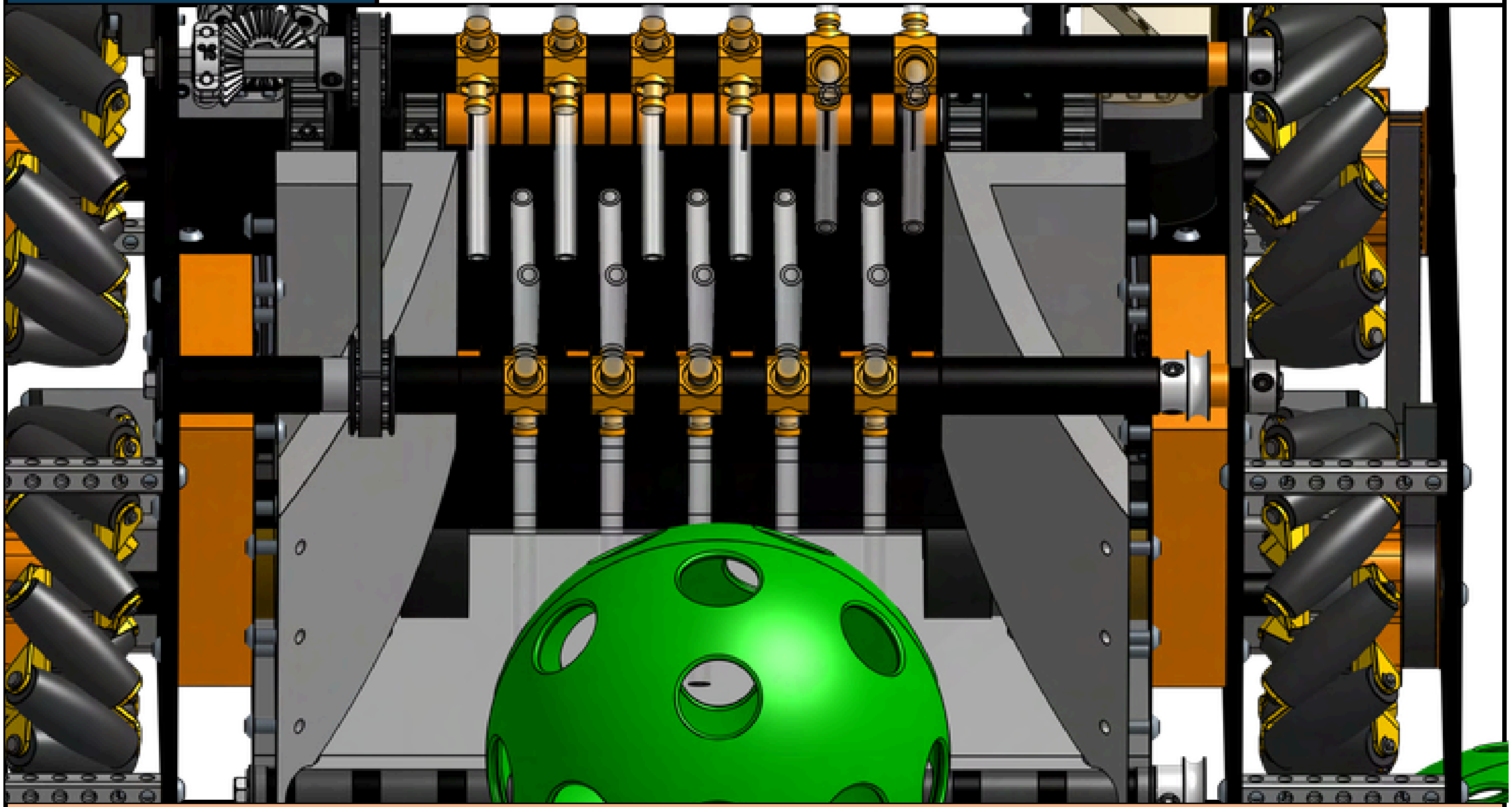
Type	Simplicity	Size	Functionality	Cost	Total
Spinning Rubber Band 2 stages	4.0	4.0	4.0	4.0	**16.0**
Spindexer into ramp	2.0	3.0	5.0	1.0	**11.0**
Ramp	4.0	2.0	3.0	5.0	**14.0**

We rated our designs on 5 criteria from 1 to 5 (1 being the worst and 5 being the best). The best design was determined by the intake with the most points.

Version 1.0

Brainstorm	CAD	Prototype	Build	Review

Date	*09/16/25- CAD Week 1*
Intake CAD V1	
PURPOSE	Our initial intake was made from 2 stages of surgical tubing. One issue we faced was how to transmit power from the motor to the system so that it would move. **We decided to use bevel gears to allow us to position the motor on the plates.** This would allow the motor to stay secure. We also decided that we would want to belt the two stages of the tubing. Finally, **our intake also had bottom rollers at the bottom** from which we planned to use to power our bottom rollers using **either an o-ring or a twisted rubber band.**

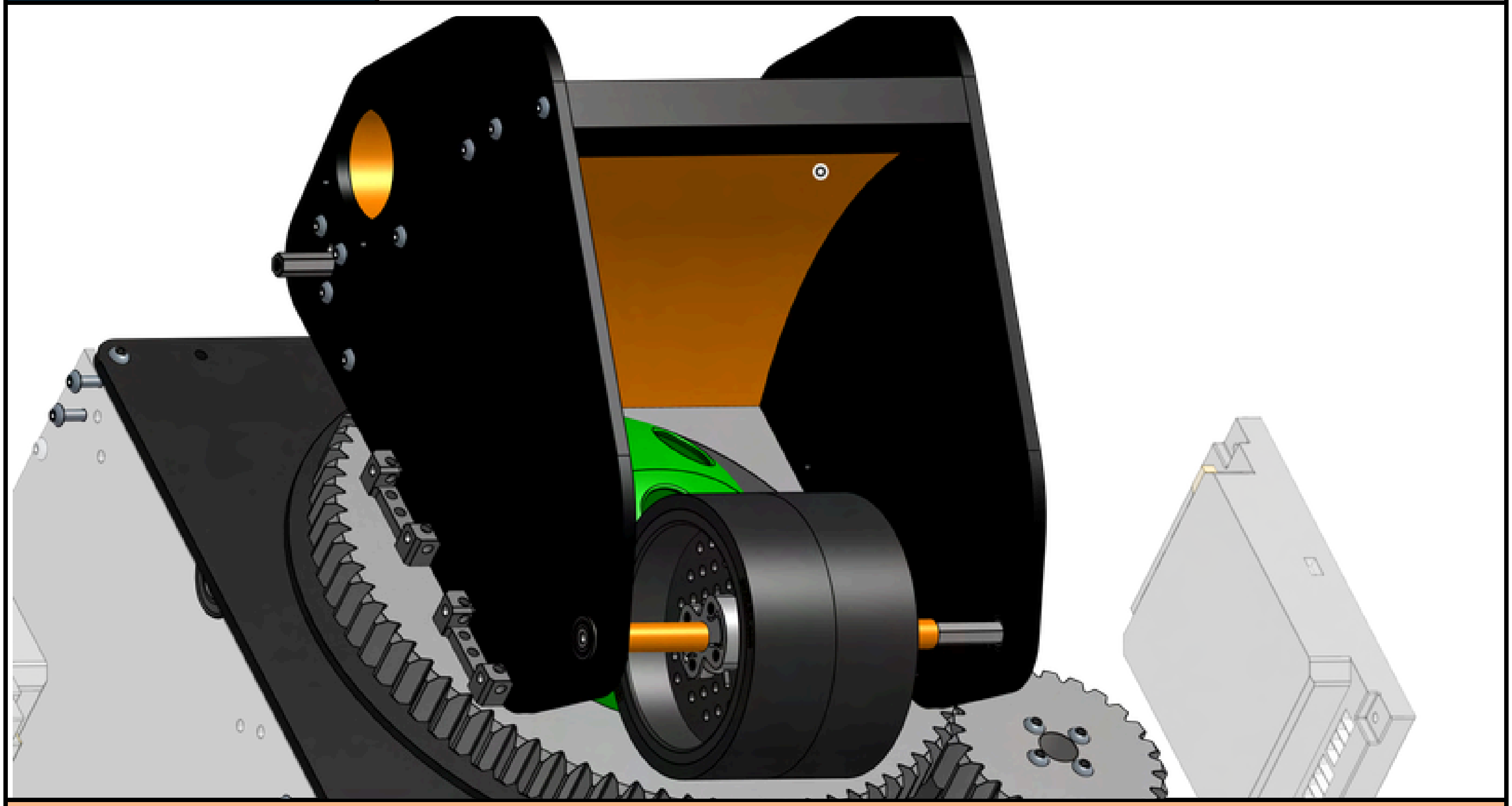


Above is our first intake iteration which has a 2 stage intake with counter rollers on the bottom the further allow the artifact to be able to be easier to go into the transfer

Version 1 .0

Brainstorm	CAD	Prototype	Build	Review

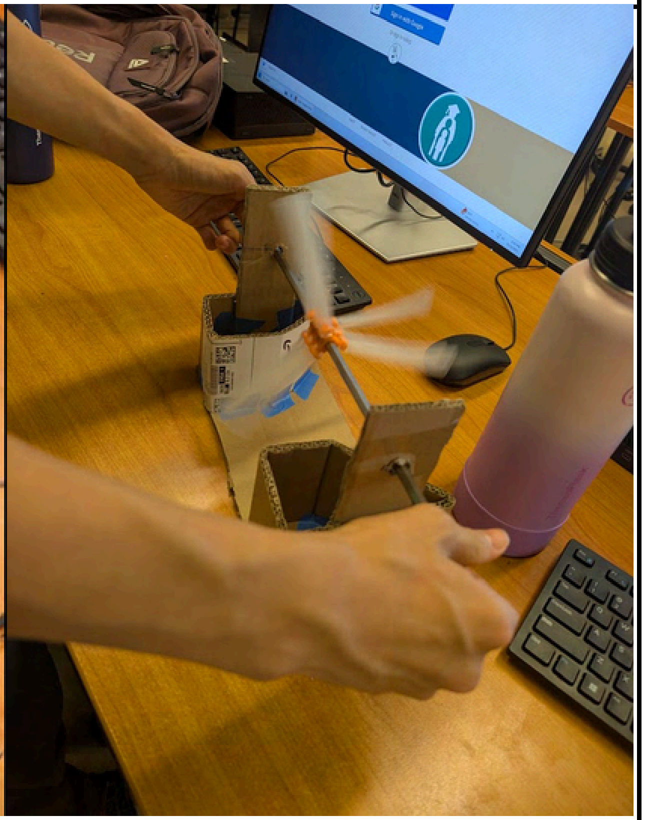
Date	*09/18/25- CAD Week 1*
Shooter Cad V1	
PURPOSE	Our outtake was made up of **two flywheels in the center** with **2 motors in the back** that would be **belted to the flywheel** in the front. We also decided that we would want to add a turret into our design to allow us to shoot at any angle. Another aspect about our first iteration of our shooter was we geared the turret **instead of the conventional belting method as belts can often skip on turrets pulleys** which can lead to the motor not giving an accurate position for the turret. Instead, we used **3D Printed gears** which would slip much less, making the turret more accurate.



Above is the first iteration of our outtake, featuring a geared turret and a dual motor powered flywheel.

Version 1 .0

Date	*09/19/25- CAD Week 1*
Intake Prototyping V1	
PURPOSE	We wanted to see how our intake would roughly work in real life to see if it would be worth to use it. **Thus, we used cardboard to help test the intake idea**. Although it showed promising results, later analysis of the cad showed us that **the tubing was too long to be able to fit within the 18 inch constraint and it was very much prone to tanglement due to the tubing being very close** to each other. Thus, we decided to work on creating a new intake that would be able to **stay within the 18x18x18 constraint** and would also be able to **score without the risk of tangling.**

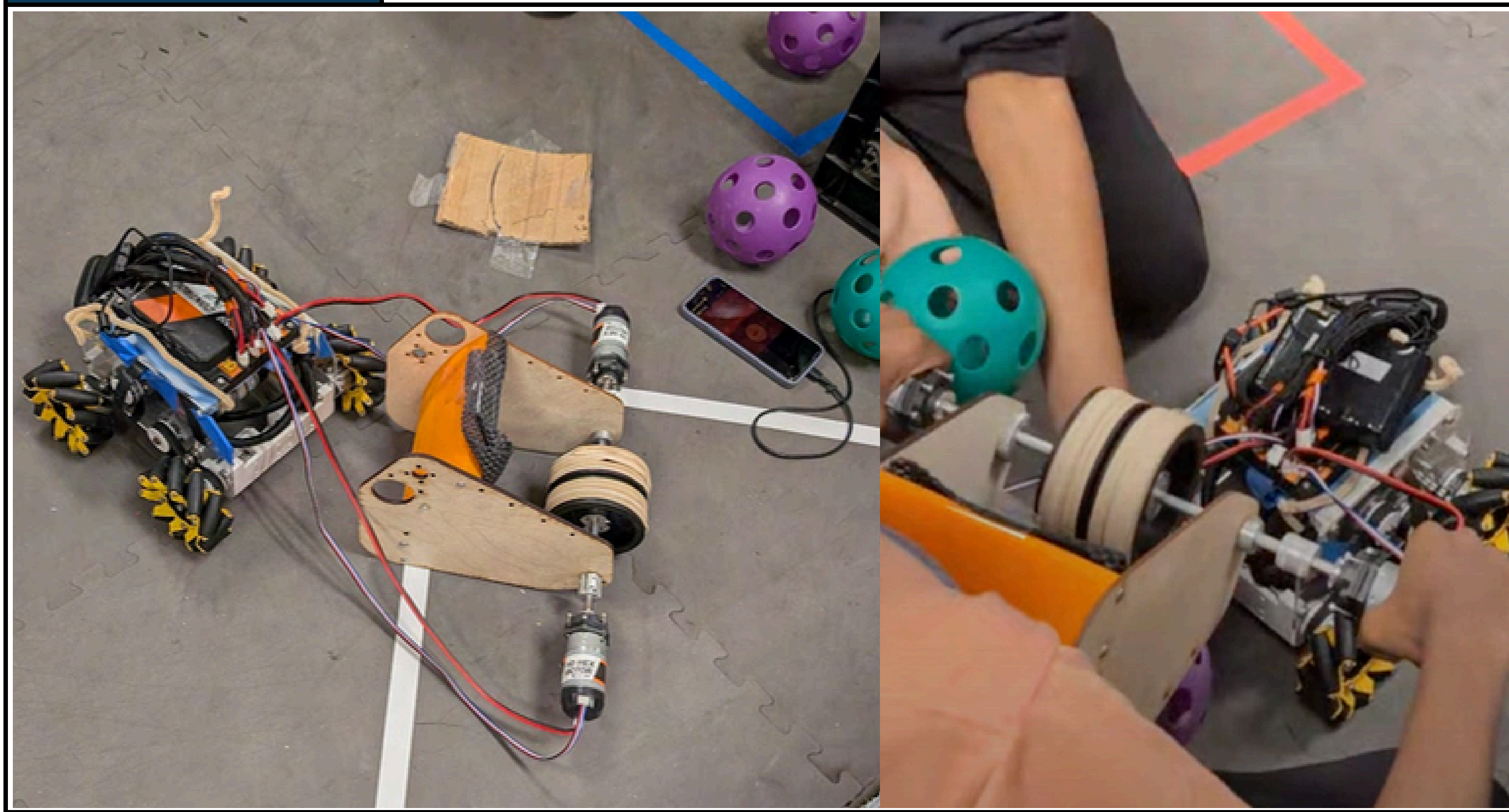


We roughly tested our intake idea using cardboard and the main material and surgical tubing to test how it could hold up in an actual game.

Version 1 .0

Brainstorm	CAD	Prototype	Build	Review

Date	*09/22/25- CAD Week 2*
Shooter Prototyping V1	
PURPOSE	We had finally created our outtake design and decided to test its capability. Our outtake's prototype was **made up of VEX 60A Flex Wheels** and **rubber bands to improve compression**. Our reasoning for using these wheels was that we were waiting for parts to arrive and **wanted to see the capacities of VEX Wheels in FTC due to their major usage in VEX VRC**, a game that has **been for the most part a shooter game**. We decided to **test it from the farzone and were surprised to see that it was working decently well.** One part that we felt we could improve upon was the overall compression of the artifact when shooting it as well as the backspin of the artifact after it would land in the goal. We also realized that we would need to also tune the velocity as the current 6000 rpm motors were leading us to overshoot from the far zone.

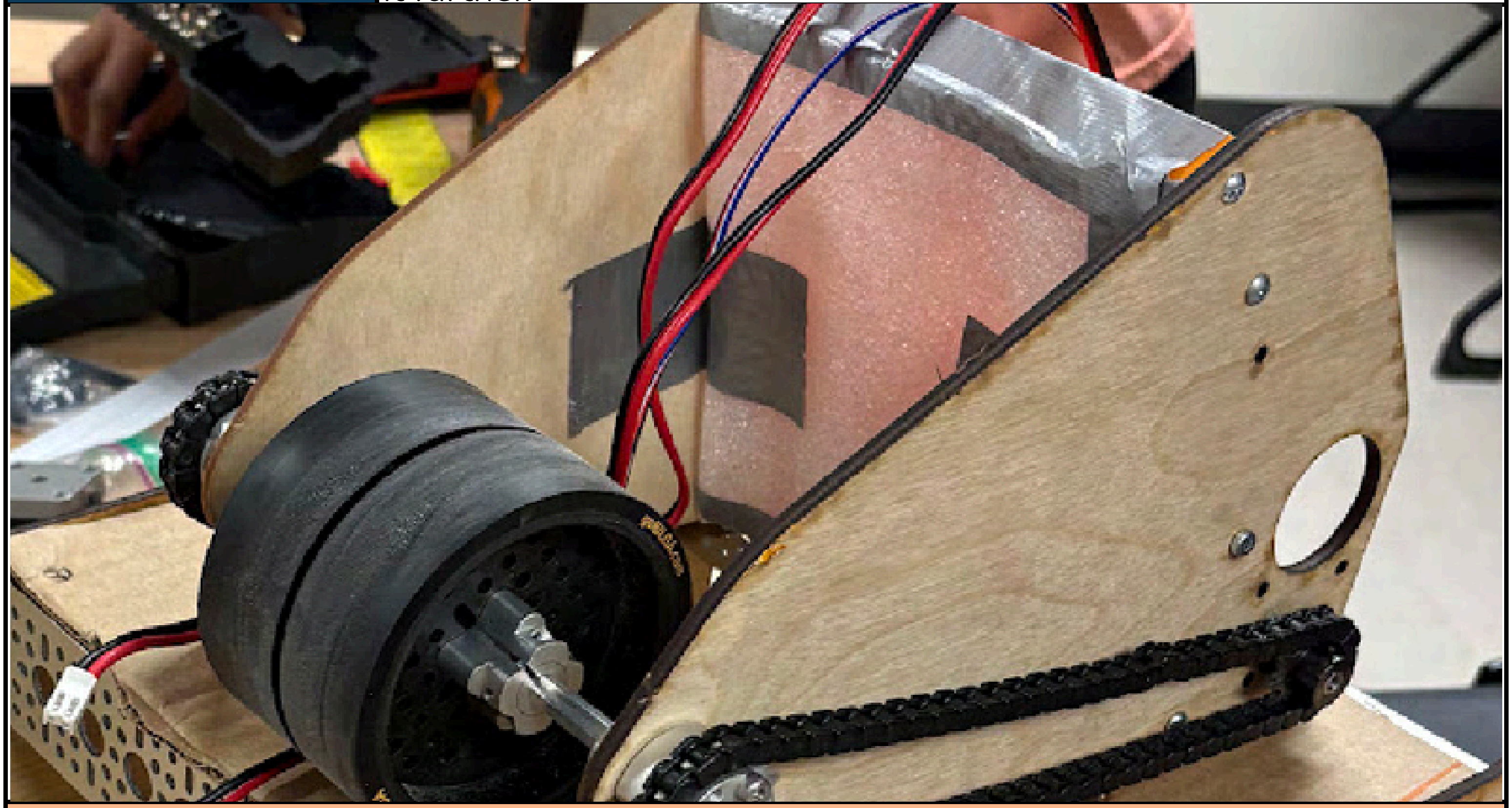


We tested our Outtake prototype using 2 6000 rpm REV Motors, 60A Flexwheels with rubber bands, and cardboard to act as the ramp.

Version 1.0

Brainstorm | CAD | Prototype | **Build** | Review

Date	*09/24/25- CAD Week 2*
Shooter Build V1	
PURPOSE	Using the data collected from our prototyping, we decided to create our actual shooter, using our prototype data. From our prototyping, we realized that we needed more compression on our shooter. To alleviate this, we decided to order **30A Rhino Wheels from Gobilda** and add foam onto the back of our shooter to create more compression which would lead to greater distance and accuracy overall. Also, **we decided to order higher quality and newer motors from Gobilda**. Finally, we decided to use the **sprocket and chain** we had from our prototyping as we found that it would work well and was **much cheaper for now compared to buying new belts and pulleys**. Finally, we decided to use two Rhino Wheels placed next to each other as it allowed us to have more surface area on the artifact allowing us to shoot it further.



Above is our Outtake that we built for our robot.

Version 1 .0

Brainstorm | CAD | Prototype | **Build** | Review

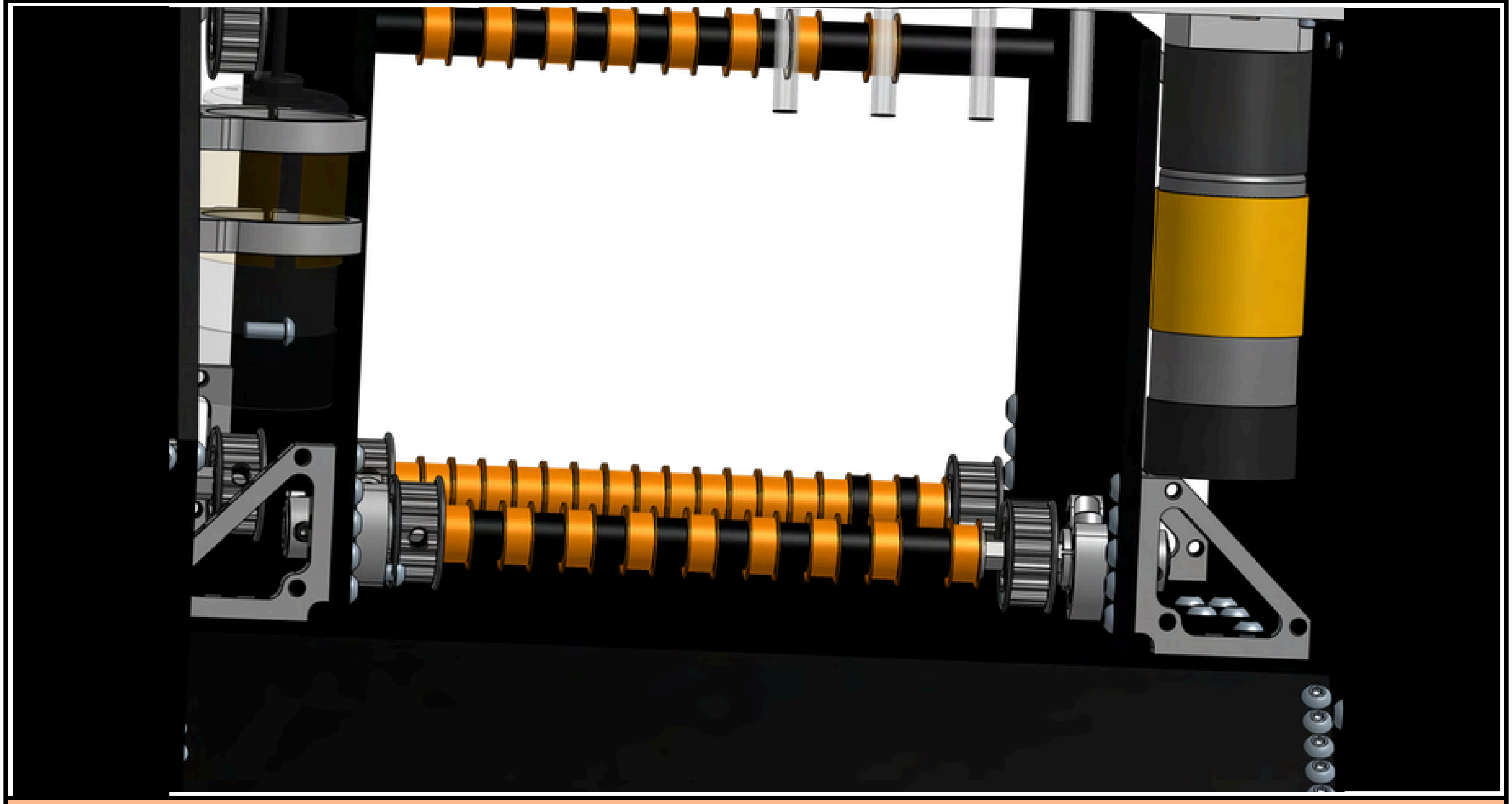
Date	*09/25/25- CAD Week 2*
Turret V1	
PURPOSE	We then began to work on our turret. At the start of the season, we were debating whether to even implement a turret from the start, but after discussing the pros and cons, we decided to create one. **Our turret was made of a bearing stack, meaning that it would have a plate that would rotate between the bearings** and by powering the system with a motor and a type of power transmission, **we could achieve up to 270 degrees of rotational freedom**. Also, we decided to use 3D Printed gears for the power transmission and spacers inside the bearing stack to make the bearing stack more secure. **Finally, we realized that we would need to CNC the plate that would be rotatable** as that would be the one that would be rotating on and we needed a plate that was exactly 4mm to fit perfectly between the bearing stack.

Pictures of our bearing stack turret are pictured above.

Version 1.0

Brainstorm	**CAD**	Prototype	Build	Review

Date	*10/01/25- CAD Week 3*
Transfer V1	
PURPOSE	For our transfer, we originally wanted to create a transfer based on rubber bands that would spin in multiple stages. **Our idea consisted of 2 stages belted** together all powered by a single motor that would also be used to power the intake. **This was inspired a bit from the design of team 11115 Gluten Free** for their own transfer as from we had seen, it seemed like a **faster way to transfer compared to other ideas that were brought up.** One concern that we had before CADding this design was the worry that the rubber bands would either snap from overtensioning or wouldnt be effective from undertensioning.



Above is our first Transfer design CADded together

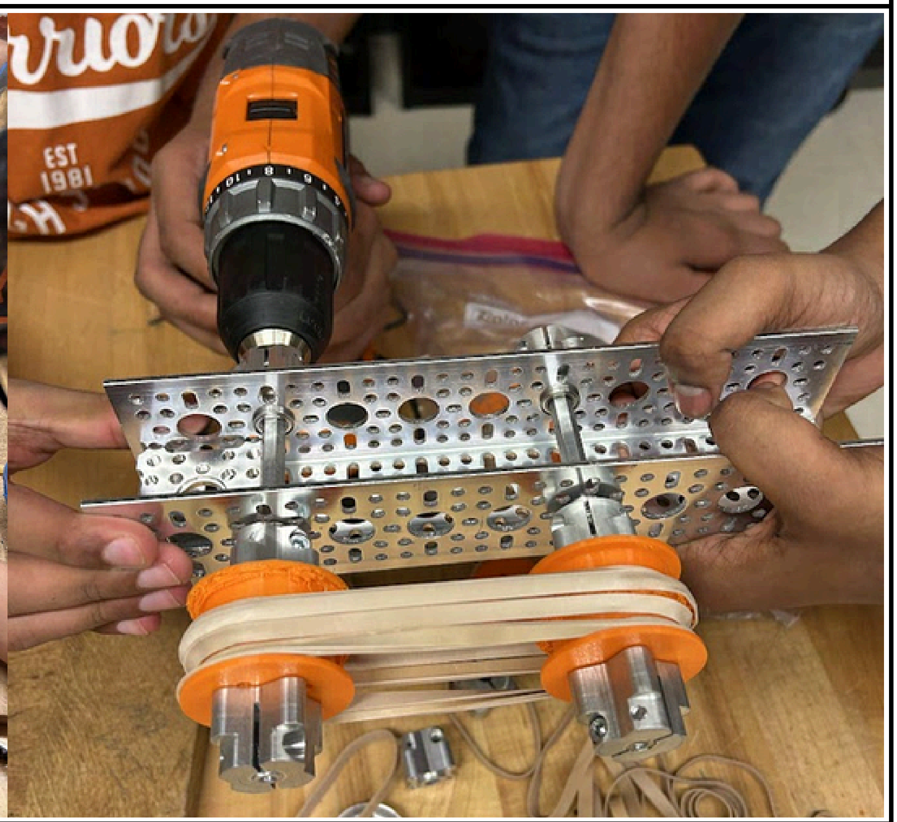
Version 1 .0

Brainstorm	CAD	**Prototype**	Build	Review

Date	*10/06/25- CAD Week 3*

Transfer V1

PURPOSE	We continued to work on the transfer and wanted to see how it would work. **We used spools from ITD, along with rubber bands. We used hyper hubs to clamp the spools**. We then used a drill to power the system, to see hw it could work in real life. After testing, **we found that the rubber band system could rapidly transfer the artifacts to the turret.** But, we realized that since the transfer was made in 2 stages; one being almost completely horizontal, and one being almost completely vertical, we realized that using this set up could lead the artifact to be unable to transfer consistently and rapidly. Thus, we decided that although it was a good concept, **we needed a much more consistent alterntaive** that would be **reliable** for a while.

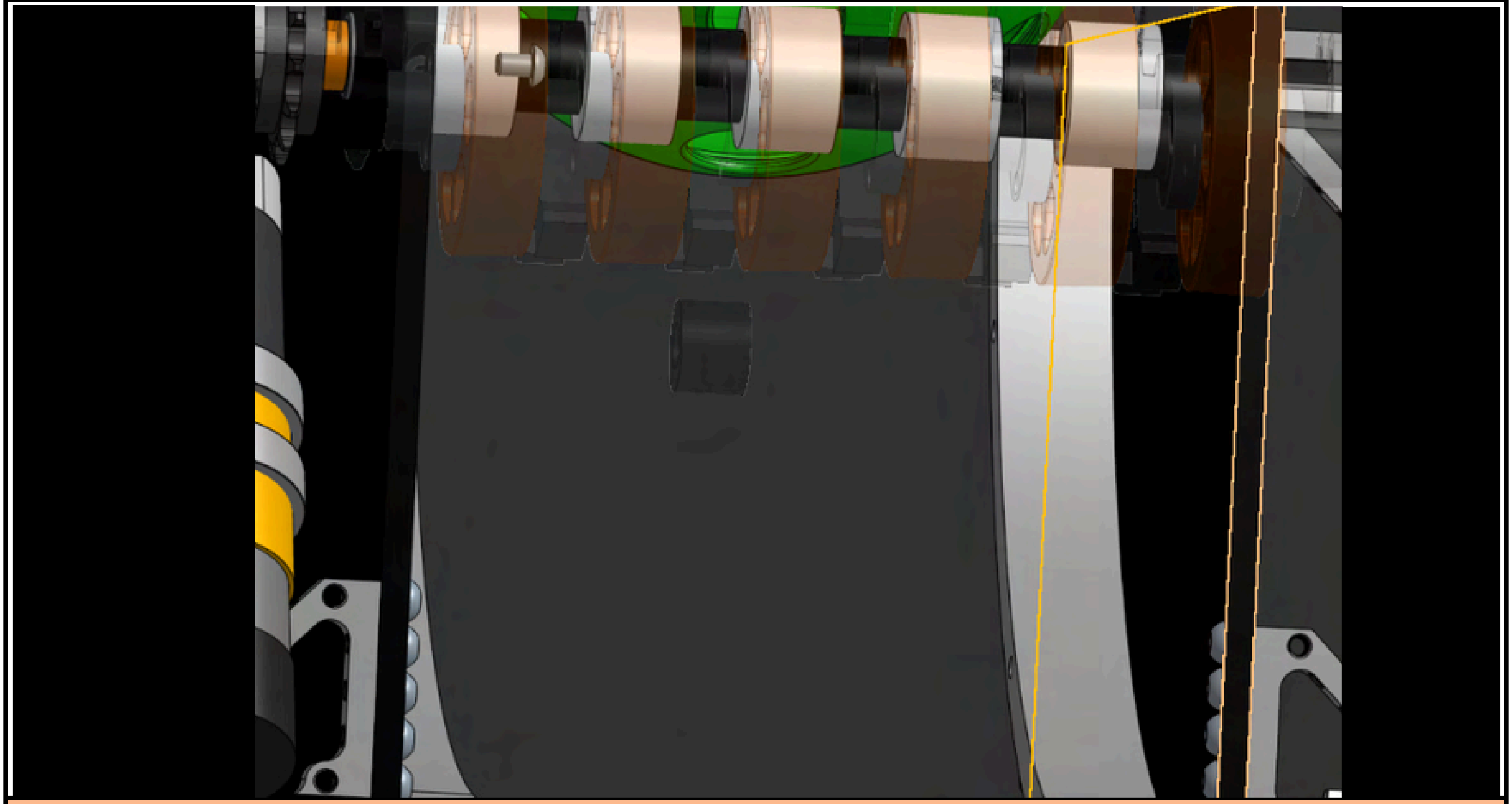


We decided to test the capacities of a rubber band transfer to see its capacities in a potential design

Version 1 .0

Brainstorm	CAD	Prototype	Build	Review

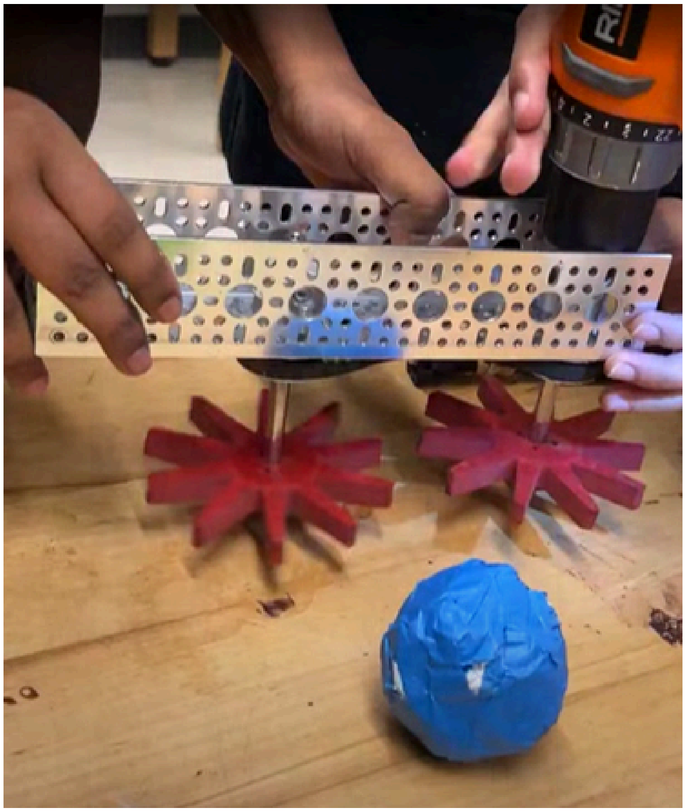
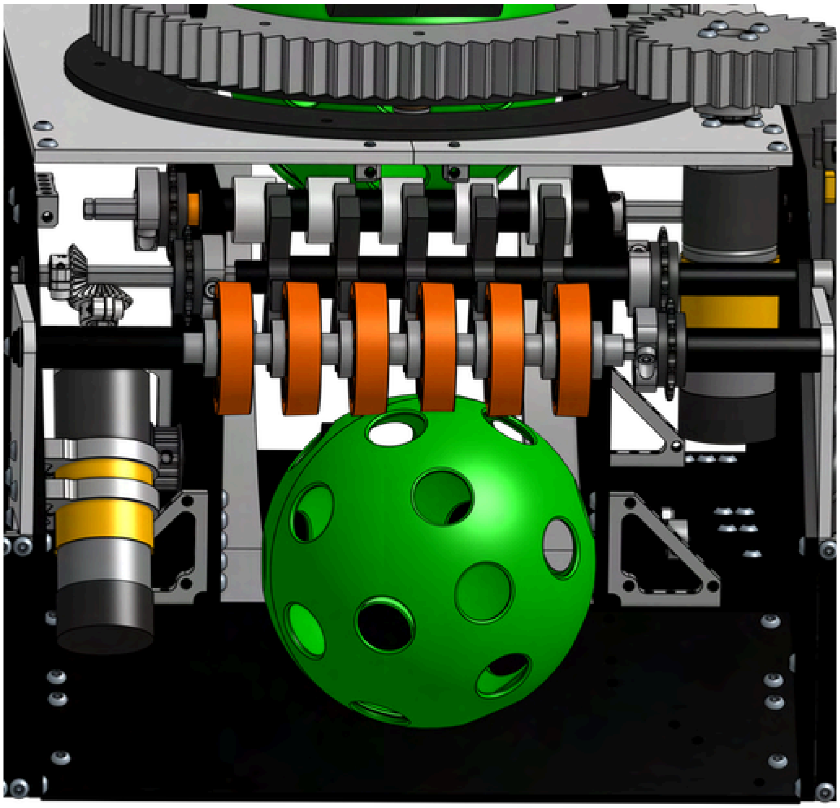
Date	*10/10/25- CAD Week 3*
Transfer V2	
PURPOSE	After our idea with the rubber band failed, we decided to try a simpler idea. We decided to change our transfer to a ramp type transfer which would be much simpler. **We also decided to use TPU rollers in between**. We then decided to use sprockets to power this mechanism using the intake motors power. We felt that this solution was much simpler than what we had before and would be abel to work much better. **Our ramp was 3D Printed from PETG filament** and was attached to our transfer plates which were located inside the actual robot plates.



Our newly created ramp transfer design.

Version 1.0

Date	*10/10/25- CAD Week 3*
Intake V1.25	
PURPOSE	After testing our first iteration of our intake, we realized we would need a new type of intake. **We decided to use compliant intake pieces**; pieces which we had experimented with for intake after kickoff. Originally we didn't see its potential compared to surgical tubing, but after testing, we realized that **compliant pieces would be much better for intaking** as they were shorter and able to pick up just as well. We also wanted for the artifact to be able to transfer through the robot to the turret and so we **added a second stage of boot wheels** that would be chained to the compliant wheel intake. We also decided to **keep the geared motor and ramp from V1**.

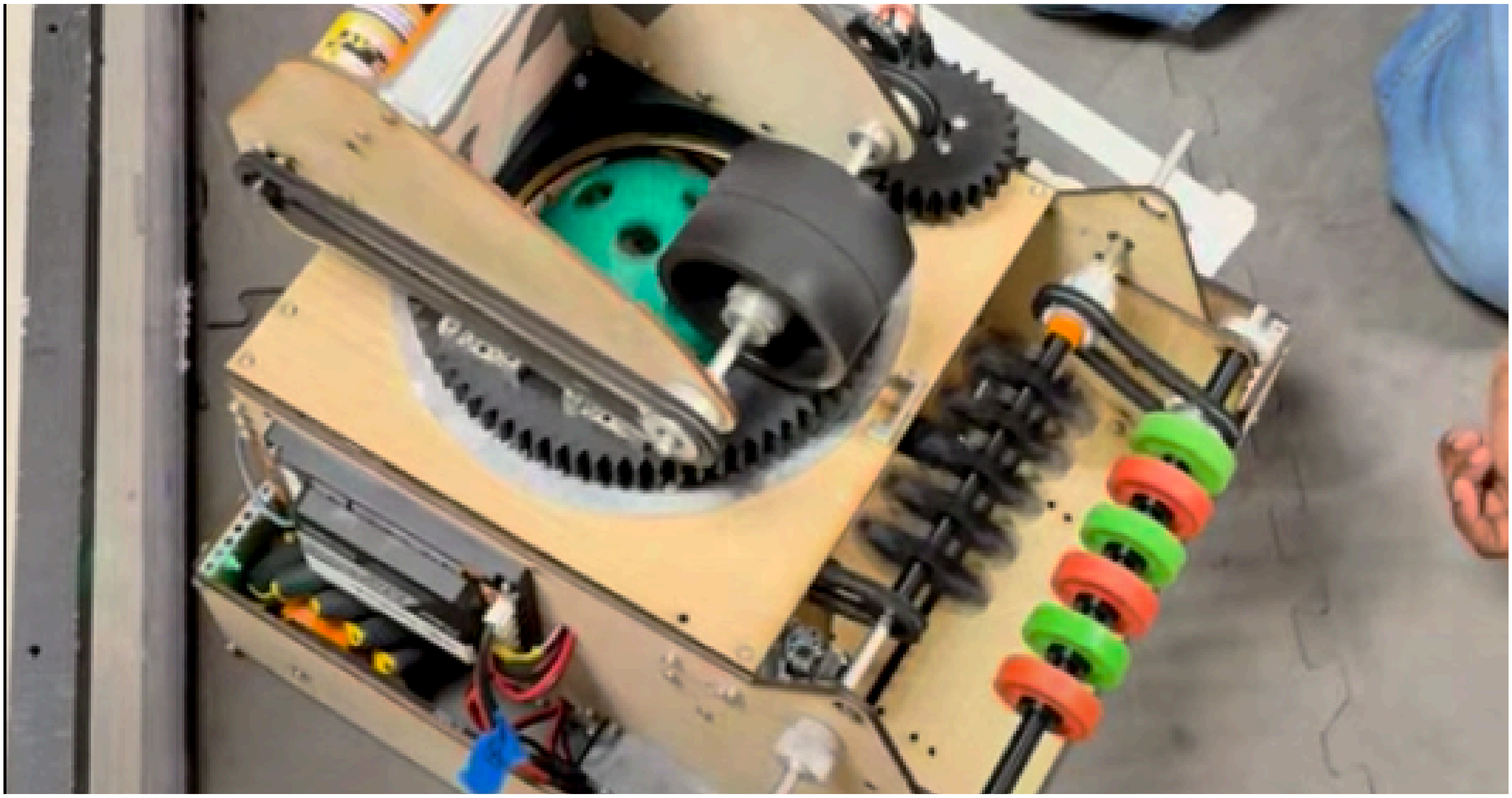



(Left) We had previously experimented with complaint intake pieces the week after kickoff and now felt they would be more appropriate. (Right) the Cad of the new idea for intake.

Version 1 .0

Brainstorm	CAD	Prototype	Build	Review

Date	*09/25/25- CAD Week 2*
Intake V1.25	
PURPOSE	We finally assembled together our new intake system. When we testing it, **the intake and ramp seemed to work very well**. However, with more intense testing, we realized that the artifacts would still get stuck between the transfer plates and the ramp at every few artifacts. Even after changing the angles from which we intaked at, **the intake was still very prone to jamming**. We realized that in order to improve our robot, **we would need to add guides** that would be able to reduce the amount of jamming that was occuring with the current iteration of the intake.

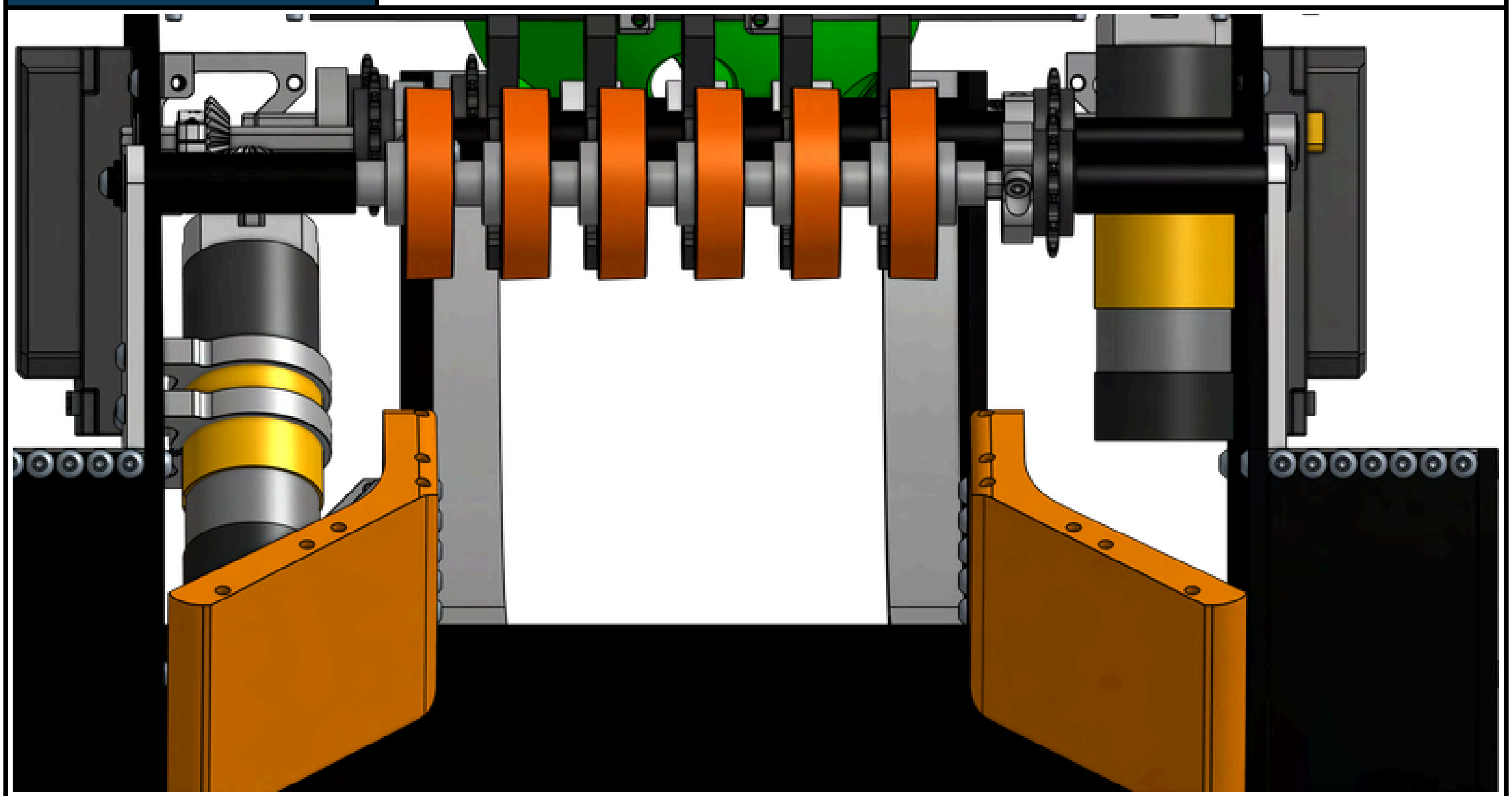


Picture of our intake above

Version 1 .0

Brainstorm	CAD	Prototype	Build	Review

Date	*10/10/25- CAD Week 3*
Intake V1.5	
PURPOSE	From our previous intake design, we decided that in order to better intake game elements, we would need to have some sort of guides to make the artifacts go to the transfer much more smoothly. Thus, we decided to add guides. **These guides were slightly more than 2.5 inches tall; just above half the height of the artifact**. This guide was thought be to a much better alternative to the previous ramp iteration. Also, we decided to **drill holes into the ramp to which we would use to screw the intake guides into place**. We hoped that this change would lead to faster cycle times.

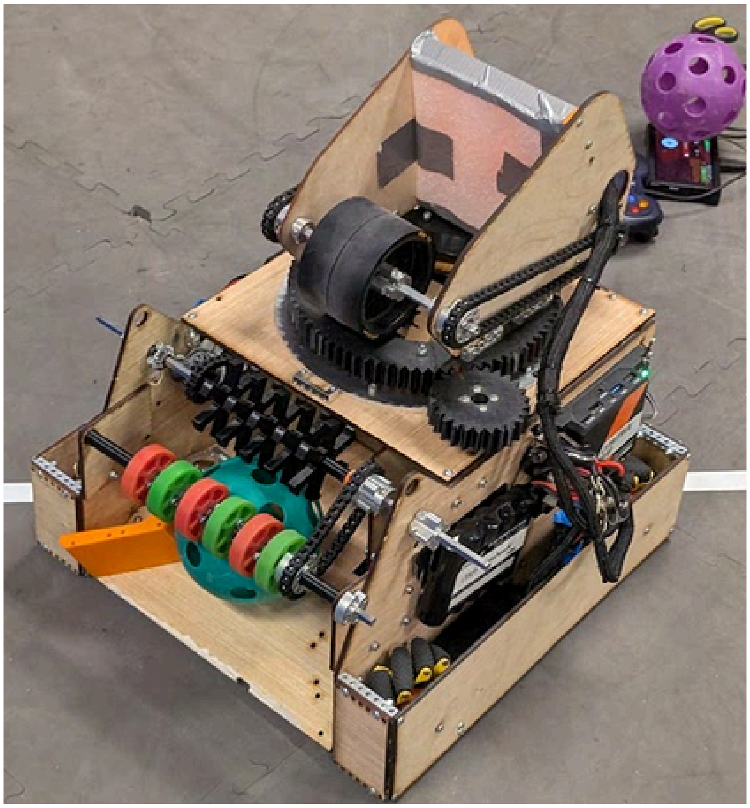
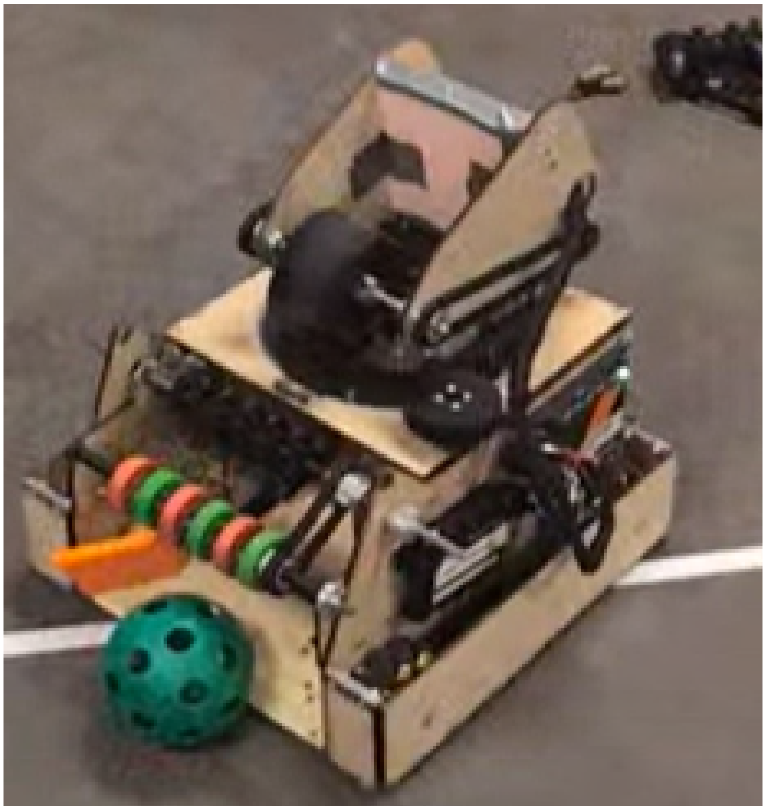


Above is our V3 Intake fully cadded out. We hoped that this new design would lead to less jams and be more efficient at intaking.

Version 1 .0

Brainstorm	CAD	Prototype	**Build**	Review

Date	10/15/25- CAD Week 4
Intake V1.5	
PURPOSE	After CADing the new guides, we **finally attached them to our ramp**. At first, they seemed to be doing their job. The artifacts were not getting jammed between the transfer and intake anymore and artifacts were smoothly able to transfer. However, after our testing, **we measured the new dimensions of the robot and realized that the dimensions of the robot exceeded 18 inches**. We tried our best to reCAD it to be within 18 inches, but realized it **was impossible due to the high angle of the wooden ramp which didnt allow the guides to either fit or be effective at guiding** the artifacts. Thus, we needed to once again redesign our intake.

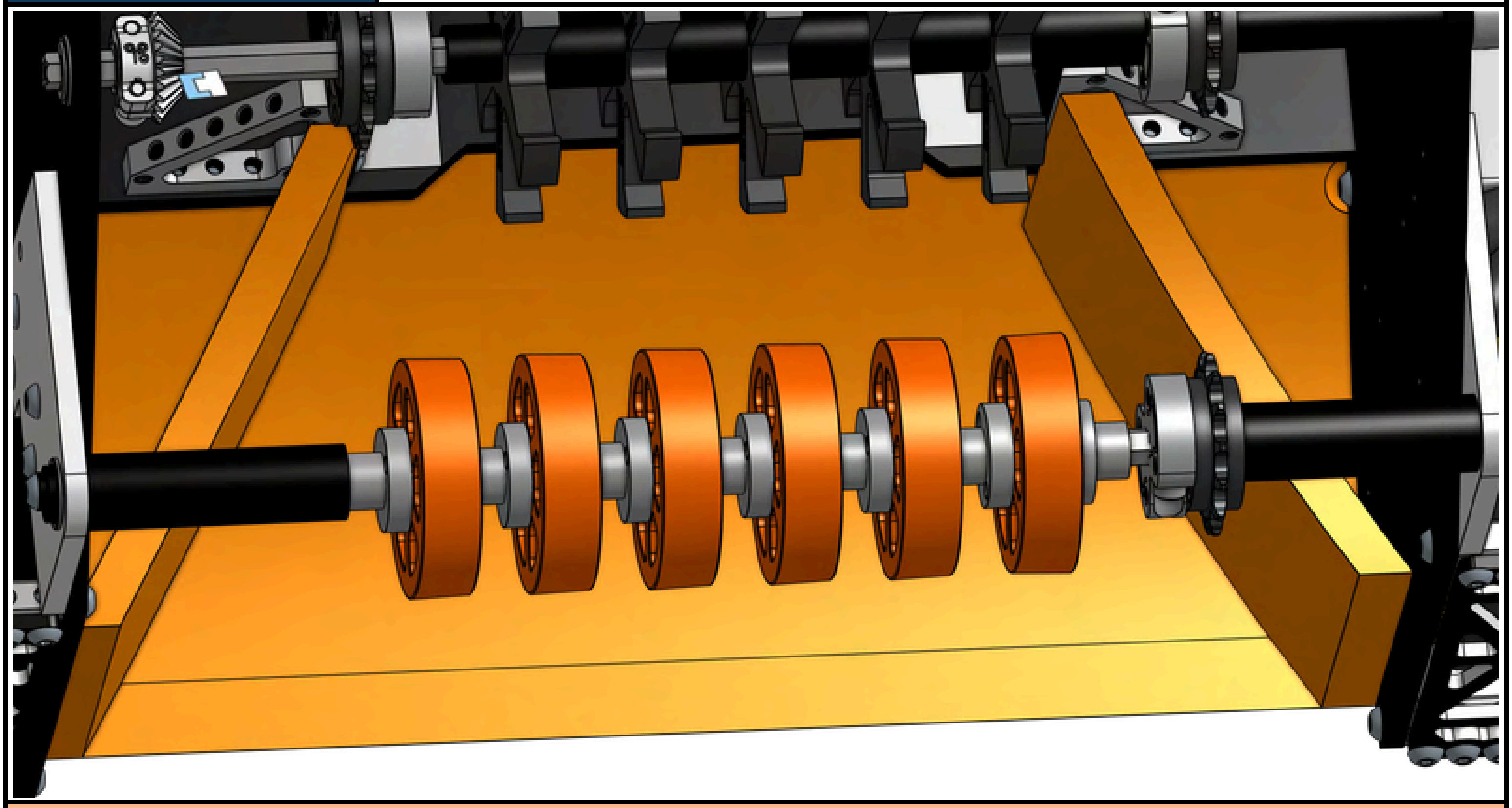



Picture of our intake above

Version 1 .0

Brainstorm	CAD	Prototype	Build	Review

Date	*10/17/25- CAD Week 4*
Intake V1.75	
PURPOSE	After testing our previous ramp and guides, we decided to create a completely new ramp, one that would be more curved yet would be able to **fit within the 18 by 18 by 18 inch constraint**. We hoped that by **curving the ramp more and making the incline less steep from the start**, we could score much faster and intake much faster. Our new **ramp was made out of PETG filament** and was printed with the guides directly implemented into the new ramp already. We also realized that **by having a greater curve on the ramp, the artifacts could roll farther into the robot initially** making it easier to intake the artifacts using our compliant intake.



Our V4 Intake Ramp displayed above

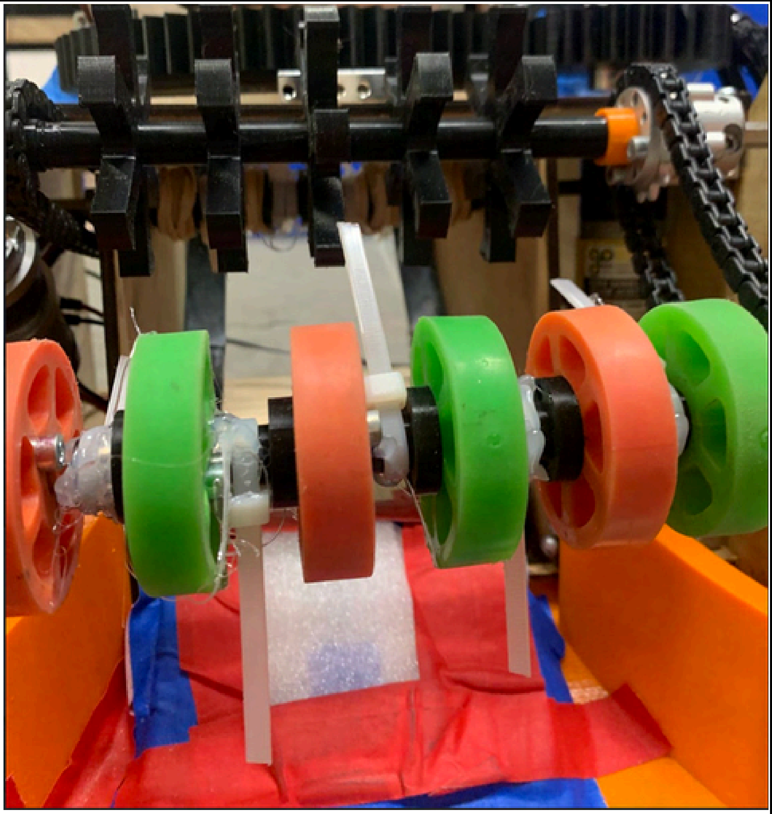
Version 1 .0

Brainstorm | CAD | Prototype | **Build** | Review

Date	*10/18/25- CAD Week 4*

Intake V1.75

PURPOSE	After originally CADding the ramp, we were excited to test it out. However, we realized that we would need to implement some changes to make it work better. One of our main issues is the **compliant intake didn't allow us to "grip" artifacts as well** when trying to intake. To alleviate this, we decided to **glue zipties onto the intake axle to allow our intake to have more reach** and help compensate for the disadvantage that was created from using compliant wheels. After testing, **we realized that this change would allow us to have much better autonomous consistency** and a much better robot overall.

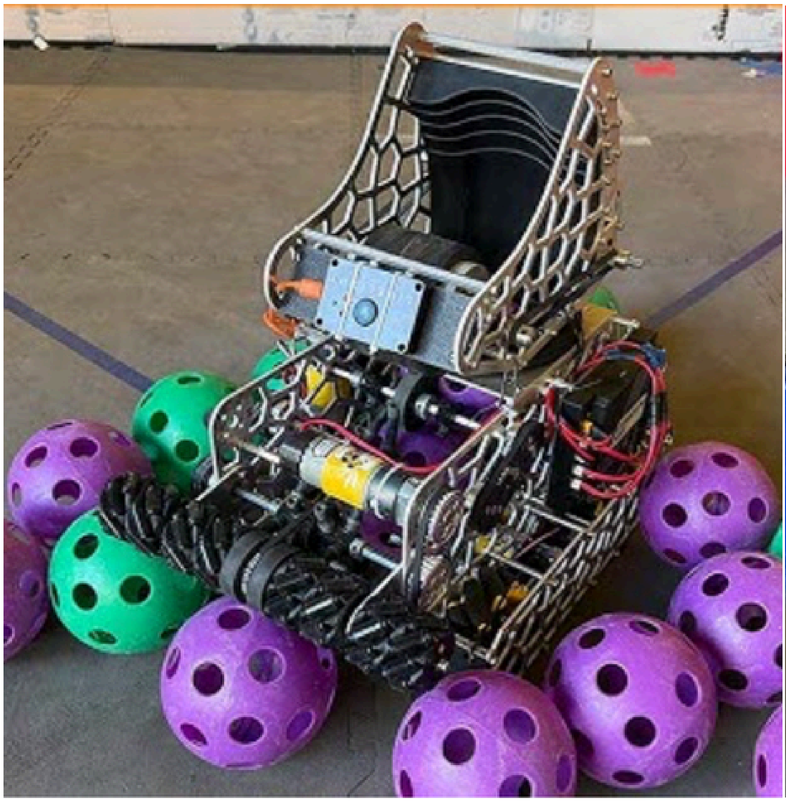


Pictures of our fully built intake including the newly added zipties above.

Version 2 .0

Brainstorm | CAD | Prototype | Build | Review

Date	*11/16/25- Version 2 Brainstorming Week 1*
Robot V2	
PURPOSE	From our competitions, we had soon realized that our current robot had **many flaws** in terms of the speed that the robot was able to cycle at. From our failures in our competition we realized that our robots **potential in terms of scoring was only about 120**. Thus, we knew we had to redesign our robot to have a higher scoring capacity and be able to cycle artifacts faster. Additionally, we wanted to be able to **create a more consistent robot overall**. We felt that there had been so many design flaws in our first iteration that a **redesign was necessary** to be able to continue to perform well.



Some of the robots we looked at for inspiration to plan a better V2 robot. Watching other teams and seeing what worked well for them was integral to our own success as it told us what would work well and what didn't

Decision Matrix

Brainstorm | **CAD** | **Prototype** | **Build** | **Review**

Date	***11/17/25- Brainstorming Week 1***
PURPOSE	We already knew some of the options we wanted to make our V2. So we decided to compare them side to side to see which one would be the best.

Intake Decision Matrix

Type	Simplicity	Size	Functionality	Cost	Total
Rubber Band Intake	5.0	3.0	5.0	3.0	**16.0**
Vector Wheel Intake	1.0	4.0	4.0	5.0	**14.0**
Current Intake	5.0	1.0	2.0	4.0	**12.0**

We rated our designs on 5 criteria from 1 to 5 (1 being the worst and 5 being the best). The best design was determined by the outtake with the most points.

Decision Matrix

Brainstorm | CAD | Prototype | Build | Review

Date	*11/19/25- Brainstorming Week 1*
PURPOSE	We already knew some of the options we wanted to make our V2. So we decided to compare them side to side to see which one would be the best.

Outtake Decision Matrix

Type	Simplicity	Size	Functionality	Cost	Total
Adjustible Hood shooter on a turret	4.0	4.0	5.0	5.0	**18.0**
Current Shooter	5.0	1.0	3.0	2.0	**11.0**
Mechanically linked two wheel shooter	1.0	2.0	5.0	2.0	**10.0**

We rated our designs on 5 criteria from 1 to 5 (1 being the worst and 5 being the best). The best design was determined by the outtake with the most points.

Decision Matrix

Brainstorm | CAD | Prototype | Build | Review

Date	*11/21/25- Brainstorming Week 1*
PURPOSE	We already knew some of the options we wanted to make our V2. So we decided to compare them side to side to see which one would be the best.

Transfer Decision Matrix

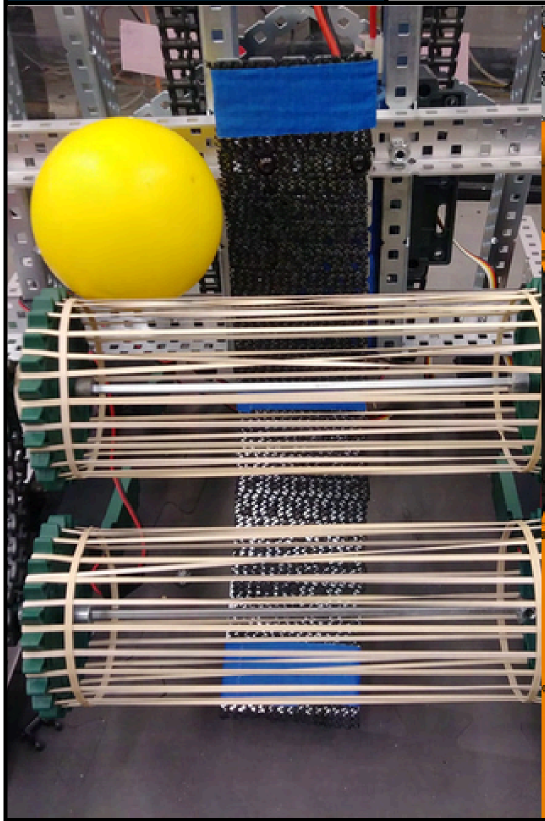
Type	Simplicity	Size	Functionality	Cost	Total
Layers of surgical tubing and ramp remade	4.0	5.0	5.0	4.0	**18.0**
Gecko Wheels Layers	5.0	2.0	2.0	3.0	**12.0**
Current Transfer	5.0	2.0	2.0	2.0	**11.0**

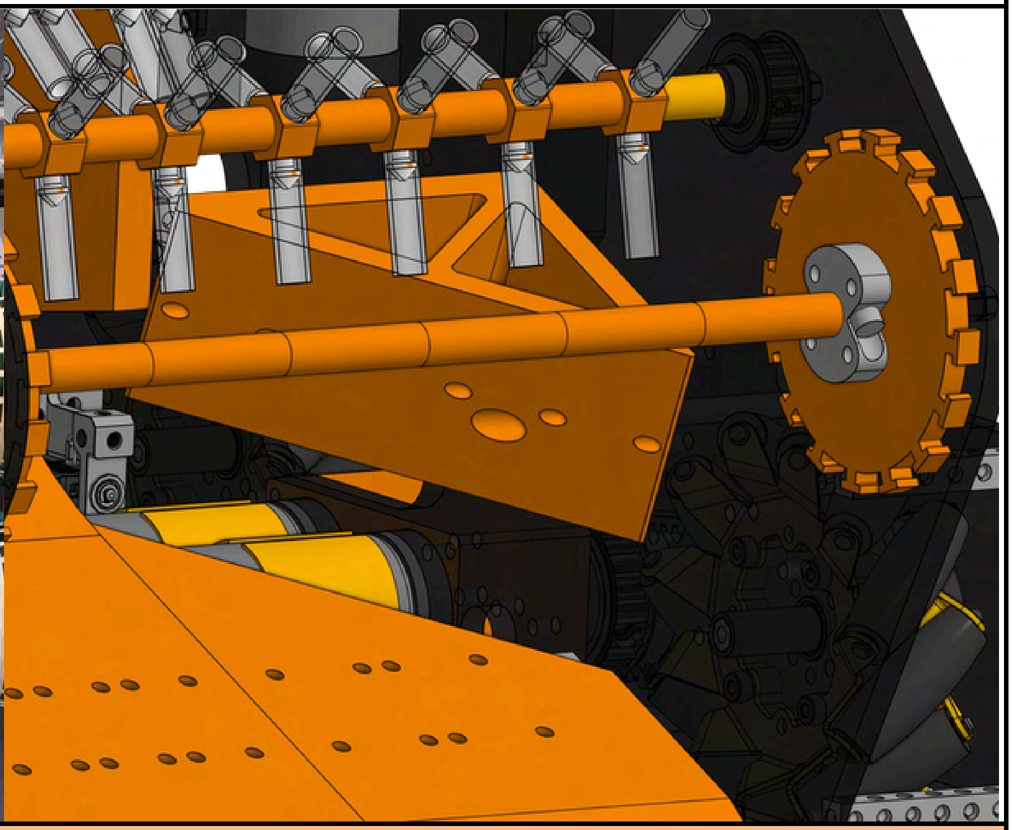
We rated our designs on 5 criteria from 1 to 5 (1 being the worst and 5 being the best). The best design was determined by the outtake with the most points.

Version 2 .0

Brainstorm	CAD	Prototype	Build	Review

Date	*11/23/25- CAD Week 1*
Intake V2	
PURPOSE	For our remade Intake, we settled on a Rubber band intake, **inspired from VEX robots** that had implemented it. The benefit of using a Rubber Band Intake was that the rubber bands made it bend to the **shape of the artifact and gave much more grip when pulling the artifact in**. One drawback that we analyzed from the videos was that this type of intake was **very easily breakable** due to the weakness of rubber bands. Thus, although it would be costly, we decided to **pursue the idea due to its successful implementation in both FTC and VEX**.





(Left) Our inspiration for our intake design. (Right) our intake designed in cad(without the rubber bands)

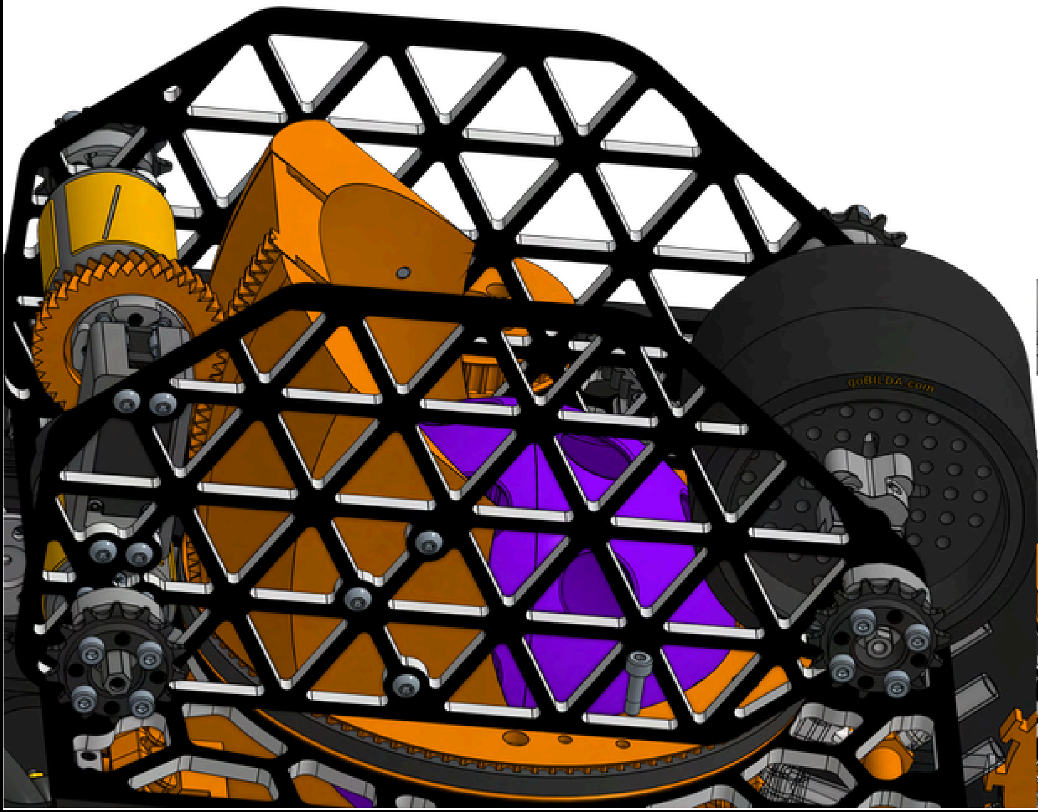
Version 2 .0

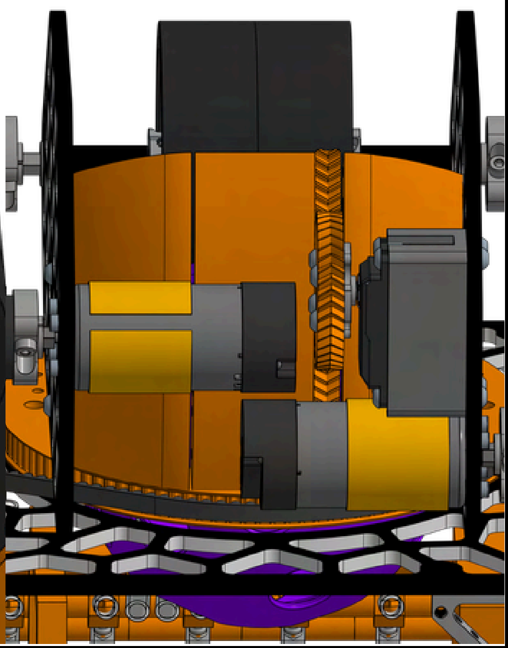
Brainstorm	CAD	Prototype	Build	Review

Date	*11/24/25- CAD Week 1*

Shooter V2

PURPOSE	We also decided to remake our shooter to be better. Our main goal from the start had been to be able to achieve auto lock by our December competition. **auto lock meant that your turret was able to slowly move to adjust for your robots movements** and theoretically allowed you to shoot from anywhere. This would make almost all **your shots consistently go in**. In order to achieve this, besides turret, we wanted to be able to make an adjustable hood. **This would allow for the angle the hood would shoot from** to be able to **change with the distance** we were shooting from.



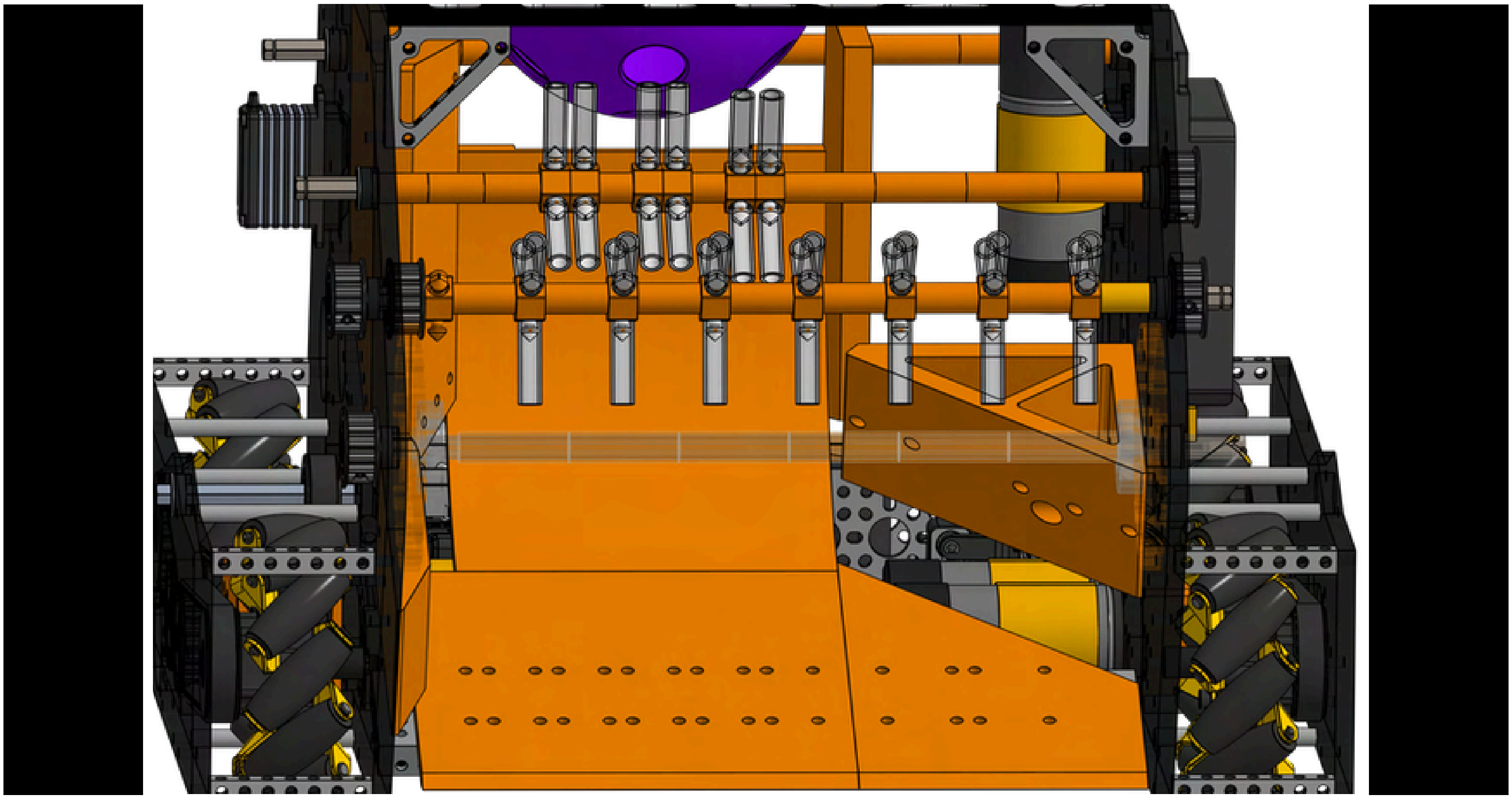


Our design idea for our new shooter. we hoped by adding a hood, our shooter would be more accurate.

Version 2 .0

Brainstorm	**CAD**	Prototype	Build	Review

Date	*11/25/25- CAD Week 1*
Transfer V2	
PURPOSE	We also attempted to remake our transfer system. This time around, we wanted **more stages** the ensure that the artifacts would transfer all the way through to the shooter outtake. One issue that we faced earlier in our robot was **that the artifact was unable to go through to the shooter** causing a few seconds delay between the transfer and the outtake. This **led to our cycles being much slower which was affecting our ability in getting the ranking points we needed to rank highly**. By having a more curved ramp geometry at the end and by designing to have our tubing there, we were able to **improve our cycle times drastically.**

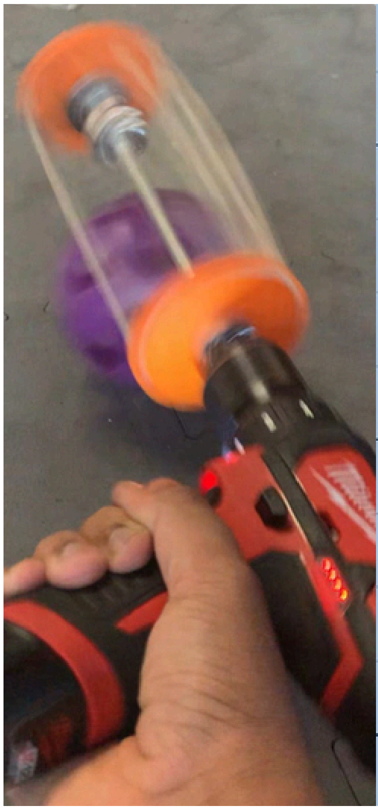


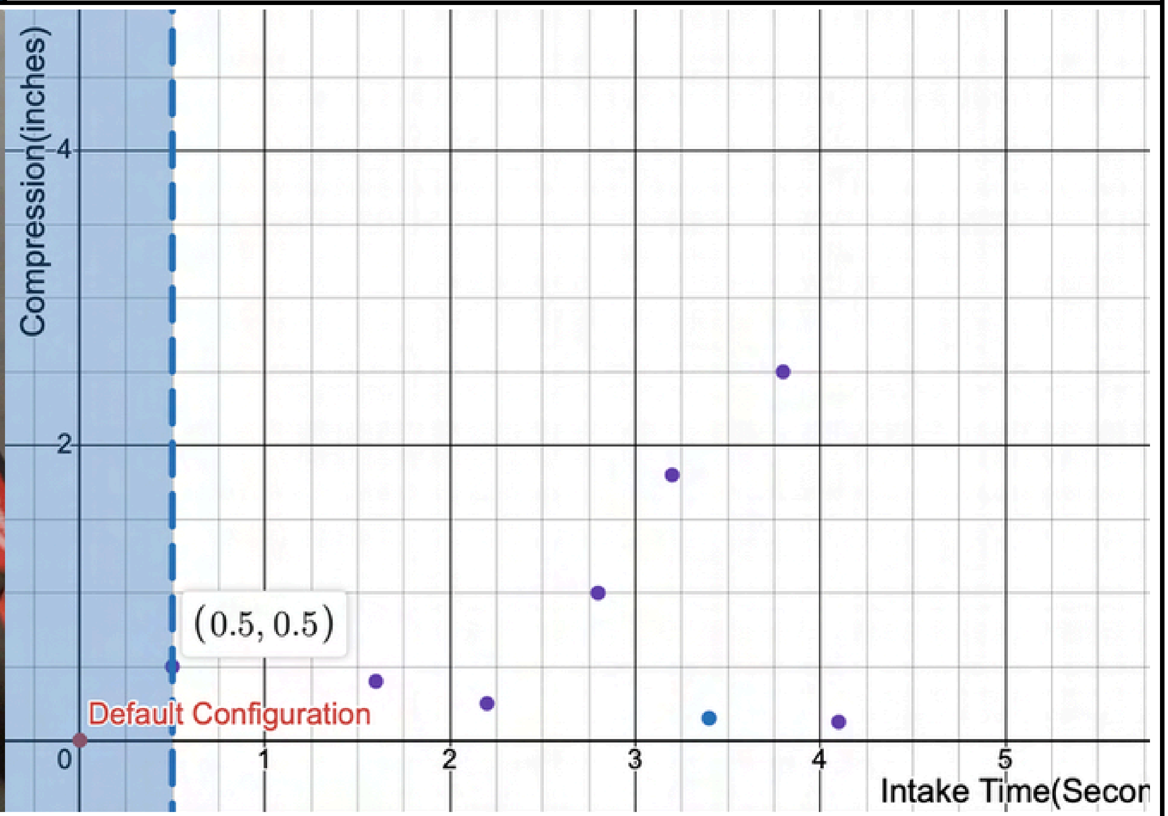
Our newly designed Transfer Geometry meant to help the artifacts travel faster from the intake to the shooter. This geoemtry change allowed our transfer to be much faster.

Version 2 .0

Brainstorm | CAD | **Prototype** | Build | Review

Date	*11/26/25- CAD Week 1*
Intake V2	
PURPOSE	Since we had heard about the high chances of rubber bands breaking under stress, we ourselves decided to **test the compression** that rubber bands could sustain before snapping. We attached our printed rubber band intake inserts, attached the rubber bands, and began to test. After a lot of testing, we found that **thicker rubber bands tended to snap less** and that the ideal compression for the rubber bands that would be the best for longevity and speed would be around half an inch of compression. The issue we faced was that **if the artifact was over compressed it would transfer slowly and if it was under compressed the artifact could slip and be ineffective**. Thus, we tested to find an optimal value, and adjusted our cad to be able to meet that value more accurately.



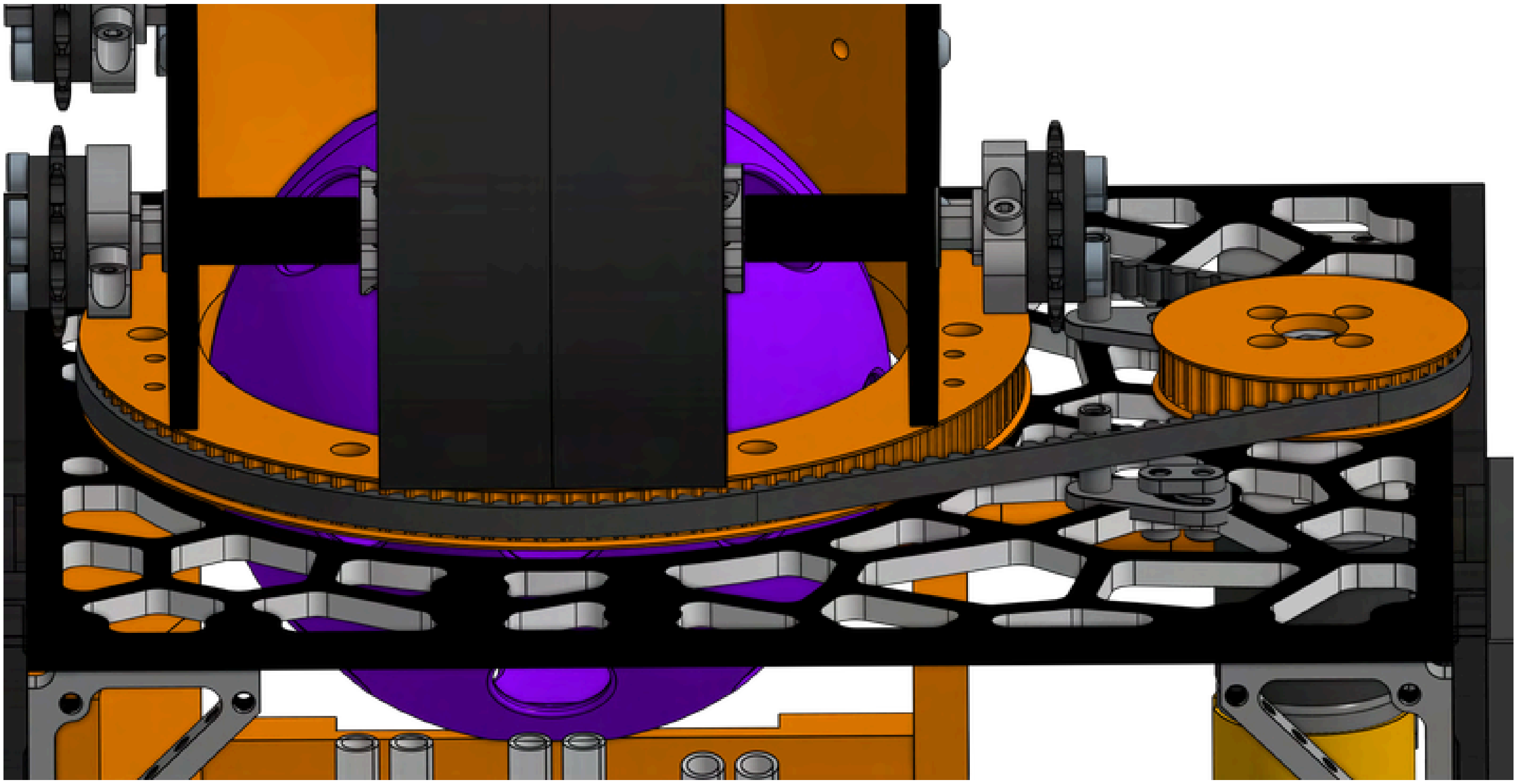


Above are pictures of us testing and recording the compression on the intake and the time that it took to fully move the artifact out of the intake system from first contact. **(0.5,0.5) optimal.**

Version 2 .0

Brainstorm | **CAD** | Prototype | Build | Review

Date	*11/27/25- CAD Week 1*
Turret V2	
PURPOSE	We also decided to recad our turret. Before we had done a geared system using **big gears to run our turret**. Now, we realized that the big gears had made it difficult for software to set positions for the robot's turret and had **backlash**. Thus, we decided to **switch to a belted system to drive the turret**. Using a belt system would allow us to more accurately measure the values of the turret in different positions which was pivotal in making the turret able to **"auto lock"** on targets. Also, we decided to have the turret bearings rotate on the wood itself. Another change that we made was making the **diameter of the turret itself bigger** so that the artifact could fit through more easily.

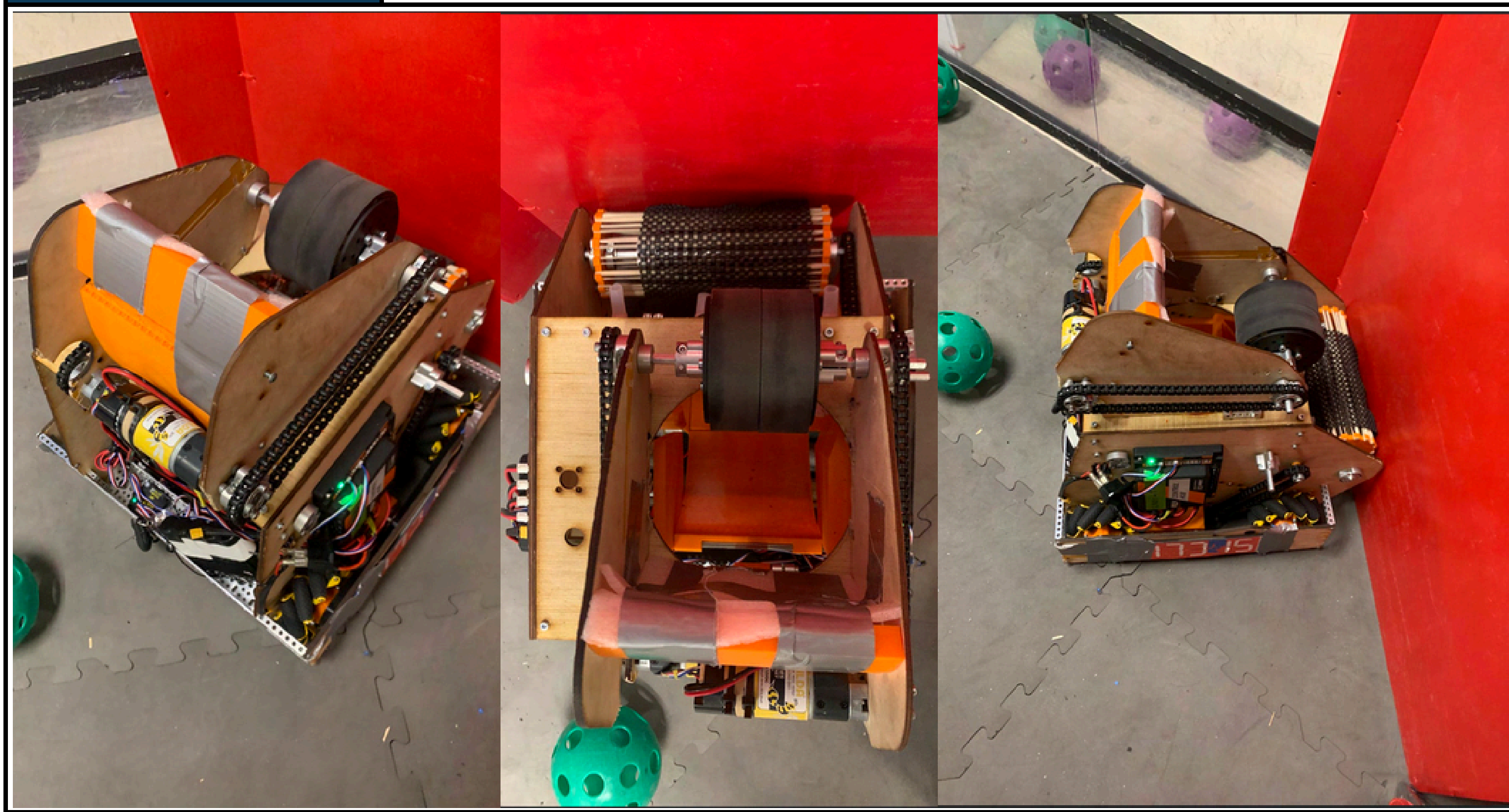


Picture of our newly made turret system. This turret sysyem would hopefully allow us to achieve auto lock onto the goals from wherever on the field.

Version 2 .0

Brainstorm | CAD | Prototype | **Build** | Review

Date	***11/28/25 - 12/2/25 Build Week 1***
Robot V2	
PURPOSE	Over the next 4 days, we built together our robot. Due to the proximity of our next competition to when we had began to brainstorm, we knew we needed to assemble fast. Soon, while testing our robot, we realized that **two major problems** were occurring, the **turret was very inconsistent in its smoothness**, and the **shooter was shooting straight up in the air despite changing the hood angle**. We did some debugging and realized that we had miscalculated the **wheel size that we needed to use. Instead of 72mm, we used 96mm** for our shooter's wheels. However, since there wasn't enough time to get new wheels, we used the old one.



Picture of our robot construction above for our League Meet 3.

Version 2 .0

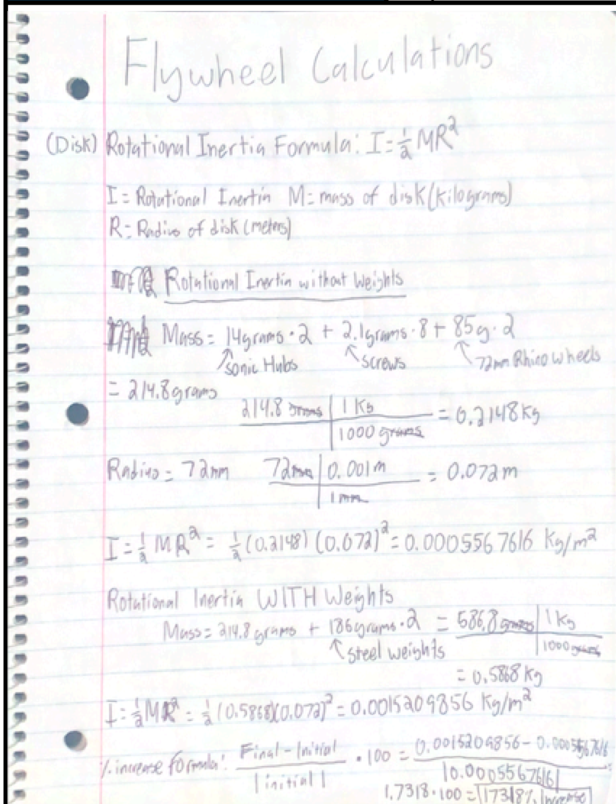
Brainstorm	CAD	Prototype	**Build**	Review

Date	*11/28/25 - 12/4/25 Build Week 1*

Robot V2

PURPOSE

Along with the changes that we made listed above, we also decided to add flywheel weights that we had recently ordered. We ended up using **VEX flywheel weights** to also increase our rotational inertia. Increasing our rotational inertia would allow us to have **less backspin(less bounce) on our shots**. Despite the tradeoff on a longer spin up time(time for flywheel to get up to top speed), it was a positive one, a **greater rotational inertia** would allow us to be more consistent and the **longer spin-up time was able to be accounted for by more driver practice**. Thus, for long term gain, we decided to add weights to our flywheel.



Flywheel Calculations

(Disk) Rotational Inertia Formula: $I = \frac{1}{2}MR^2$

I = Rotational Inertia M = mass of disk (kilograms)
R = Radius of disk (meters)

Rotational Inertia without Weights

Mass = 14grams · 2 + 2.1grams · 8 + 85g · 2
Sonic Hubs, screws, 72mm Rhino wheels
= 214.8grams

$\frac{214.8 \text{ grams}}{} \cdot \frac{1 \text{ Kg}}{1000 \text{ grams}} = 0.2148 \text{ Kg}$

Radius = 72mm $\quad \frac{72\text{mm}}{} \cdot \frac{0.001\text{m}}{1\text{mm}} = 0.072\text{m}$

$I = \frac{1}{2}MR^2 = \frac{1}{2}(0.2148)(0.072)^2 = 0.0005567616 \text{ Kg/m}^2$

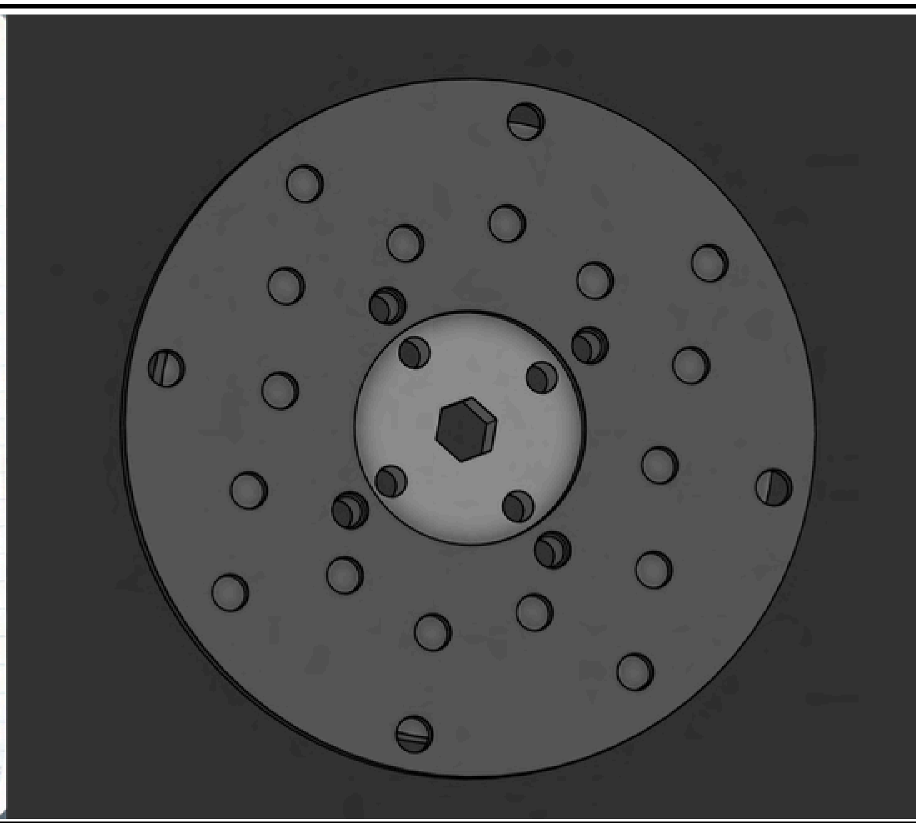
Rotational Inertia WITH Weights

Mass = 214.8 grams + 186grams · 2 = $\frac{586.8 \text{ grams}}{} \cdot \frac{1 \text{ Kg}}{1000 \text{ grams}}$
steel weights
= 0.5868 Kg

$I = \frac{1}{2}MR^2 = \frac{1}{2}(0.5868)(0.072)^2 = 0.0015209856 \text{ Kg/m}^2$

% increase formula: $\frac{\text{Final} - \text{Initial}}{|\text{initial}|} \cdot 100 = \frac{0.0015209856 - 0.0005567616}{|0.0005567616|}$

$1.7318 \cdot 100 = \boxed{173.18\% \text{ Increase}}$

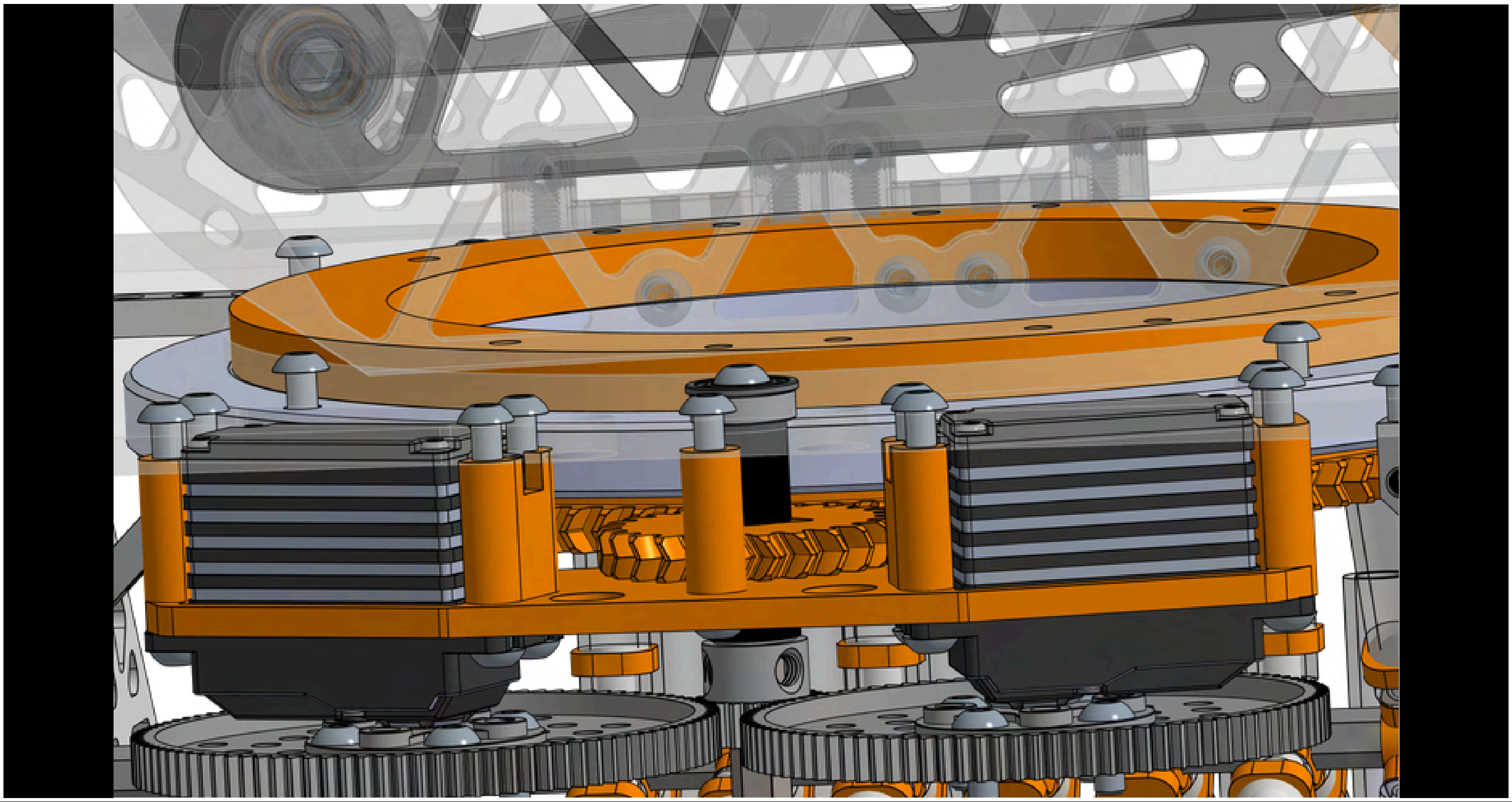


By doing the calculations, we found that by adding only TWO flywheel weights, we would increase our rotational inertia by **173.18%** of its original value without weights, proving how beneficial weights were for our flywheel's accuracy.

Version 3 .0

Brainstorm	**CAD**	Prototype	Build	Review

Date	*12/12/25 CAD Week 2*
Turret V3	
PURPOSE	For our league championship, we decided to simply continue to iterate on our mechanisms. One aspect that we added back was our turret. However, we used **an Andymark bearing** compared to the traditional custom bearing stack turret. This allowed us to have **consistency with the movability of the turret** which would allow our software team to tune the turret to auto lock much better. Another change we made was **driving our turret using 2 servos and a gear system to transmit the power of two metal gears to a singular, smaller gear** which would drive the overall turret in a system of two more printed gears.

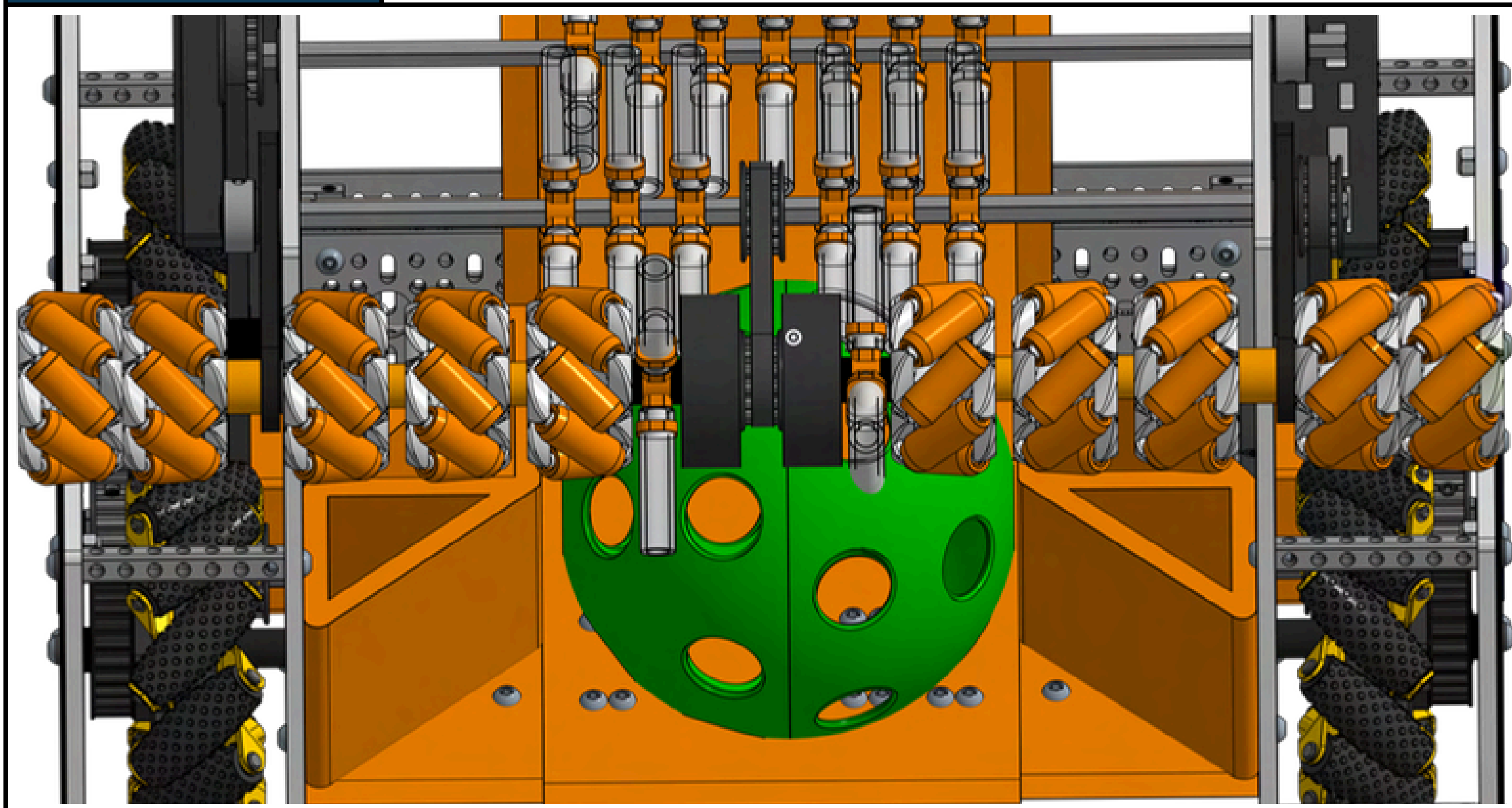


Above is the gear relations from our two servos to our turret. This usage of servos saved us a motor slot which was used in our transfer instead.

Version 3 .0

Brainstorm	**CAD**	Prototype	Build	Review

Date	12/14/25 CAD Week 3
Intake V3	
PURPOSE	Using inspiration from other designs, we decided to try using **vector wheels, wheels that were slanted at a 45 degree angle similar to a mecanum wheel**. These wheels allowed us to draw all the artifacts in a general area towards the center on our intake, where we placed two gecko wheels. This setup allowed us to almost **"pull" the artifacts in**, similar to a **whirlpool**. We also decided to make this mechanism as a 4bar linkage, a non powered bar which **naturally sits at an optimal position, to be decided through testing**. This allowed it to pull artifacts well, with the ability to adjust it later if needed.

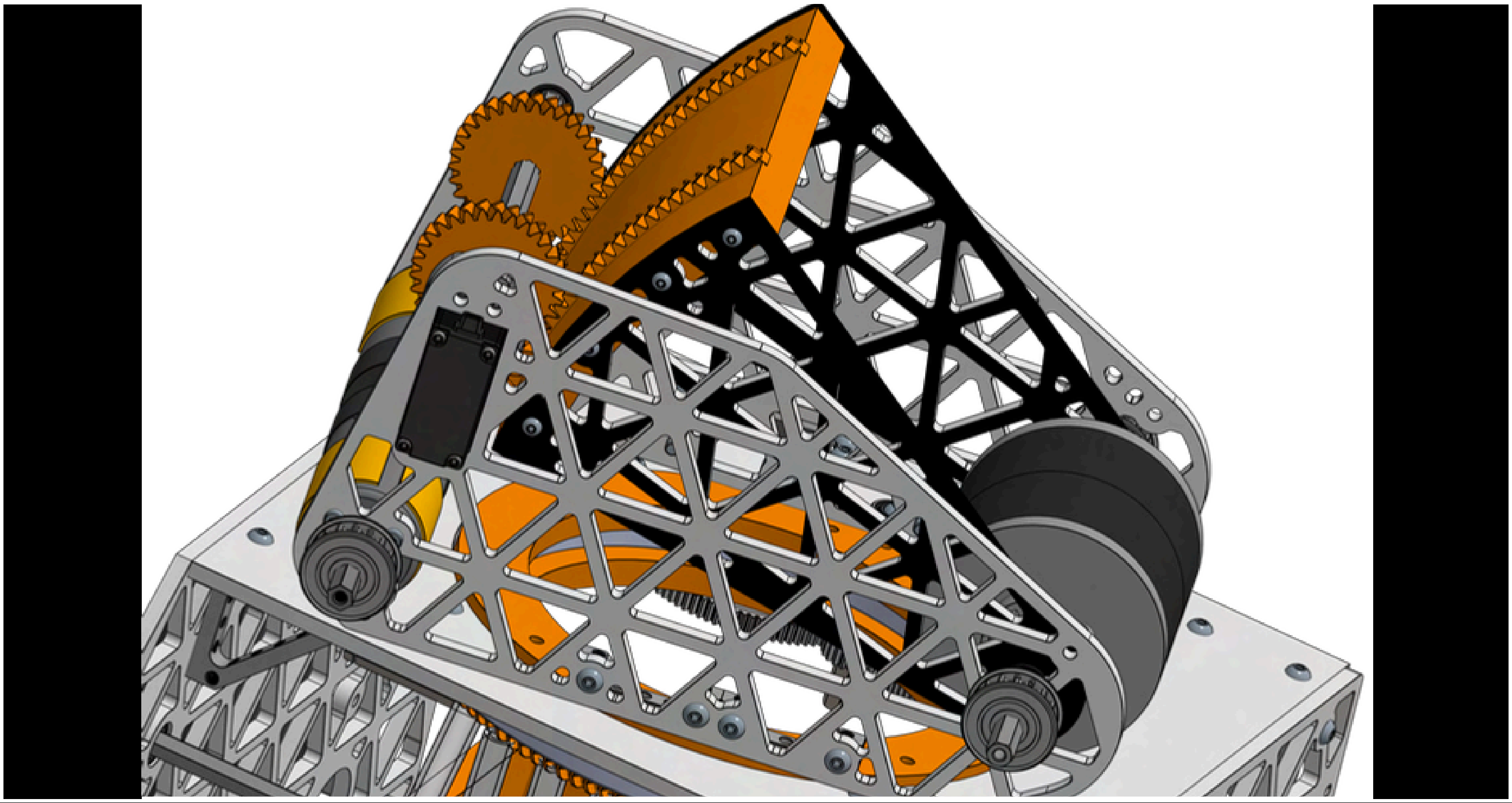


Our vector wheeled intake design in cad. In theory, the vector wheel intake would be belted to the first stage of transfer from which both were driven by the same motor.

Version 3 .0

Brainstorm	CAD	Prototype	Build	Review

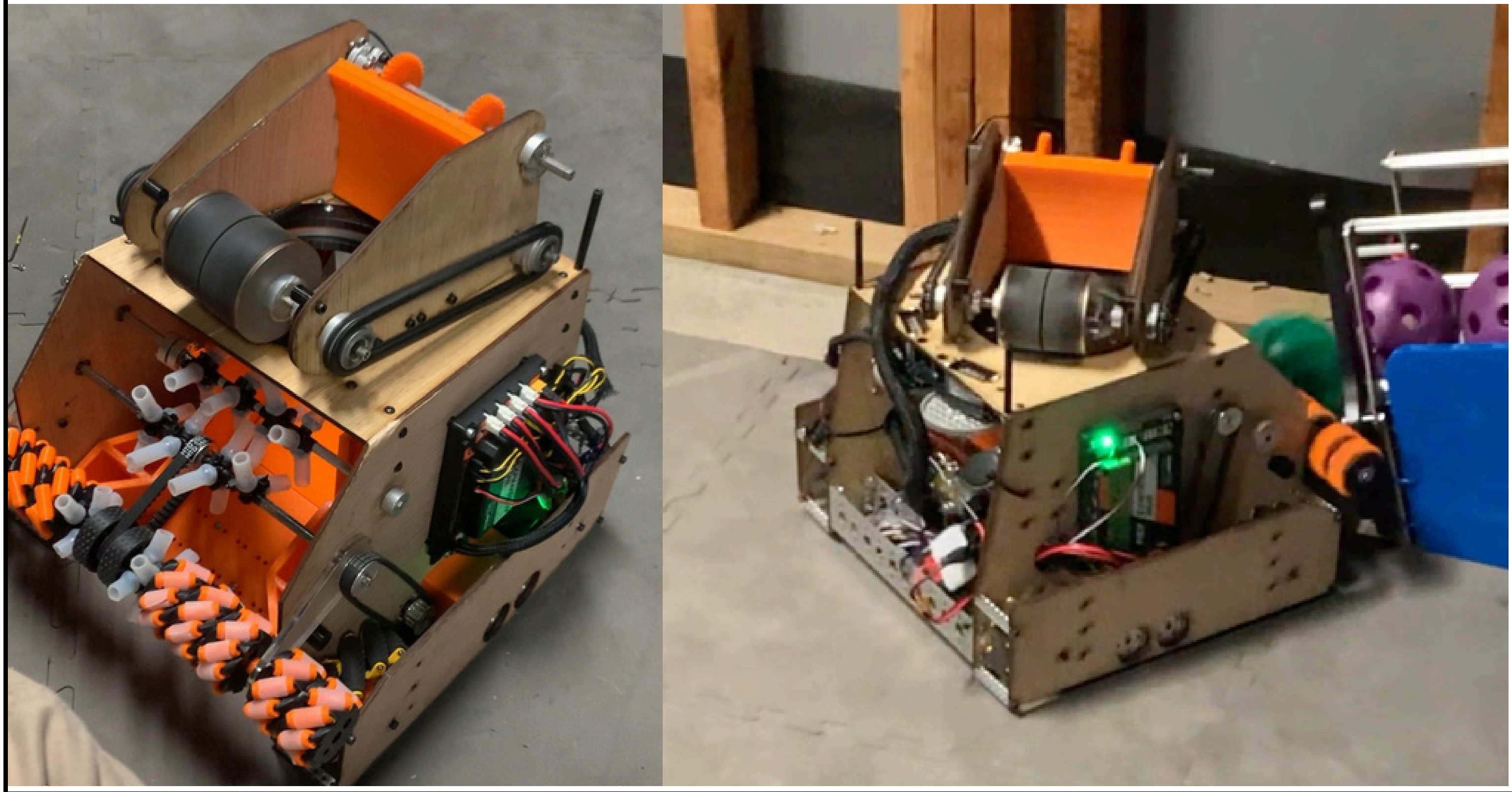
Date	*12/14/25 CAD Week 3*
Outtake V3	
PURPOSE	We also decided to redo our outtake geometry. Although the same in concept, we replaced the bearing based hood with a completely seperate part hood. **In our new hood geometry, we separated the adjustable hood from the base shooter.** This allowed us to be able to adjust our hood separate from the main geometry and allowed us to create a more rigid hood. **Our new design used manufactured wood for the hood which was much more stable and rigid compared to the original design** which relied on bearings and 3d print. **Due to the 2nd design being more rigid**, and working better overall, we decided to stick with it.



Above is our newly designed Outtake geometry

Version 3 .0

Date	***12/19/25 - 12/28/25 Build Week 1-2***
Robot V3	
PURPOSE	After 2 weeks, we were able to fully assemble together our new robot despite setbacks from the holidays where the entire team couldn't meet. **One addition we made after finishing building were a hardstop on the shooter hood** and **black standoffs to act as channels for our turret wiring**. Throughout the build process, we faced many challenges. One of which was the issue of **screw sinkage** which had greatly hurt our robots that we had designed in the past because the screws would sink inside the wood and damage its strength. **Instead, we decided to sink washers into the wood** which would **damage the wood less**.

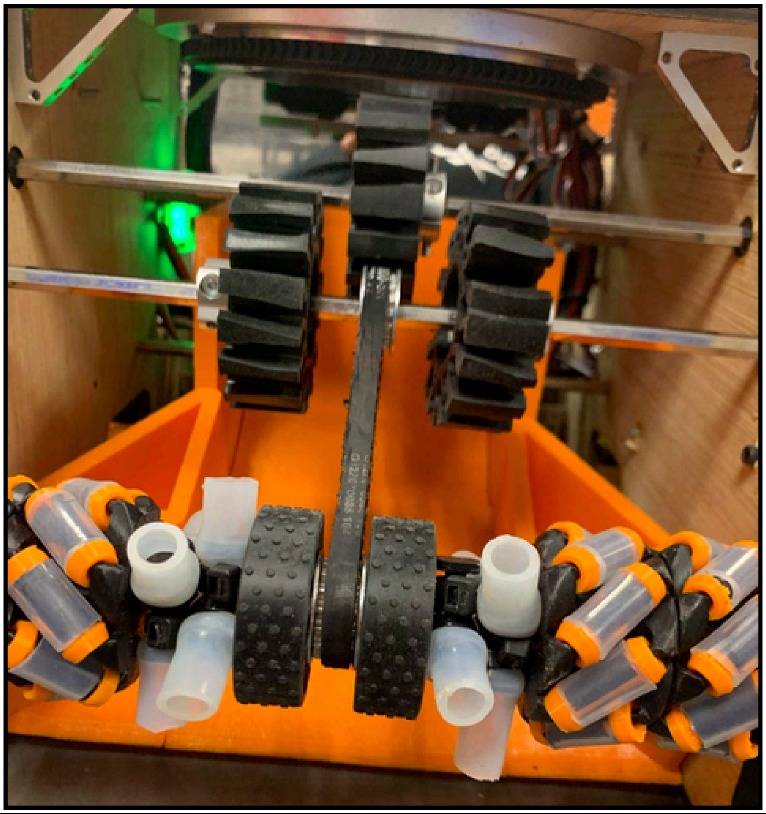


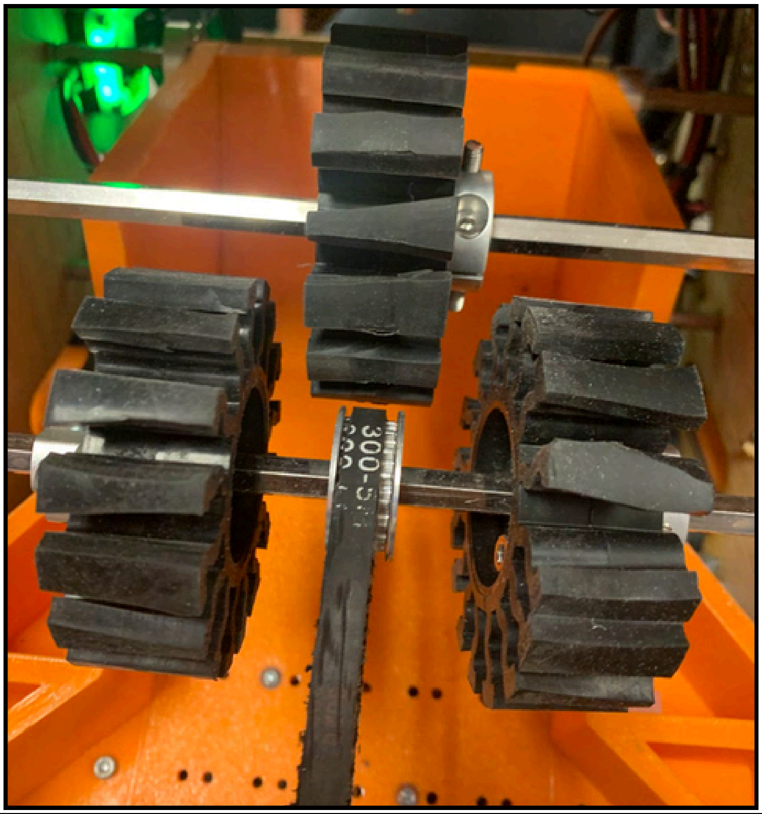
(Left) Sinking washers into the wood, then attaching screws to reduce damage to the wood.
(Right) Practicing how our robot would intake in a real game with our new robot.

Version 3 .0

Brainstorm	CAD	Prototype	**Build**	Review

Date	*01/10/25 Build Week 3*
Robot V3	
PURPOSE	We realized through many runs that the **tubing was expensive** to continue to replace as many of the tubing cores would break and the tubing itself would fall off. We realized that if we continued with this approach, it would be costly and waste a lot of budget and material. **We wanted to switch to something that would break less often like gecko wheels**, but the issue with gecko wheels were that their **intake radius was higher than tubing.** To try and get the best of both worlds. **we decided to cut the geckos to create a slightly lower radius** while maintaining good structure of the geckos.





Above the newly adjusted geckos, by cutting the geckos, we cut almost 4mm in terms of total radius when the geckos wheels would spin which would give the transfer more flexibility.

Version 3 .0

Date	***02/05/25 Build Week 7***
Robot V3	
PURPOSE	After allowing software to continue to run tests with the robot, we quickly ran into an error, it became clear that despite our attempts, the vector wheels greatly lacked grip causing slippage when attempting to grab onto to artifact. **To alleviate for this, we switched them out for gecko wheels in the middle** to ensure we would be able to have a **better grip**. Despite vector wheels being initially presumed to be good, **we soon realized a combination would need to be used due to the vector wheels proneness to breaking very easily**. We also would add **grip tape** to our ramps base to allow balls to flow in easier.

Newly switched intake using gecko wheels. Before by using vector wheels, the plastic material of vector wheels had made it very difficult to maintain a constant compression on balls due to the rollers being angled. This made it very difficult to intake at our competition.

Version 3 .0

Brainstorm | CAD | Prototype | Build | Review

Date	*02/09/25 Build Week 8*
Robot V3	
PURPOSE	Today was the first day after our competition so before we made any changes we took a bit of time to understand **what went well** at the competition as well as **aspects of the robot that we could improve** upon from the last competition. Overall, we reaized that servo turret had been very consistent and realized that day that we should attempt to **cater more to our software teams capacity if we wanted to succeed**. Another issue we had faced at our League Championship was **jamming issues on the intake which killed our cycles**. So, we began to brainstorm how to alleviate these issues and ensure these issues wouldn't come up again.

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Version 3 .0

Brainstorm	**CAD**	Prototype	Build	Review

Date	***02/10/25 Build Week 8***
Intake blank and Outtake blank	
PURPOSE	First change we made was in CAD. Since we had to play in the Semi Area Championship in one week, we decided to priortize consistency. We **decided to make our turret stationary**. We had made this decision beforehand before our lm3 due to a lack of time which had worked well. Once again, since we had only a week till our competiton and our turrret wasnt working well.Thus, for our competiton, we made it stationary.

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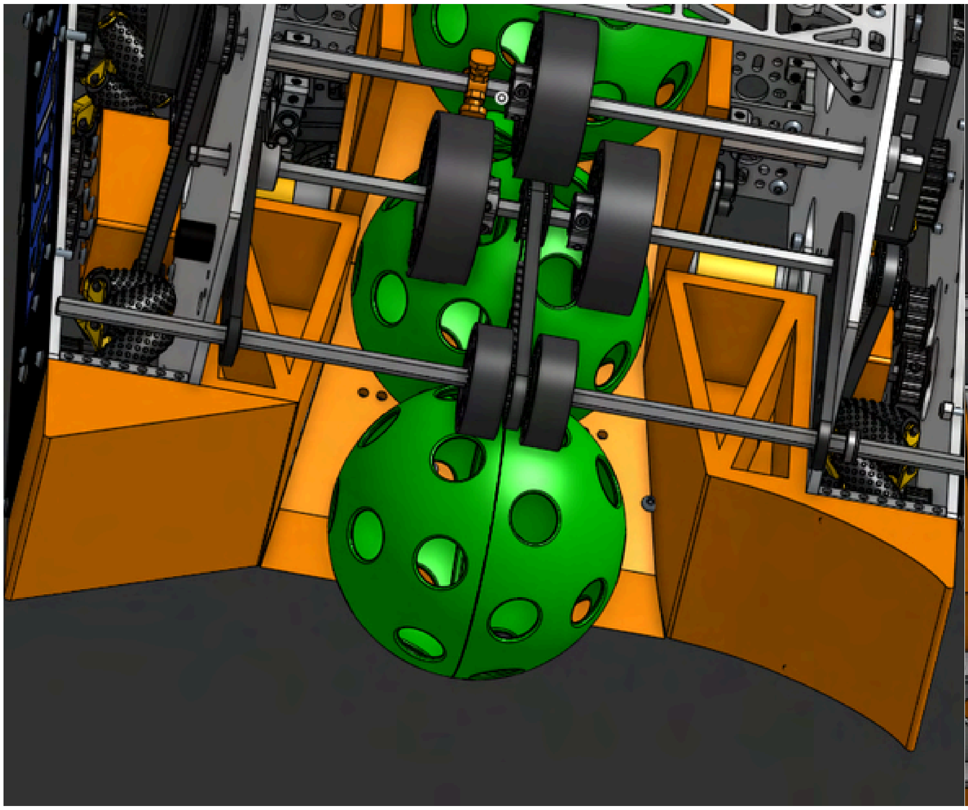
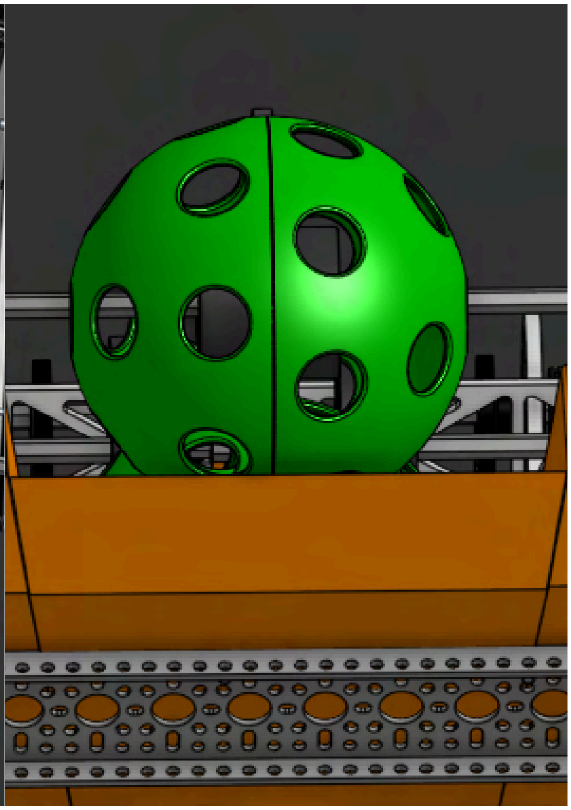
Version 3 .0

Brainstorm	**CAD**	Prototype	Build	Review

Date	02/11/25 Build Week 8
Intake blank and Outtake blank	
PURPOSE	Another change we made was on our intake guides. Before the symmetry of them had **led to jams when intaking two balls. We thus changed it to be asymmetrical**, a change we had seen on other teams facing **similar issues such as Turtle Walkers**. This asymmetrical ramp geometry would make it much less likely we would face jams. **Along with our guides, we decided to extend our intake ramp further out** to make it easier for balls to contact the intake. We hoped by extending the ramp it would lead to faster intaking times, an issue we faced with our intake at our league championship that **we hoped to fix before the Semi-Area Championship.**





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Version 3 .0

Brainstorm | CAD | Prototype | **Build** | Review

Date	***02/12/25 Build Week 8***
Robot V3	
PURPOSE	Another change that we made was changing how we had **countersprung our intake**. Before, we had **used Surgical Tubing** since that was the option we had at hand. Although Surgical Tubing worked well short term. **It wouldnt last long due to being very fragile and breaking on contact**. We decided to replace it with **Bungee cord, a popular option that we had seen used for counterspringing on misumi slides for both the Into the Deep and Centerstage Season**. The bungee cord was much more **resilient to hits and held its tension much better than the surgical tubing** we had used before. We also added Copper Tape to the back of our hood to reduce shooter backspin.

Picture of our newly updated intake counterspringing(Left) and Copper Tape on our outtake(Right). Copper Tape had been shown to reduce backspin on the outtake and thus we decided to try it for ouselves to see its effects.